

Spectra Manager™

Spectra Analysis

Software Manual



Preface

This instruction manual serves as a guide for using this software. It is intended to instruct first-time users on how to properly use the software, and to serve as a reference for experienced users.

Before using the software, read this instruction manual carefully, and make sure you fully understand its contents. This manual should be easily accessible to the operator at all times during software operation. When not using the software, keep this manual stored in a safe place. Should this instruction manual be lost, order a replacement from your local JASCO distributor.

Note: With this software you can use the same graphic user interface to analyze a wide variety of data from various spectroscopic instruments. This manual explains all the functions offered by this software using data from a JASCO spectrometer. We have tried to ensure that all functions are explained clearly for users of any JASCO instrument compatible with this software, but if you cannot find an explanation for a specific function please contact your local JASCO representative.

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Notation Used

The following notational conventions are used throughout this manual:

General Notation

Notation	Meaning
[Measurement] menu [Parameters...] command	Names of menus, commands, and text boxes are enclosed in square brackets '[]', followed by a description indicating whether the function is a menu, command, text box, etc. Shortcut keys used to select menus or commands are underlined.
<OK>, <Cancel>	Names of buttons are enclosed in angular brackets '< >'.

Keyboard Operations

Notation	Meaning
Shift Ctrl	The key is enclosed in a square and shown in boldface.
Alt , F	Keys that are to be pressed in succession are separated by commas. In the example shown on the left, the Alt key is to be pressed and released, followed by the F key.
Shift + →	Keys that are pressed simultaneously are separated by a "plus" sign. In the example shown on the left, press the → key while holding down the Shift key.

Mouse Operations

Notation	Meaning
Point	Move the mouse pointer to the specified item.
Click	Quickly press and release the mouse button.
Double-click	Click the mouse button twice in rapid succession.
Drag	Point to an item, click and hold down the mouse button. Move the mouse with the button held down, and release the button when the pointer is in the desired position.

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1. Introduction

1.1 How to Use This Manual

This manual is for the JASCO [Spectra Analysis] program that runs on *Microsoft Windows XP*. It has four sections including this one. The sections are summarized below.

1. Introduction

This chapter describes how to use this manual and provide an overview of the [Spectra Analysis] program. Read this section before using the program.

2. Introduction to [Spectra Analysis] program

This chapter provides detailed descriptions of the procedures for starting the [Spectra Analysis] program, opening saved spectra, finding peak position, and saving/printing results.

For first-time users of the [Spectra Analysis] program, follow the procedures in Section 2 to become familiarized with the [Spectra Analysis] program.

3. [Spectra Analysis] Program Reference

This chapter is about the menu commands in detail for the [Spectra Analysis] program. Refer to it as necessary.

4. Appendix

This appendix provides technical information and file types for the [Spectra Analysis] program.

1.2 Overview of the [Spectra Analysis] Program

[Spectra Analysis] is a common program to the entire range of JASCO analytical instruments including UV/Vis, near-infrared, infrared, fluorescence, Raman, polarimetry and circular dichroism spectrometers. This program can analyze spectrum data and time-course data measured by the analytical instruments.

The main functions of the [Spectra Analysis] program are summarized below. Other optional functions are also available depending of the instrument used to measure data. Refer to Section 3 "[Spectra Analysis] Program Reference" for details.

- (1) File menu: saves/opens spectra, and prints data.
- (2) Edit menu: copies spectra to the Clipboard or between views and deletes spectra.
- (3) View menu: changes scales, colors, and fonts of spectra. Shows or hides results of peak processing.
- (4) Processing menu:
 - Baseline Correction
 - Smoothing
 - Noise Elimination
 - Deconvolution
 - FFT Filter (noise elimination)
 - Interpolation and Truncation
 - Arithmetics (between spectra or a spectrum and a constant)
 - Deviation
 - Kramers-Kronig Transformation
 - Peak Find
 - Peak Height
 - Peak Area
 - Full Width at Half Maximum
 - Subtraction
 - X and Y Unit Conversion

Note: All of the screen shots in this manual show examples of FT-IR spectrometer data. As a result, the abscissa (y-axis) shows either absorbance or transmittance.

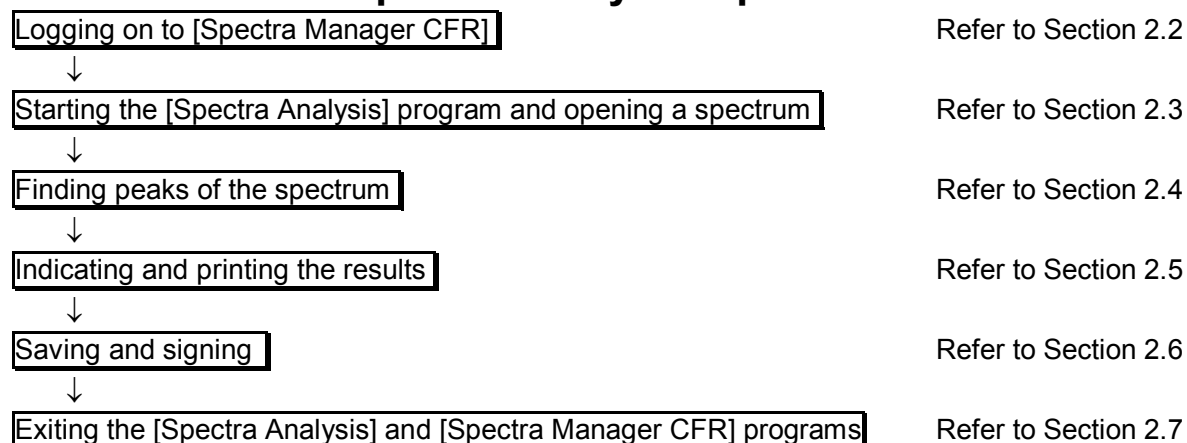
2. Introduction to [Spectra Analysis] Program

This chapter provides detailed descriptions of the procedures for starting the [Spectra Analysis] program, opening saved spectra finding peak position, and saving/printing results.

For first-time users of the [Spectra Analysis] program, follow the procedures in this section to understand the overview of the [Spectra Analysis] program.

Note: This section is described based on the fact that the initial settings have already been performed by Administrator of the [Spectra Manager CFR]. Refer to the [Administrative Tools] manual for the initial settings.

2.1 Overview of Spectra Analysis Operation



2.2 Logging on to [Spectra Manager CFR]

- (1) Select [Program] - [JASCO] - [Spectra Manager] from the *Windows* [Start] menu or double-click the [Spectra Manager] icon on the *Windows* desktop. The following dialog box is displayed (Fig. 2.1).

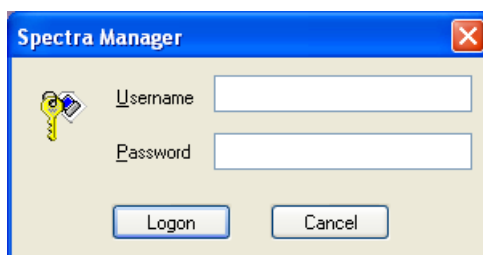


Figure 2.1 Logging on to Spectra Manager CFR

- (2) Type "Administrator" in the [Username] text box and click the <Logon> button, leaving the [Password] text box blank. The [Spectra Manager CFR] program is started (Fig. 2.3). Clicking the <Cancel> button exits [Spectra Manager CFR].

Note 1: If the current password has expired, or the user is logging on to the program for the first time, the following dialog box is displayed (Fig. 2.2). Enter a new password to continue the logon procedure.

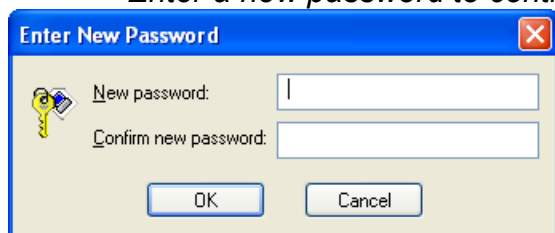


Figure 2.2 [Enter New Password] dialog box

Note 2: If an incorrect password is entered five times consecutively during a logon attempt, the user is locked out and an error message is displayed. If this happens, an Administrator will need to unlock the user by assigning a new password to the user.

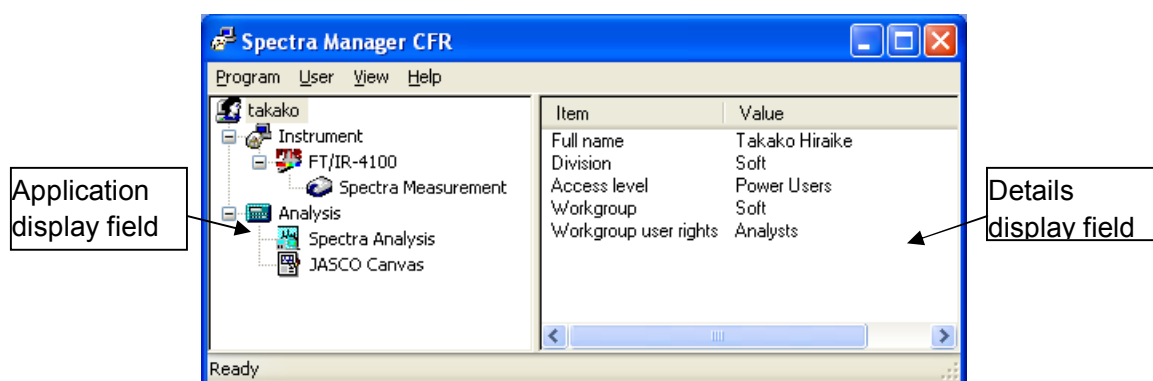


Figure 2.3 [Spectra Manager CFR] window

Left field of the window (application display field) shows a list of instruments, measurement programs and analysis programs in hierarchical order (a tree), and right field (details display field) displays any details of the selected item.

Note: The contents of the window differ depending on the user logged on.

2.3 Starting [Spectra Analysis] program and opening spectra

(1) Starting the [Spectra Analysis] Program

- 1) Click [Spectra Analysis] displayed in the application display field. Details display field shows detailed information of the [Spectra Analysis] (Fig. 2.4). The upper part of the details display field shows a list of file names in the selected folder and the lower part shows the contents of the selected file in the upper part.

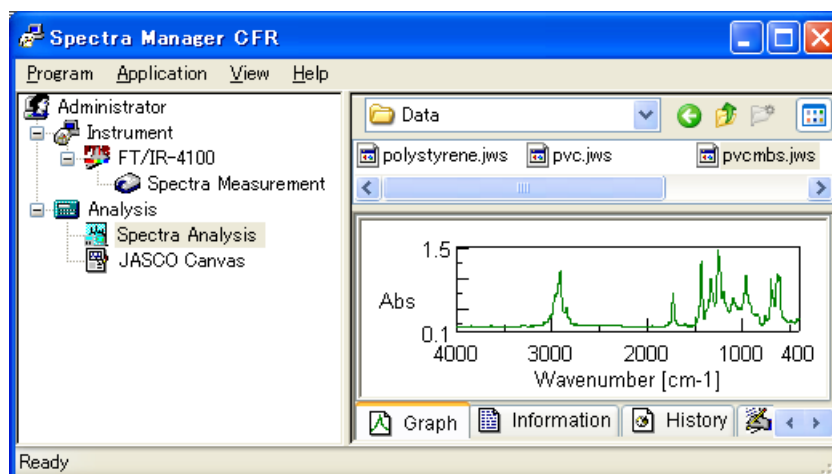


Figure 2.4 [Spectra Manager CFR] window (when [Spectra Analysis] is selected).

- 2) Select [Application] – [Start] to start [Spectra Analysis] (Fig. 3.5).

Note 1: Programs can be started by double-clicking the program name.

Note 2: Programs can be started by opening data. After selecting a data file, select [Application] – [Open Data] to open the file. Programs can also be started by double-clicking a corresponding file.

Note3: In the [Spectra Manager CFR] window, files can be searched by search function. Refer to the [Administrative Tools] manual for details.

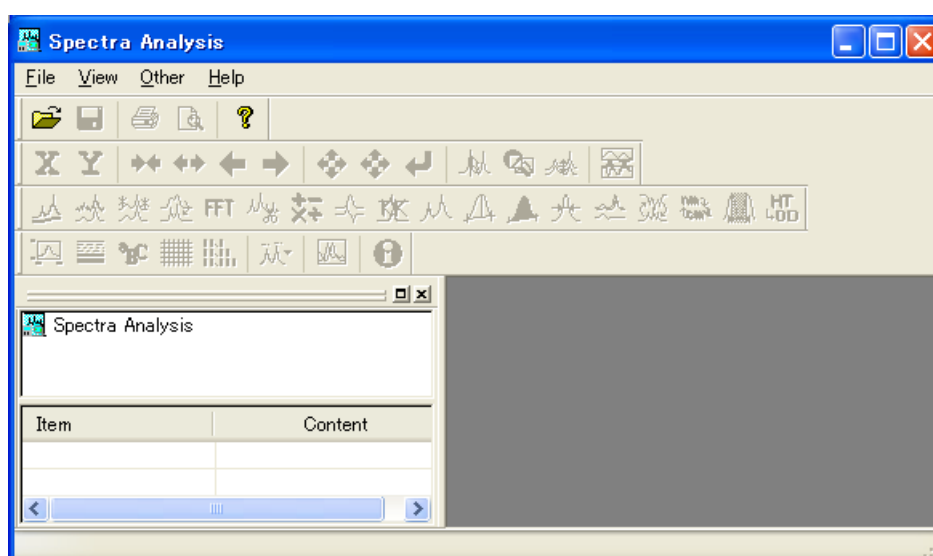


Figure 2.5 [Spectra Analysis] window

(2) Opening a spectrum

- 1) Select [File] - [Open...] from the main menu to display the [Open Data] dialog box (Fig. 2.6).

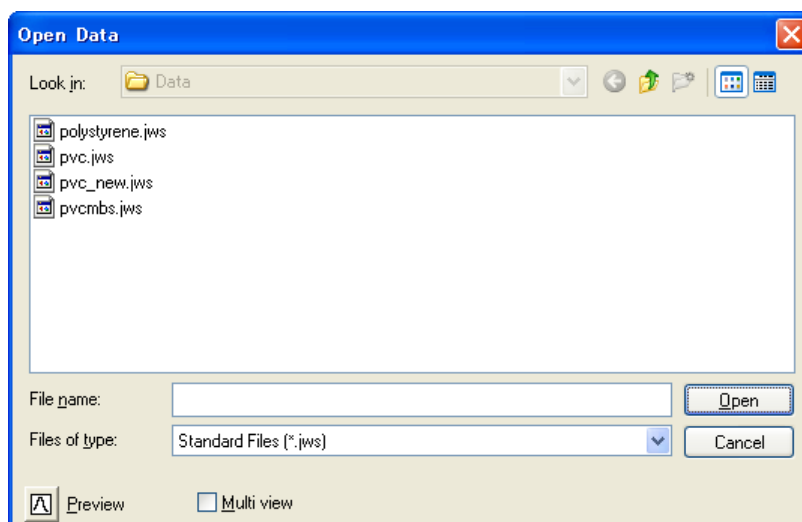


Figure 2.6 [Open Data] dialog box

Note: Select [File] - [Import...] to open old format standard files (.jws).*

- 2) Select the target spectrum file and click the <Open> button. The spectrum file opens and the spectrum is displayed in the View.

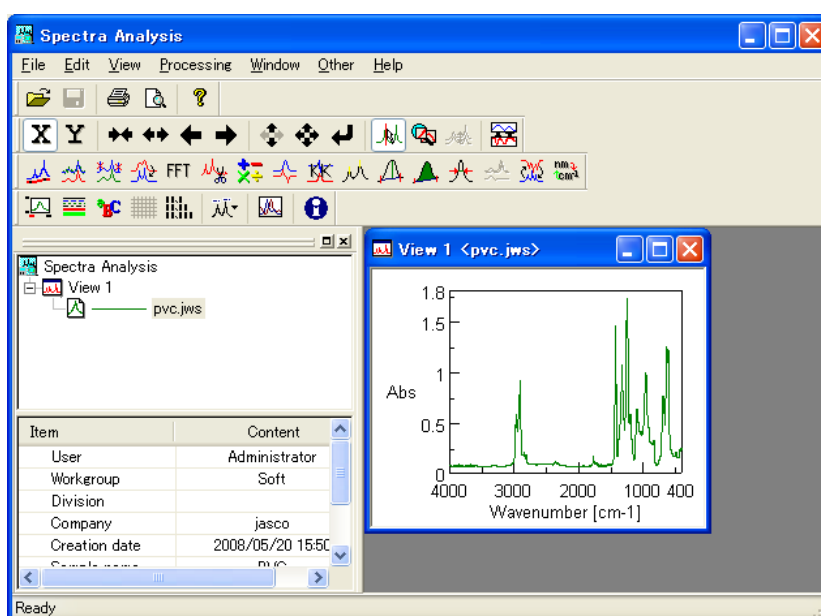


Figure 2.7 [Spectra Analysis] window

The [Spectra Analysis] window consists of two main components: a dockable [Work Space] window and the view windows. The [Work Space] window shows the names of open files and numbered views in a tree-like hierarchy, and spectra are displayed in the view windows. (Fig. 2.7)

Two or more windows may be opened simultaneously to display multiple spectra. Each of these windows is referred to as a "View". In addition, multiple spectra can be overlaid in a single View.

2.4 Finding Spectrum Peaks

- (1) Select [Processing] - [Peak Processing] - [Peak Find...] to display the [Peak Find...] dialog box (Fig. 2.8).

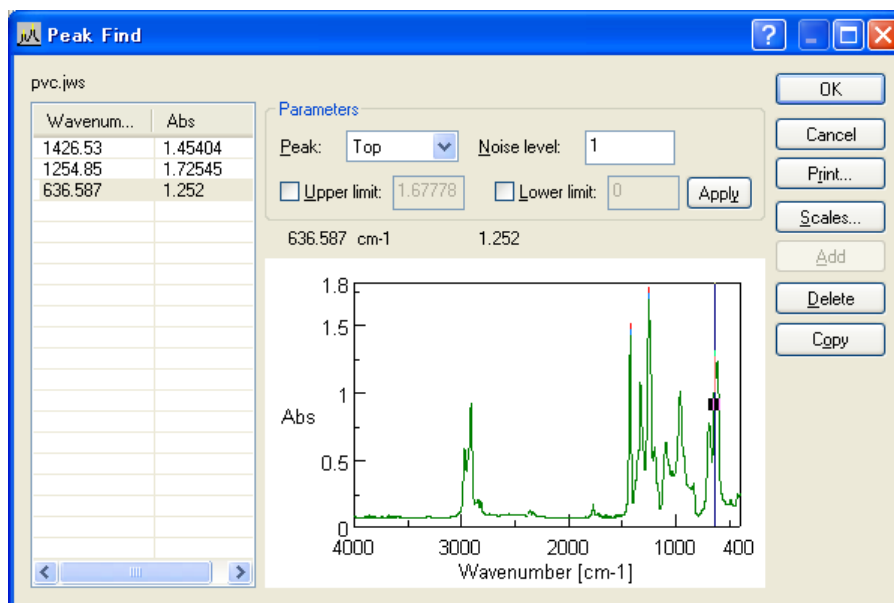


Figure 2.8 [Peak Find] dialog box

The [Peak Find] dialog box consists of three main components: a peak table, parameters, and the spectrum display area.

(2) Setting parameters and finding peaks

- 1) Select "Top (peaks)", "Bottom (valleys)", or "Both (peaks and valleys)" from the [Peak] drop down list.
- 2) Input a value in [Noise level] text box. Peaks less than the input value are regarded as noise.
- 3) Select the [Upper limit] and [Lower limit] check boxes to input values, if necessary. The values can also be set by dragging the limit setting bar displayed in the spectrum display area.
- 4) Click the <Apply> button. Peaks are detected with the parameters and results are displayed in the peak table. Peak positions are indicated with short bars.

(3) Adding peaks

- 1) Drag the long bar to the target peak position to activate the <Add> button.
- 2) Click the <Add> button to add the peak to the peak table.

(4) Deleting peaks

- 1) Select a target peak to be deleted. Select peak in the peak table or move the long bar to the peak position to activate the <Delete> button.
- 2) Click the <Delete> button to delete the peak from the peak table.

(5) Copying peak table

Click the <Copy> button to copy the values in the peak table to the Clipboard. For instance, the copied data can be pasted on the Memo.

(6) Writing the results of Peak Find to memory

Click the <OK> button. The results of Peak Find on the spectrum are written to memory.

2.5 Indicating and Printing the Results

(1) Indicating results of Peak Find on the spectrum

Select display mode from [View] - [Peak Indicators] to display the peak indicators on the spectrum. The indicators in Fig. 2.9 are displayed with the [Bar, X] mode.

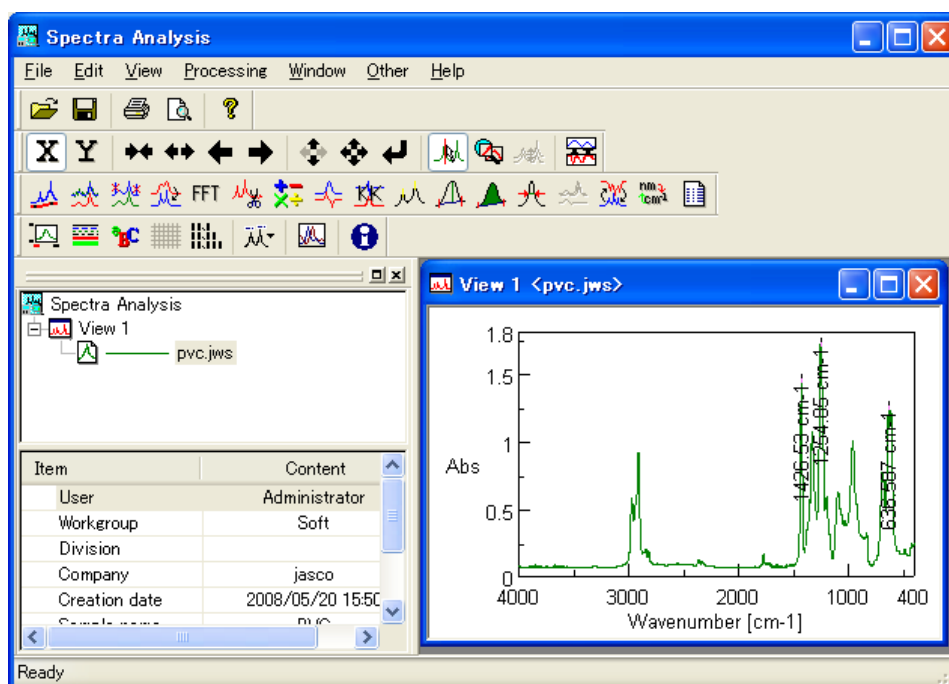


Figure 2.9 [Spectra Analysis] window (spectrum with peak indicators)

(2) Displaying print images

1) Select [File] – [Print Preview] to display the [Print Preview] dialog box (Fig. 2.10).

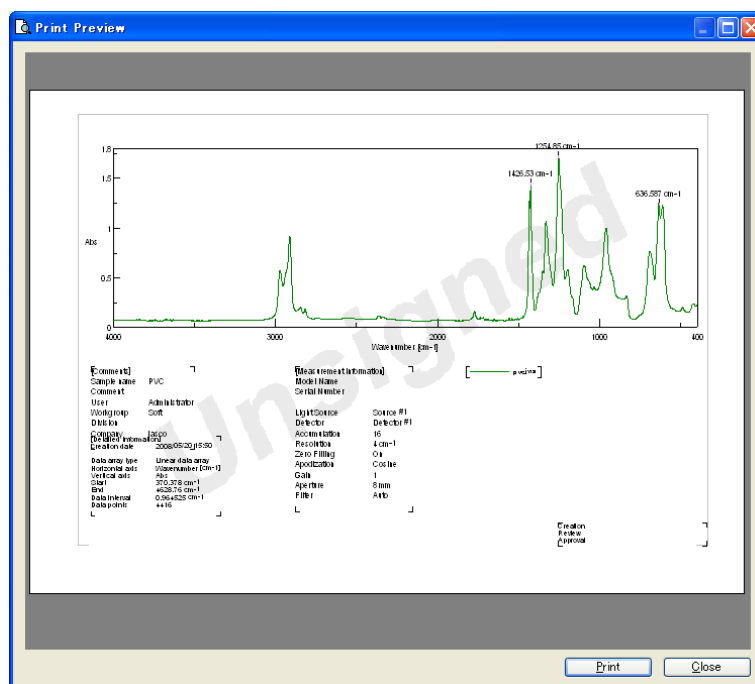


Figure 2.10 [Print Preview] dialog box

(3) Printing results

Click the <Print> button after checking the print image. Spectrum and results of Peak Find are printed.

2.6 Saving and Signing

(1) Select [File] – [Save As] to display the [Save As] dialog box (Fig.2.11).

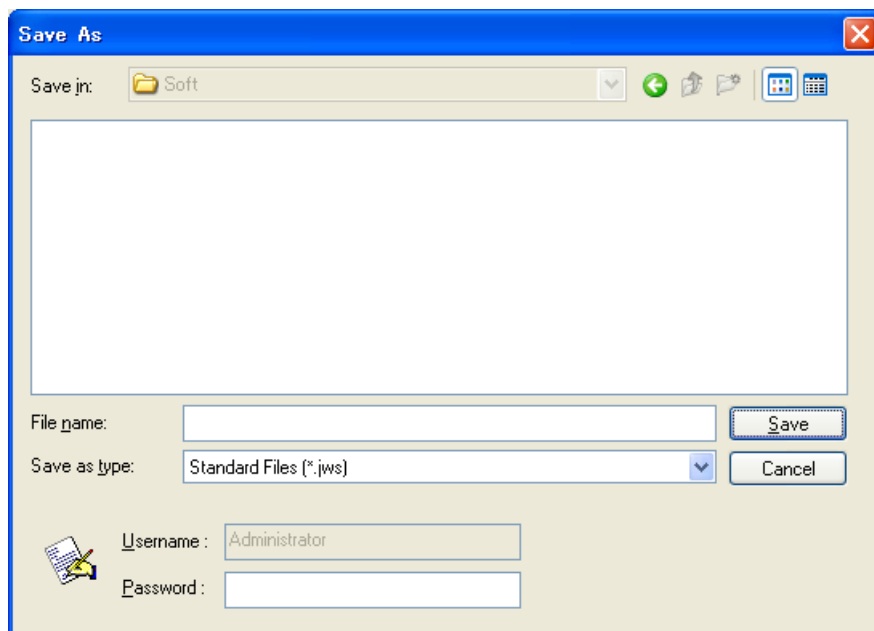


Figure 2.11 [Save As] dialog box

Note: In the CFR version, a spectrum can only be saved in the Workgroup folder and its subfolders. Folder settings can be performed in [Administrative Tools] only by Administrators. Refer to the [Administrative Tools] manual for details.

(2) After typing a file name and password, click the <Save> button.

Note: To perform “Approval” and “Review” signatures on a saved spectrum by Managers, display the [Spectra Manager CFR] window. Refer to the [Administrative Tools] manual for details.

2.7 Exiting [Spectra Analysis] Program and [Spectra Manager CFR]

(1) Exiting [Spectra Analysis] program

Select [F]ile - [E]xit to exit the [Spectra Analysis] program and return to the [Spectra Manager CFR] window,

Note: When spectra have not been saved, the following message is displayed. Save spectra, if necessary. A message will be displayed for each unsaved spectrum.

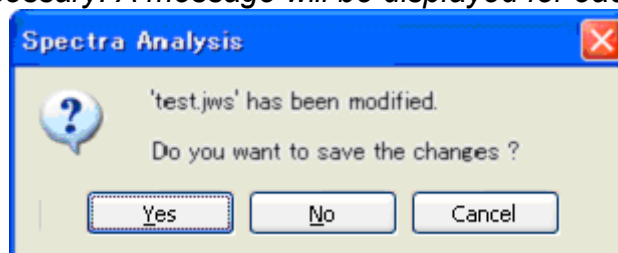


Figure 2.12 Error message when spectra have not been saved

(2) Exiting [Spectra Manager CFR]

Select [P]rogram - [E]xit to close the [Spectra Manager CFR] window and return to *Windows*,

(3) Shutting down *Windows*

Shut down *Windows* according to the usual procedure.

3. [Spectra Analysis] Program Reference

3.1 Program Window and Menus

3.1.1 Program Window

Select [Spectra Analysis] on the [Spectra Manager CFR] window to start the [Spectra Analysis] program (Fig. 3.1).

Note: When parameters in measurement programs are set to [Send to Analysis], the [Spectra Analysis] program runs automatically and opens a new view to display the measured spectrum.

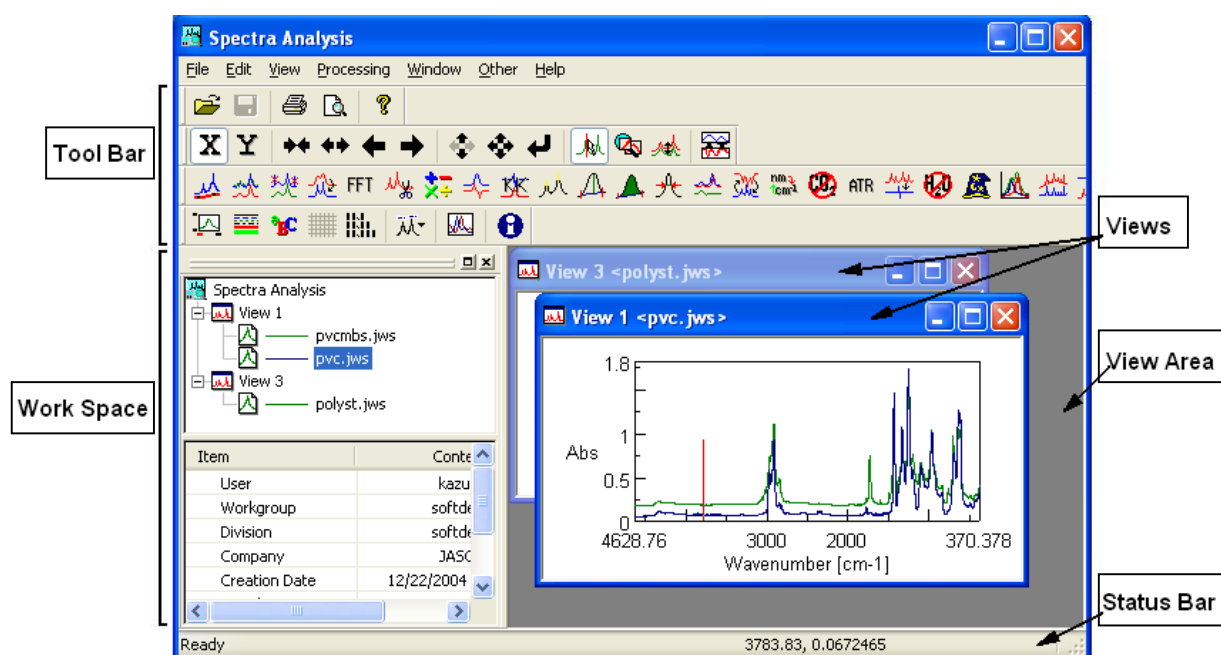


Figure 3.1 [Spectra Analysis] window

Work Space	The workspace is divided vertically into two panels by a bar, one panel displays view numbers and associated filenames, and the other displays information about the currently active spectrum.
View Area	Views are displayed in separate windows in this area.
Views	A window that displays a single spectrum (or overlaid spectra). The filename of the active spectrum is displayed in the title bar of the View window. (If a spectrum has not been saved to file, the title will be displayed as Memory-#)
Toolbars	The toolbars contain shortcuts to frequently used functions. The toolbar buttons for data processing functions can be customized.
Status Bar	Displays the x and y data values of the currently selected position in the active spectrum. To select a position, move the mouse pointer to the desired position over the spectrum and left-click the mouse.

3.1.2 Menu List

[File] menu	
<u>O</u> pen...	Opens one or more spectra files.
<u>O</u> verlay...	Overlays one or more spectra files to the active view.
<u>C</u> lose	Closes the active view.
<u>S</u> ave	Saves the active spectrum.
<u>S</u> ave <u>A</u> s...	Saves the active spectrum with a new filename.
<u>I</u> mport...	Imports a spectrum in a format other than the standard Spectra Manager format.
<u>E</u> xport...	Saves data in the JCAMP-DX or text format.
<u>O</u> pen <u>V</u> iew...	Opens an existing view information file in a new view.
<u>S</u> ave <u>A</u> ctive <u>V</u> iew...	Saves the active View with a new name.
<u>P</u> rint...	Prints the active spectrum.
<u>P</u> rint <u>P</u> review...	Displays a print preview of the active spectrum.
<u>P</u> rint <u>S</u> etup...	Designates the target printer and print parameters.
<u>E</u> xit	Exits the [Spectra Analysis] program.

[Edit] menu	
<u>C</u> opy as <u>P</u> icture	Copies a spectrum to the <i>Windows</i> clipboard as a picture.
<u>C</u> opy as <u>B</u> itmap	Copies a spectrum to the <i>Windows</i> clipboard as a bitmap.
<u>C</u> ut <u>S</u> pectrum	Removes the currently selected spectrum from the active view and places it onto the Spectra Analysis clipboard.
<u>C</u> opy <u>S</u> pectrum	Copies the currently selected spectrum in the active view to the Spectra Analysis clipboard.
<u>P</u> aste <u>S</u> pectrum	Pastes the contents of the Spectra Analysis clipboard into the active view.
<u>D</u> elete <u>S</u> pectrum	Deletes the currently selected spectrum in the active view.
<u>S</u> elect <u>A</u> ll	Selects/clears all spectra in the active view.
<u>C</u> omments...	Comments regarding a spectrum may be added or edited using this window.

[View] menu		
<u>T</u> oolbars	<u>M</u> ain	
	<u>V</u> iew	
	<u>D</u> ata <u>P</u> rocessing	
	<u>V</u> iew <u>C</u> ommand	
<u>S</u> tatus <u>B</u> ar	Shows or hides the status bar.	
<u>W</u> ork <u>S</u> pace	Shows or hides the Work Space.	
<u>S</u> cales...	Changes the scale of the active view.	
<u>P</u> atterns...	Selects the line style and color of the active view.	
<u>F</u> onts...	Selects the font settings and text orientation for the active view.	
<u>G</u> ridlines...	Adds/removes vertical and/or horizontal grid lines to the active view.	
<u>S</u> tyles...	Changes the scale settings for the active view.	
<u>H</u> orizontal <u>A</u> xis <u>U</u> nits...	Changes the quantity or units displayed as the horizontal axis.	
<u>I</u> nformation...	Displays the spectrum information.	
<u>N</u> ormalize	Normalizes the scale of each spectrum displayed in the active view.	
<u>P</u> eak <u>I</u> ndicators...	Designates how the peak value or peak position of the active spectrum should be displayed.	

[Processing] menu

Corrections	Baseline...	Corrects a spectrum using a specified baseline.
	Smoothing...	Smooths the spectrum.
	Noise Elimination...	Eliminates noise for which the cause is known.
	Deconvolution...	Separates overlaid peaks.
	FFT Filter...	Eliminates interferometric noise from a spectrum.
	Truncate Data...	Truncates data from the side of a spectrum.
Operations	Arithmetic...	Performs arithmetic operations between spectra or between a spectrum and a constant.
	Derivatives...	Calculates derivatives for a spectrum.
	K-K Transform...	Performs Kramers-Kronig Transformation.
Peak Processing	Peak Find...	Finds spectrum peaks (or valleys).
	Peak Height...	Calculates the peak height and peak height ratio.
	Peak Area...	Calculates the peak area and peak area ratio.
	FWHM...	Calculates the full width at half maximum value.
Subtraction...	Calculates a difference spectrum.	
X Unit Conversion...	Converts the x-axis units of the spectrum.	
Y Unit Conversion...	Converts the y-axis units of the spectrum.	
Common Options	Curve Fitting	Transfers a spectrum to the [Curve Fitting] program.
	Data Dump	Displays values of measurement data.
	Data Accumulation	Averages data overlaid in the same View.
	Arithmetic between Data	Performs arithmetic operations between data overlaid in the same View.
	Channel Conversion	Converts channels of data in a View and picks out data having a desired channel from overlaid spectra or multiple channels in a View.
	Arithmetic between Channels	Performs arithmetic operations between channels displayed in the same View.
IR Options	Run KnowItAll	Transfers a spectrum to the search software (Sadtler) (Not available in Spectra Manager CFR)
	Data Comparison	Compares spectra to identify the measured data.
	ATR Correction...	Converts the absorption intensity relationship between the high wavenumber and low wavenumber range of a spectrum.
	Interferogram to Spectrum...	Converts an interferogram (IF) into a single-beam spectrum.
	Spectrum to Interferogram...	Converts a single-beam spectrum into an interferogram (IF).
	Auto Baseline...	Automatically performs baseline correction.
	H2O Reduction	Eliminates the absorption bands of water vapor in a spectrum.

	CO2 Reduction	Eliminates the absorption bands of carbon dioxide in a spectrum.
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[Window] menu

<u>C</u> ascade	Cascades all open views.
T <u>i</u> le <u>H</u> orizontally	Arranges window as non-overlapping horizontal tiles.
T <u>i</u> le V <u>e</u> rtically	Arranges windows as non-overlapping vertical tiles.
A <u>r</u> range <u>I</u> cons	Arranges minimized window icons at the bottom of the window..
C <u>l</u> ose All	Closes all views.

[Other] menu

C <u>u</u> stomize <u>T</u> oolbar...	Customizes the [Processing] toolbar.
<u>O</u> ptions...	Changes the view and print style options.
<u>D</u> ecimal Places...	Sets the number of decimal places of the vertical and horizontal axes.

[Help] menu

<u>C</u> ontents	Outlines the contents of the Help function.
<u>S</u> earch Topic	Searches for a specific topic within the Help function.
<u>A</u> bout...	Displays information about the version of the [Spectra Analysis] and product license information.

3.1.3 Toolbars

The toolbars contain buttons for frequently used functions, such as saving files, changing scale settings and performing data processing. The buttons are of four basic types and are thus arranged in four separate toolbars: Main, View Toolbar, Processing and View Command. Each of these toolbars can be hidden or displayed independently. In addition, the Processing toolbar buttons can be customized if necessary.

Note: Refer to 3.7.1 [Customize Toolbar] to learn more about customizing the Processing toolbar.

Main

	Opens one or more spectra files. Shortcut to [File]-[Open...].
	Saves the active spectrum to disk. Shortcut to [File]-[Save] or [File]-[Save As...].
	Prints the active spectrum. Shortcut to [File]-[Print...].
	Displays a print preview of the active spectrum. Shortcut to [File]-[Print Preview...].
	Displays version information about the program. Shortcut to [Help]-[About].

View

	When selected, the following buttons can be used to modify the x-axis scale settings.
	Reduces the spectrum horizontally.
	Expands the spectrum horizontally.
	Moves the spectrum to the left.
	Moves the spectrum to the right.
	When selected, the following buttons can be used to modify the y-axis scale settings.
	Reduces the spectrum vertically.
	Expands the spectrum vertically.
	Moves the spectrum up.
	Moves the spectrum down.
	Resets the vertical axis to auto scale.
	Resets the scales of both axes so that all measured data is displayed in the current View.
	Returns to default scale settings.
	Tracing mode: The mouse is used to display spectrum x and y coordinate data.
	Zoom mode: The mouse is used to zoom in on a region of the spectra.
	Offset mode: The mouse is used to offset overlaid spectra vertically within a View.
	Changes the active spectrum within an overlaid View.

Data Processing	
	[Baseline]
	[Smoothing]
	[Noise Elimination]
	[FFT Filter]
	[Interpolation/Truncation.]
	[Arithmetic]
	[Deconvolution]
	[Derivatives]
	[K-K Transform]
	[Peak Find]
	[Peak Height]
	[Peak Area]
	[FWHM]
	[Subtraction]
	[X Unit Conversion]
	[X Unit Conversion]
View Command	
	[Scales]
	[Patterns]
	[Fonts]
	[Gridlines]
	[Styles]
	[Peaks]
	Toggles between Overlay/Normalize. Normalize is used to display overlaid spectra using independent auto scaling.
	Displays spectrum information. Shortcut to [View]-[Information].

3.1.4 View Area Basic Operations

3.1.4.1 Zooming in on a spectrum using the mouse

Click the  button on the toolbar. The  button is displayed as depressed and the mouse changes to zoom mode. When the mouse pointer is moved over a spectrum view while in this mode, the mouse pointer changes into a crosshairs. At one corner of the desired range, press and hold the left mouse button. Then, while continuing to hold left mouse button, drag the pointer diagonally to specify the desired range. When the mouse button is released, the selected area is rescaled to fill the entire view. Click the  button to return to the original scale.

Note 1: The coordinates of the mouse cursor are displayed on the status bar. These can be used as a reference for performing the scale expansion.

Note 2: This method does not set the edges of the scale to exact values. To change the scale limits to specific values, select [View]-[Scales...] from the main menu.

Note 3: To revert to the zoom settings that were used immediately prior to zooming in, right-click the mouse button.

3.1.4.2 Displaying spectra coordinates

To display coordinate data at an arbitrary point within the spectrum, click the  button on the toolbar. The  button is displayed as depressed and the mouse is set to trace mode. Move the mouse pointer over an arbitrary point on the spectrum and left-click to display the data corresponding to the position indicated on the status bar.

Note: If the mouse button is held down while the mouse pointer is moved arbitrarily, the status bar is updated as the mouse moves.

3.1.4.3 Overlaying spectra using the mouse

The following is the procedure for overlaying spectra by dragging from one view to another using the mouse. This can be done when two or more views are open.

Note: Although multiple spectra can be opened in each view, only eight patterns (colors, line styles) can be defined in each view. Thus, when more than eight spectra are overlaid in a single view, the patterns will be repeated from the ninth spectrum on.

- (1) Select [Window]-[Tile Horizontally] from the menu. All open Views are tiled (displayed side by side).
- (2) Click in the View containing the spectra to be moved. The View is activated.
- (3) Moving the mouse pointer over the spectrum changes the mouse pointer into the shape of a tweezers . Pressing the left button on the mouse changes the mouse pointer to . Drag the mouse and release the button over the View in which you want to overlay the spectrum. (This procedure is called “drag and drop”.) The spectrum is moved from the original view and is overlaid in the View. If the original view only contained a single spectrum, the view is closed automatically.

Note 1: Overlaying spectra automatically updates the vertical axis scale. The horizontal axis does not change.

Note 2: Overlaid spectra can be split into separate views by dragging and dropping a spectrum outside of the view.

*Note 3: When dragging the spectrum, holding the **Ctrl** key down while pressing the mouse button changes the mouse pointer to . The spectrum is then copied by dragging and dropping, with the original view left unchanged.*

3.1.4.4 Changing the active spectrum

When multiple spectra are overlaid in a View, operations such as file saving, spectrum copying and data processing are performed only on the active spectrum within that view. The name of the active spectrum is displayed in the title bar of the View and is highlighted in the [Work Space] window when the view is active.

Note: The unit of the vertical axis of the spectrum changes according to the active spectrum.

Active spectra can be selected in four different ways as outlined below: using the mouse, using the [Work Space] window, using the toolbar button, or using the keyboard.

(1) Using the mouse

- 1) Verify that the  button (Tracing mode) on the toolbar is depressed.
- 2) Move the mouse pointer over the spectrum to be activated. The mouse pointer turns into the shape of tweezers . Left-click to activate the selected spectrum.
- 3) Verify with the View title bar or in the [Work Space] window that the active spectrum has been changed.

(2) Using the [Work Space] window

Click on the filename in the Work Space window. The filename will be highlighted and the spectrum will become active.

(3) Using the button on the toolbar

Click the  button on the toolbar to cycle through the spectra in the view. Click the  button until the required spectrum becomes active (as shown in the [Work Space] window or in the View title bar).

(4) Using the keyboard

Pressing the [↑] or [↓] cursor keys changes the active spectrum in the active View. Use the cursor keys until the necessary spectrum filename appears in the View title bar.

3.1.4.5 Moving or copying all spectra in a View simultaneously

All spectra in a View can be moved or copied by performing a single operation when two or more Views are open.

This can be accomplished either by dragging and dropping using the mouse or by using the edit menu. The method of dragging and dropping using the mouse is as follows:

- (1) Select [Window]-[Tile Horizontally] from the menu to ensure that all Views are visible.
- (2) Click within the View, but outside of the bounding rectangle of the graph, and the mouse pointer will change to . Drag the mouse pointer over the destination View and release the button. All spectra are overlaid and the original View closes.

Note1: If the **Ctrl** key is held down while the mouse is pressed and dragged, the mouse pointer changes to  (different from above). All spectra in the View can then be copied by dragging and dropping using the mouse. That is, the original View remains unchanged.

Note2: Same procedures can be performed between Work Spaces. Click on a View number and drag the mouse pointer over the destination View number.

3.1.4.6 Vertically offsetting multiple spectra within a View

Moves spectra overlaid in a View along the y-axis so that overlapping spectra can be more easily viewed and compared.

Vertical offset parameters can be entered numerically using a dialog box or can be set visually by dragging spectra within a View using the mouse.

(1) Setting vertical offsets numerically

- 1) Click the  button on the toolbar. The  button is displayed as depressed, and the mouse enters offset mode.
- 2) Press and hold the right mouse button in a View. A red vertical cursor is displayed. To numerically set the vertical offset between spectra, a reference wavenumber from which vertical offsets are calculated must be selected. Move the vertical cursor to the appropriate location and release the button to display the following dialog box. (Fig. 3.2) Enter the reference x coordinate and offset interval, and click the <OK> button.
If zero is entered as the offset interval, the spectra will be shifted vertically to overlap at the reference x-coordinate.

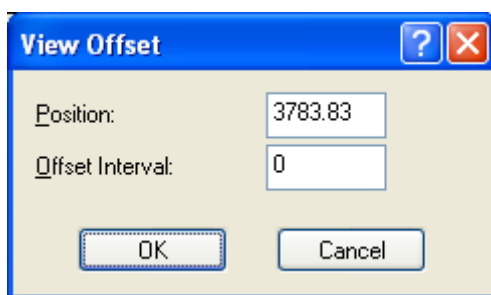


Figure 3.2 [View Offset] dialog box

(2) Dragging spectra in the View

- 1) Click the  button on the toolbar. The  button is displayed as depressed, and the mouse enters offset mode.
- 2) Move the mouse pointer onto the spectrum to be moved. The mouse pointer changes into a double-ended vertical arrow. Click and hold the left mouse button, and drag the spectrum up or down within the View as desired. Release the mouse button to confirm the offset location.

Note: To reset all offsets, hold down the **Ctrl** key and click the right mouse button.

3.2 [File] Menu

3.2.1 [Open...]

Select [File] - [Open] from the main menu to display the [Open Data] dialog box, which is used to open a spectrum (or spectra) in a new View. Multiple files can be selected simultaneously and opened in the same or separate Views.

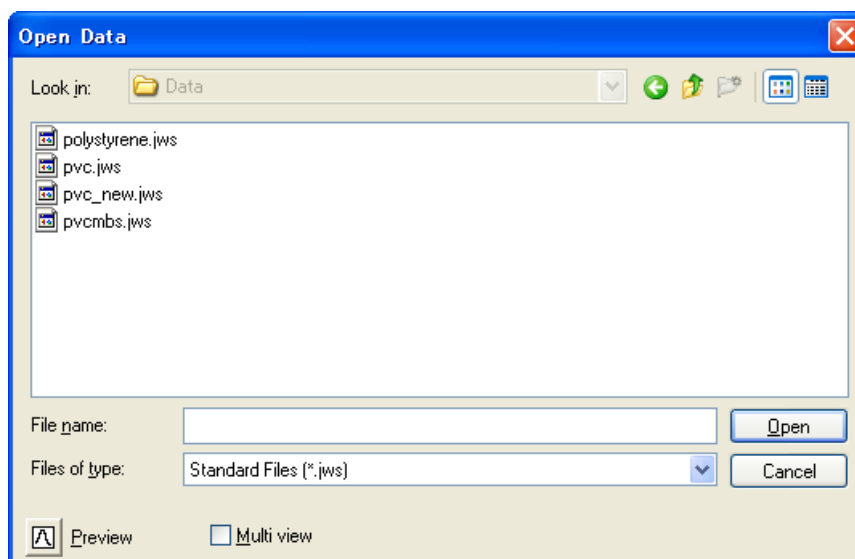


Figure 3.3 [Open Data] dialog box

[Look in]	Used to designate the drive or directory that contains the files to be opened. Available drives and directories are shown in the drop-down list. The contents of the drive or directory designated in the [Look in] drop-down list are shown in the dialog box below. Select the desired spectrum file name. To select multiple non-sequential files, click on the desired spectra file names while holding down the Ctrl key. To select any number of sequential files, click on the first and last file names of the desired list while holding down the Shift key. Spectra having different file extensions cannot be selected simultaneously unless [All files] has been selected as the file type.
[File name]	Used to directly input the desired spectrum filename(s). To select multiple files, input all filenames separated by spaces. The extension may be omitted. If an incorrect extension is entered, the default extension is substituted automatically.
[Files of type]	Lists the file types (extensions) that can be opened by the [Spectra Analysis] program. If the extension is omitted when the filename is entered, the file extension designated here is automatically appended. Note: In [Spectra Manager CFR], only Standard Files (*.jws) can be selected using the [Open Data] dialog box. To open files in other formats, use the [Import] function (see 3.2.6).
[Multi View]	Select the checkbox to open multiple files in separate Views. If [Multi View] is not selected the spectra are overlaid in a single View.

<Preview>

Click the <Preview> button to display a preview of the selected spectrum and the file information about the currently selected file. When multiple files are selected, the preview panel displays information about the last file selected.

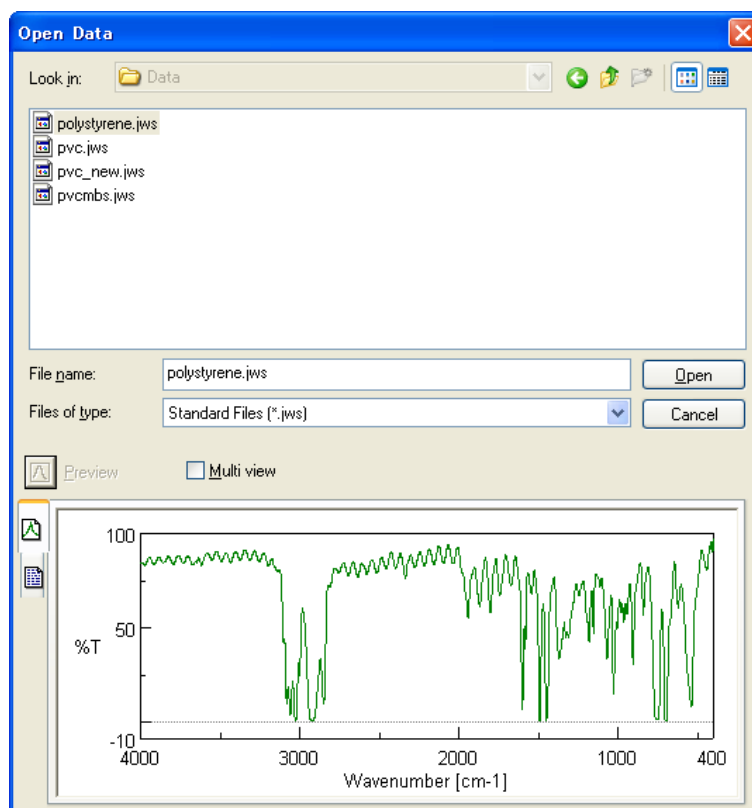


Figure 3.4 [Open Data]-[Preview]

Note: In the CFR version, the [Look in] list cannot be used to select a folder from the drop-down list. However, a subfolder within the prespecified folder can be selected.

3.2.2 [Overlay...]

Select [File] - [Overlay] to display the [Overlay Data] dialog box used to add a spectrum (or spectra) to the active View. (Fig. 3.5) Multiple files can be selected and added simultaneously.

Note: If the [Automatically scale display range when a spectrum is opened or overlaid.] checkbox is selected in the [View] tab of the [Options] dialog box ([Other] – [Options]), the vertical axis is automatically re-scaled so that all of the spectra to be overlaid are displayed, however, the horizontal axis is not changed.

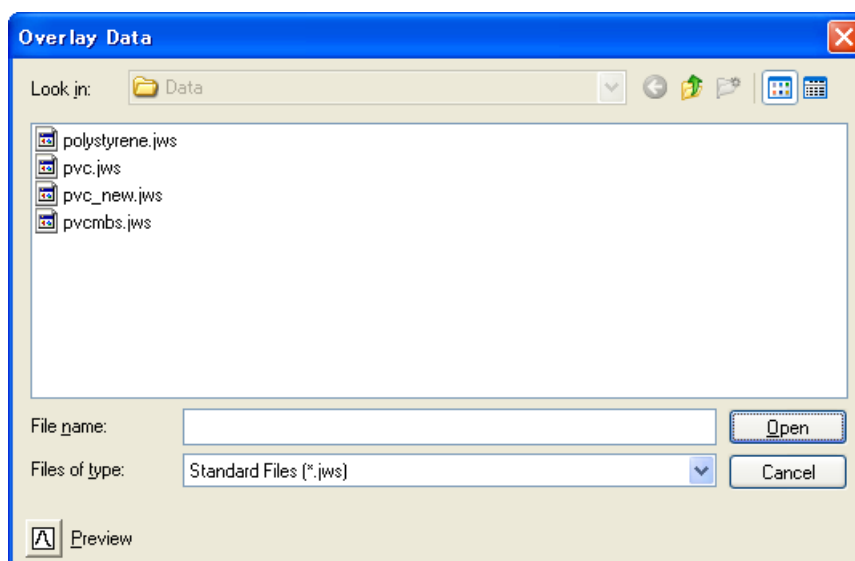


Figure 3.5 [Overlay Data] dialog box

3.2.3 [Close]

Closes the active View. The following dialog box appears when there are spectra that have not been saved. Click the <Yes> button to save the spectra before closing. Click the <No> button to close the view without saving. Click the <Cancel> button to leave the active View open.

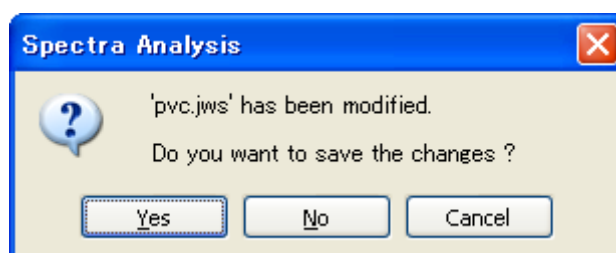


Figure 3.6 Message displayed before closing unsaved spectra

3.2.4 [Save]

Select [File] - [Save] to save the active spectrum in the active View under the current filename. The existing data are overwritten. The files that can be overwritten vary depending on the Workgroup user right and signature of the file.

Table 3.1 Privileges assigned to Save function

	Managers	Analysts	Operators
Creation signature only	√	√	√*
Creation and review signatures	√		
Creation, review, and approval signatures			

* Operators can save spectra only when the creation signature has been signed by the Operator.

3.2.5 [Save As...]

Select [File] - [Save As] to display the [Save As] dialog box, which is used to save the active spectrum in the active View using a new filename and/or in a new location. (Fig. 3.7)

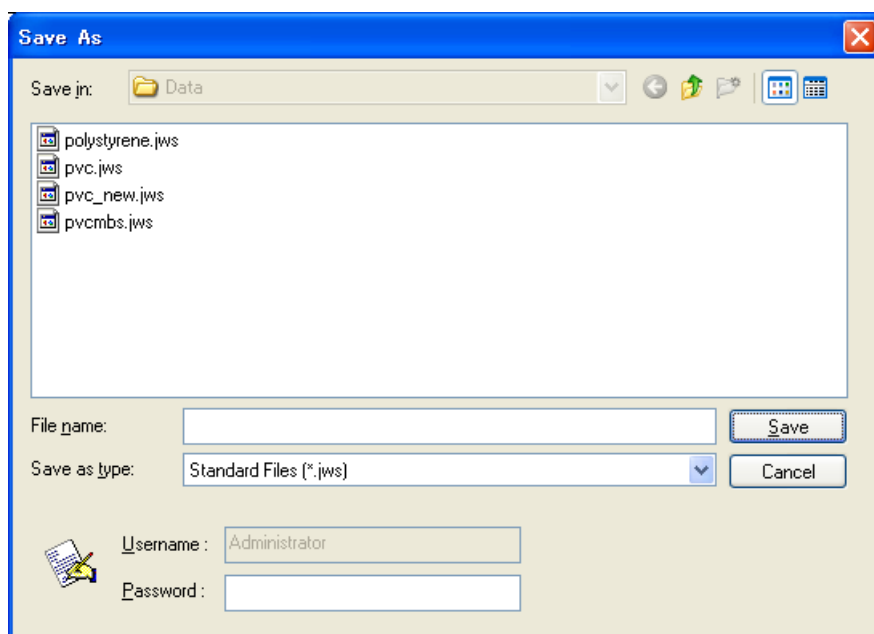
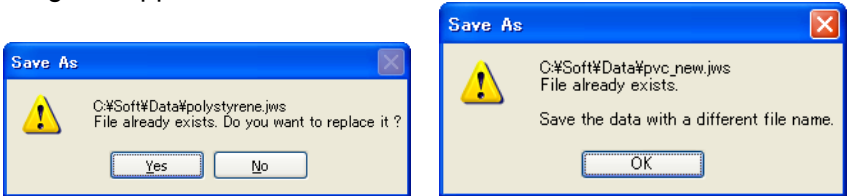


Figure 3.7 [Save As] dialog box

[Save <u>i</u> n]	Used to designate the destination drive or directory. Available drives and directories appear in the drop-down list. The contents of the drive or directory designated in the [Save <u>i</u>n] drop-down list are shown in the dialog box. (Fig. 3.7) To use an existing filename click on the desired filename.
[File <u>n</u> ame]	Used to directly input the desired spectrum filename. If the extension is omitted, the designated extension is added automatically. If an existing filename is designated and the <Save> button is clicked, the following dialog box appears.  Figure 3.8 Dialog box displayed to overwrite an existing filename <i>Note: Spectrum can be overwritten according to Table 3.1 “Privileges according to Save function”.</i>
[Save as <u>t</u> ype]	Lists the file types under which a file can be saved. If the extension is omitted when the file name is input, the file extension designated here is added automatically. <i>Note: In Spectra Manager CFR, files can only be saved in the Standard Files (*.jws). To save files in the JASCO-DX or ASCII formats, use the [Export] function (See 3.2.7).</i>
[<u>U</u> sername]	Displays the username logging on to the [Spectra Manager CFR] and cannot be changed.
[<u>P</u> assword]	In the [Spectra Manager CFR], saving spectrum requires electronic signature. Type password of the current User.

Note: In the CFR version, the [Look in] list cannot be used to select a folder from the drop-down list. However, a subfolder within the prespecified folder can be selected.

3.2.6 [Import...]

In Spectra Manager CFR, [Import] is used to open a spectrum with a format other than the standard Files (*.jws). (Fig. 3.9)

Note : In Spectra Manager CFR, this function can also be used to open a old format file that is saved in a folder other than that of a logged-on Workgroup.

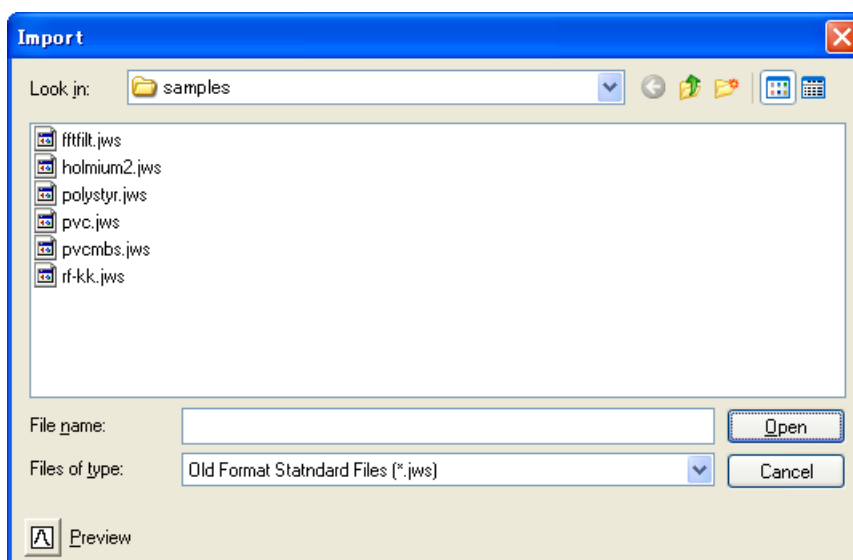


Figure 3.9 [Import] dialog box

The operating procedure is as described in 3.2.1 “[Open...]”. However, the file types described below can be selected.

[Files of type]	Lists the types of files that can be opened by the Spectra Analysis program. If the extension is omitted when the filename is entered, the file extension designated here is automatically appended. When [All files] is selected, .jws is the default extension. The following file types can be selected: <table border="0"> <tr> <td style="padding-right: 20px;">Old Format Standard Files (*.jws):</td><td>Data measured and calculated using the Spectra Manager ver.1.5#.</td></tr> <tr> <td>JCAMP-DX (*.dx):</td><td>Data in JCAMP-DX 4.24 format. Refer to the Appendix “JCAMP-DX Format Files” for a description of the file formats that can be used.</td></tr> <tr> <td>Text file (*.txt)</td><td>Data in text format. Refer to the Appendix “JCAMP-DX Format Files” for a description of the file formats that can be used.</td></tr> <tr> <td></td><td>All files (*.*) : Designates all files. Files not in one of the above formats or ASCII format cannot be loaded.</td></tr> </table>	Old Format Standard Files (*.jws):	Data measured and calculated using the Spectra Manager ver.1.5#.	JCAMP-DX (*.dx):	Data in JCAMP-DX 4.24 format. Refer to the Appendix “JCAMP-DX Format Files” for a description of the file formats that can be used.	Text file (*.txt)	Data in text format. Refer to the Appendix “JCAMP-DX Format Files” for a description of the file formats that can be used.		All files (*.*) : Designates all files. Files not in one of the above formats or ASCII format cannot be loaded.
Old Format Standard Files (*.jws):	Data measured and calculated using the Spectra Manager ver.1.5#.								
JCAMP-DX (*.dx):	Data in JCAMP-DX 4.24 format. Refer to the Appendix “JCAMP-DX Format Files” for a description of the file formats that can be used.								
Text file (*.txt)	Data in text format. Refer to the Appendix “JCAMP-DX Format Files” for a description of the file formats that can be used.								
	All files (*.*) : Designates all files. Files not in one of the above formats or ASCII format cannot be loaded.								

3.2.7 [Export...]

Saves data in the Old Format Standard Files, JCAMP-DX (*.dx), ASCII Files (*.txt), CSV Files (*.csv), or XY ASCII (*.txt). The operating procedure is the same as described in 3.2.5 “[Save As...]”.

Note: In Spectra Manager CFR, this function can be used to save a file in a folder other than that of a logged-on Workgroup.

3.2.8 [Open View...]

Select [File] - [Open View...] to display the [Open View] dialog box, which is used to open a View.

Note: In Spectra Manager CFR, this function can not be used.

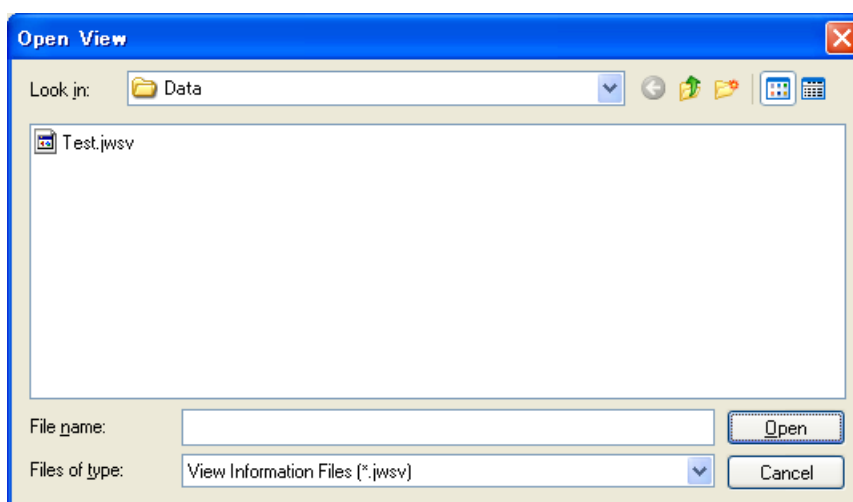


Figure 3.10 [Open View] dialog box

[Look in]	Used to designate the drive or directory that contains the files to be opened. Available drives and directories are shown in the drop-down list. The contents of the drive or directory designated in the [Look in] drop-down list are shown in the file name list.
[File name]	Used to directly input the desired filename.
[Files of type]	Sets the files to display in the filename list. Files other than the View Information Files (*.jwsv) cannot be selected.

3.2.9 [Save Active View...]

Select [File] - [Save Active View...] to display the [Save Active View] dialog box, which is used to save the active View using a new filename and/or in a new location.

Note: In Spectra Manager CFR, this function cannot be used.

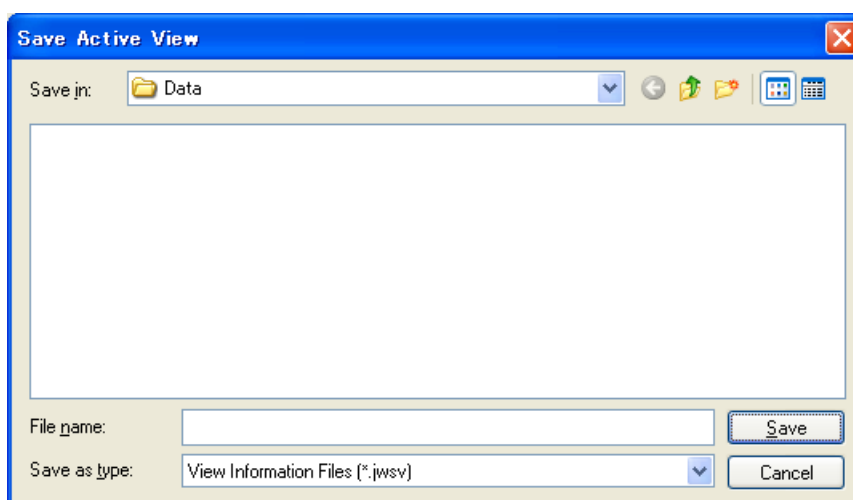
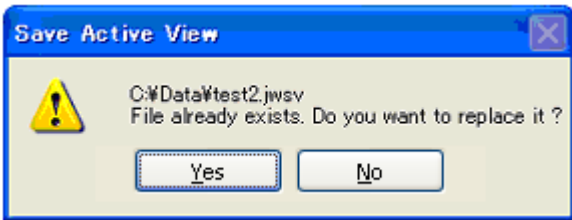


Figure 3.11 [Save Active View] dialog box

[Save in]	Used to designate the destination drive or directory. Available drives and directories appear in the drop-down list. The contents of the drive or directory designated in the [Save in] drop-down list are shown in the dialog box. (Fig. 3.11) To use an existing filename click on the desired filename.
[File name]	Used to directly input the desired View filename. If the extension is omitted, the designated extension is added automatically. If an existing filename is designated and the <Yes> button is clicked, the following dialog box appears. 
[Save as type]	Sets the files to display in the filename list. Files other than the View Information Files (*.jwsv) cannot be selected.

3.2.10 [Print...]

Select [File] - [Print...] to print the active View. The spectrum line style, frame, grid, and font, for example, are printed according to the parameters designated in [Patterns], [Fonts], and [Styles] in the [View] menu.

Note: If the [Prompt the user to type title before printing] checkbox is selected in the [Print] tab of the [Options] dialog box ([Other] – [Options]), when [File] - [Print...] is selected the dialog box shown below is displayed. (Fig.3.13)

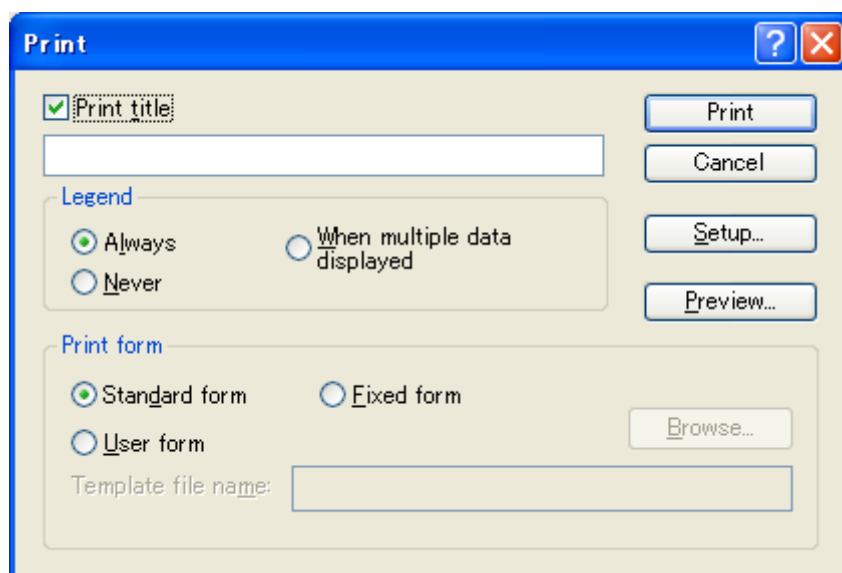


Figure 3.13 [Print] dialog box

[Print title]	Select the checkbox to print a title with the active View. Enter a title in the text field below the checkbox.
[Legend] group	Used to control whether a legend is printed for spectra.
[Always]	A legend is always printed.
[Never]	A legend is never printed.
[When multiple data displayed]	A legend is printed only when two or more spectra are displayed in the active View.
[Print form] group	Designates the print format.
[Standard Form]	Select to print using the standard format.
[Fixed Form]	Select to print using the fixed format.
[User Form]	Select to print using a template file created in JASCO Canvas. Click the <Browse...> button and designate the JASCO Canvas template file to be used.
<Setup...>	Opens the [Print Setup] dialog box used to designate the target printer and printer parameters (See Section 3.2.12).
<Preview...>	Shows a preview of what the active View will look like when printed.
<Print>	Prints the active View using the settings declared within the [Print] dialog box.
<Cancel>	Cancels the [Print] of the active View.

3.2.11 [Print Preview...]

[Print Preview...] shows a preview of what the active View will look like when printed. (Fig. 3.14)

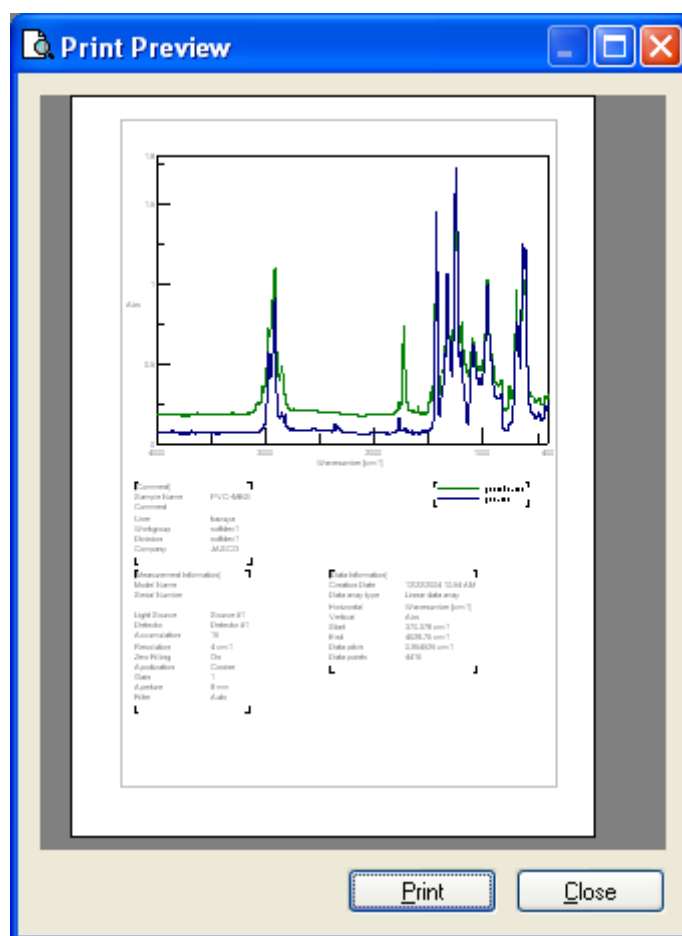


Figure 3.14 [Print Preview...] dialog box

<Print>	Prints the active View.
<Close>	Closes the [Print Preview] dialog box without printing.

3.2.12 [Print Setup...]

Select [File] - [Print Setup...] to designate the target printer and print parameters. (Fig. 3.15)

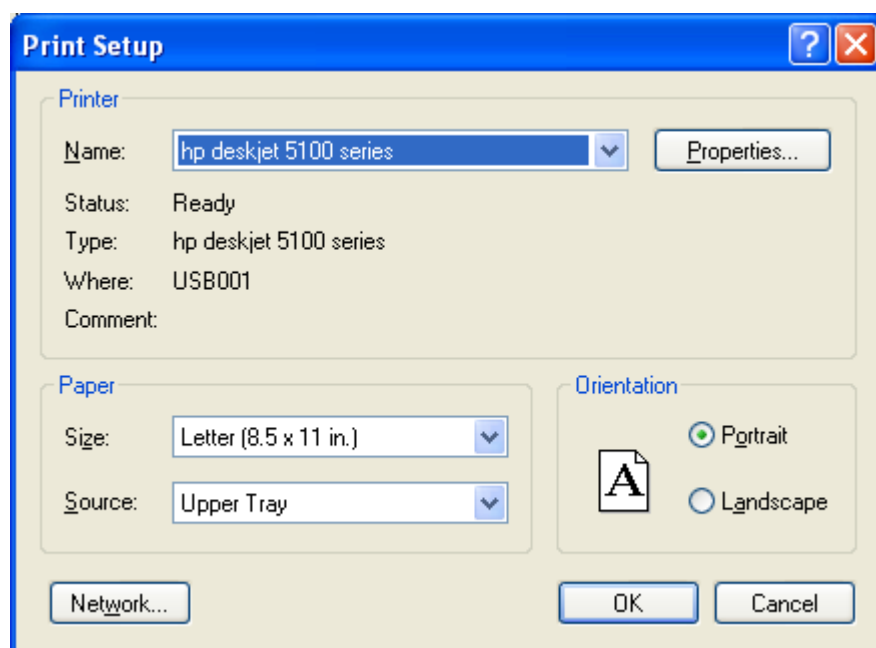


Figure 3.15 [Print Setup...] dialog box

[Printer] group	Select the desired printer from the drop-down list to select a printer other than the default.
[Orientation] group	Used to select the orientation of the paper when printing.
[Portrait]	Prints the View so that the short edge of the paper is the top of the page.
[<u>L</u> andscape]	Prints the View so that the long edge of the paper is the top of the page.
[Paper] group	Used to adjust the paper-related print properties.
[Size]	Select the paper size from the drop-down list.
[Source]	Select the paper source tray from the drop-down list.
<Network...>	Allows selection of any printer currently available to a networked PC.
<OK>	Applies the [Print Setup...] settings and closes the dialog box.
<Cancel>	Closes the dialog box without applying any changes in [Print Setup...].

3.2.13 [Exit]

Select [File] - [Exit] to exit the Spectra Analysis program. The following message appears when unsaved spectra exist. (Fig. 3.16) To save the unsaved spectra, click the <Yes> button.

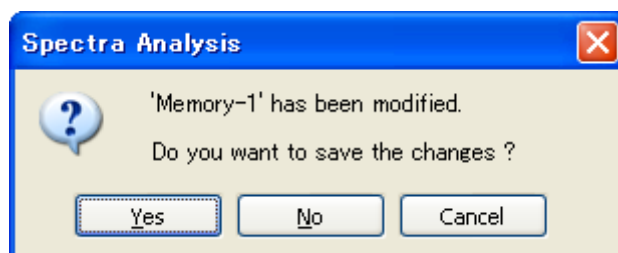


Figure 3.16 Message displayed when attempting to [Exit] with unsaved files

<Yes>	Spectra are saved and the Spectra Analysis program ends.
<No>	The Spectra Analysis program ends without saving the data.
<Cancel>	Cancels the [Exit] command and returns to the active View.

3.3 [Edit] Menu

This section describes the commands in the [Edit] menu.

3.3.1 [Copy as Picture]

Select [Edit] - [Copy as Picture] to copy a spectrum to the *Windows* clipboard as a picture. The spectrum can then be pasted into draw-type software, such as *Microsoft Draw*, for editing or pasted into a document or presentation software.

3.3.2 [Copy as Bitmap]

Select [Edit] - [Copy as Bitmap] to copy a spectrum to the *Windows* clipboard as a bitmap. The spectrum can then be pasted into paint-type software, such as *Microsoft Paint*, for editing or a document or presentation software. The image becomes distorted if the bitmap is greatly expanded or reduced. Therefore, the size of the window (spectrum) should be adjusted accordingly before copying. A clear image can be obtained if care is taken not to change the ratio between the horizontal and vertical lengths when reducing or expanding the pasted image. A large change in the horizontal and vertical ratio always distorts the image.

3.3.3 [Cut Spectrum]

Select [Edit] - [Cut Spectrum] to remove the currently selected spectrum from the active View and transfer it to the [Spectra Analysis] clipboard. When [Select All] is checked, all spectra in the active View are cut and the View is closed. Use this command or the [Copy Spectrum] command, (or the drag-and-drop feature) to place spectra from different Views into one View.

3.3.4 [Copy Spectrum]

Select [Edit] - [Copy Spectrum] to copy the currently selected spectrum in the active View to the [Spectra Analysis] clipboard. The original spectrum remains in the active View. When [Select All] is checked, all spectra in the active View are copied. Use this command to place spectra from different Views into one View.

3.3.5 [Paste Spectrum]

Select [Edit] - [Paste Spectrum] to paste the contents of the [Spectra Analysis] clipboard into the active View.

3.3.6 [Delete Spectrum]

Select [Edit] - [Delete Spectrum] to delete the currently selected spectrum in the active View. When [Select All] is checked, all spectra in the active View are deleted and the View closes.

Note: When a spectrum from a multiple spectra View is deleted or cut, the color of the remaining spectra may change.

3.3.7 [Select All]

Select [Edit] - [Select All] to select/clear all spectra in the active View. When [Select All] is checked all spectra in the active View are selected. This function can be used in combination with [Cut Spectrum], [Copy Spectrum] and [Delete Spectrum].

3.3.8 [Comments...]

Select [Edit] - [Comments] to display the [Comments] dialog box used to add or edit comments. (Fig.3.17)

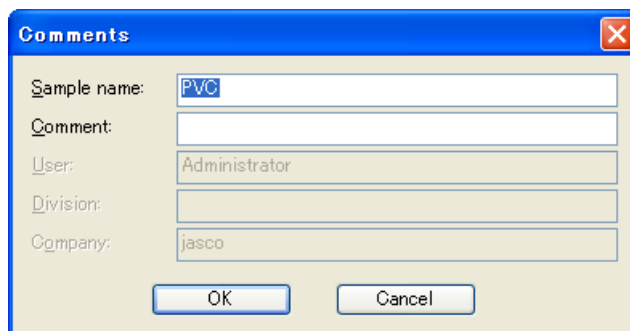


Figure 3.17 [Comments] dialog box

The [Sample name] and [Comment] fields can be filled in or edited. The maximum number of characters that can be input is 62 and 124, respectively.

Note: In the [Spectra Manager CFR], [User], [Division], and [Company] can not be edited.

3.4 [View] Menu

Sets the display format of the spectrum analysis window or View.

3.4.1 [Toolbars]

Shows or hides the toolbar.

Buttons are divided into four categories on four separate toolbars: Main, View, Data Processing and View Command. Each of these toolbars can be hidden or displayed, independently. In addition, the Data Processing toolbar buttons can be customized.

Refer to 3.7.1 [Customize Toolbar] for details.

3.4.2 [Status Bar]

Shows or hides the status bar.

3.4.3 [Work Space]

Shows or hides the Work Space.

3.4.4 [Scales...]

Select [View] - [Scales...] to display the [Scales...] dialog box used to change the scale of the spectrum/spectra displayed in the active View. (Fig.3.18)

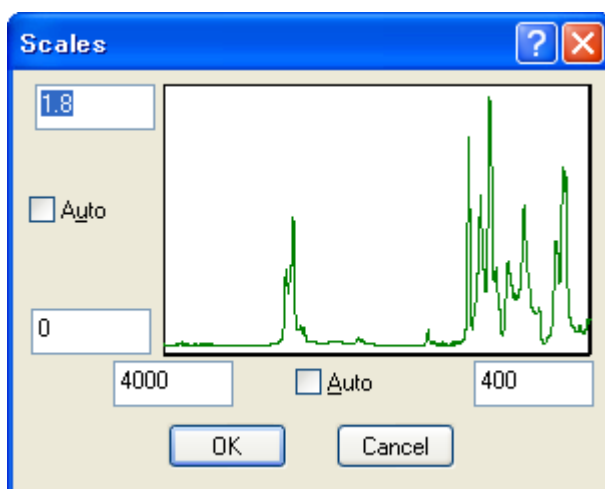


Figure 3.18 [Scales] dialog box

Horizontal scale	Enter the desired start and end values of the horizontal scale in the text boxes at the bottom of the preview pane. When [Auto] is checked, the input values are ignored and all regions are displayed.
Vertical scale	Enter the desired start and end values of the vertical scale in the text boxes at the left of the preview pane. When [Auto] is checked, the input values are ignored and the most appropriate scale for the designated horizontal range is displayed.
<OK>	Applies the changes to the [Scale] dialog box and closes the dialog box.
<Cancel>	Closes the dialog box without applying any [Scale] changes.

Note 1: When the [Apply scale settings to all views.] checkbox in the [Other]-[Options...] dialog box is selected, the scale settings are applied to all Views.

Note 2: This operation can also be performed by clicking the tool button



3.4.5 [Patterns...]

Select [View] - [Patterns...] to display the [Patterns] dialog box used to select the line style and color of the spectrum/spectra displayed in the active View. (Fig.3.19)

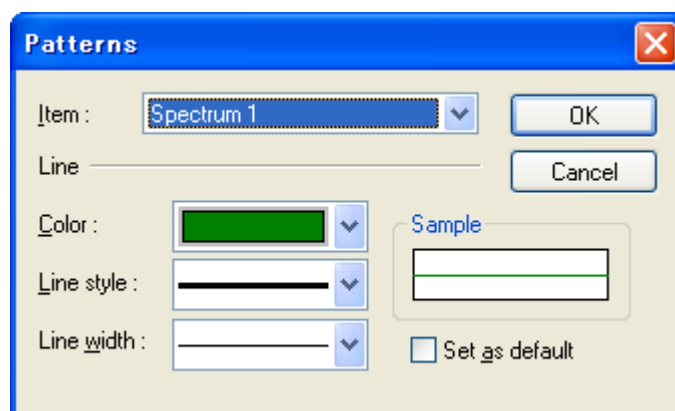


Figure 3.19 [Patterns] dialog box

[Item]	Lists items for which the color or line style can be changed.
[Color]	Select the desired color from the drop-down list. The currently designated item in the [Item] drop-down list is displayed in the selected color.
[Line style]	Shows the available line styles. The currently designated item in the [Item] drop-down list is displayed with the selected line style.
[Line width]	Shows the available line widths. The currently designated item in the [Item] drop-down list is displayed with the selected line width.
[Sample]	Displays a sample of the designated pattern.
[Set as default]	Select the checkbox to use the designated patterns for all subsequently opened Views.
<OK>	Applies the changes to the [Patterns] settings and closes the dialog box.
<Cancel>	Closes the dialog box without applying any changes to the [Patterns] settings.

Note 1: When the [Apply view settings to all spectra.] checkbox in the [Other]-[Options.] dialog box is selected, the pattern settings are applied to all Views.

Note 2: This operation can also be performed by clicking the tool button



3.4.6 [Fonts...]

Select [View] - [Fonts...] to display the [Fonts] dialog box used to select the font settings and orientation for the active View. (Fig. 3.20)

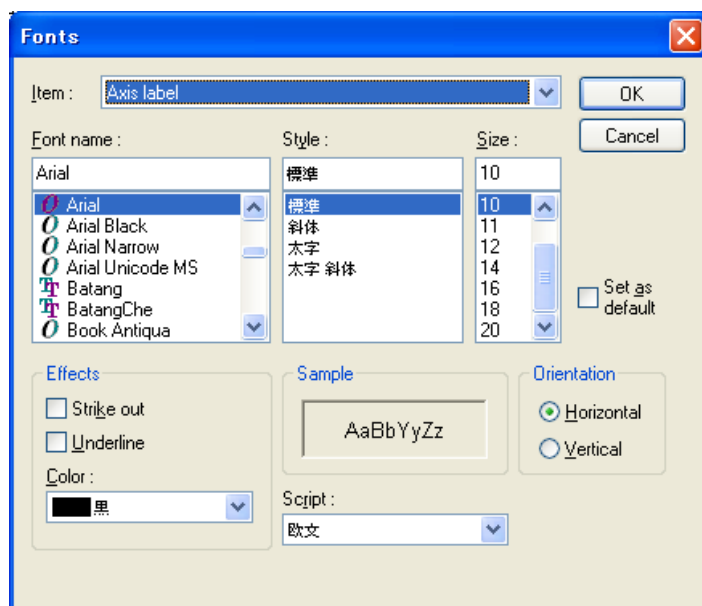


Figure 3.20 [Fonts] dialog box

[Item]	Lists the items for which the font settings can be changed. The font settings for the axis label, scale label, and peak mark can all be changed.
[Font name]	Lists available font names. The currently designated item in the [Item] drop-down list is displayed in the selected font.
[Style]	Lists available styles. The currently designated item in the [Item] drop-down list is displayed in the selected style.
[Size]	Lists available sizes. The currently designated item in the [Item] drop-down list is displayed in the selected size.
[Effects]	Strikeout and underline can be specified.
[Sample]	Displays a sample of the specified font.
[Orientation]	Selects the orientation in which the axis label is to be displayed. [Horizontal]: Horizontal to the spectrum display window [Vertical]: Vertical to the spectrum display window
[Set as default]	Select this checkbox to use the designated fonts for all subsequently opened Views.
[Script]	
<OK>	Applies the [Fonts] settings and closes the dialog box.
<Cancel>	Closes the [Fonts] dialog box without applying and changes in settings.

Note 1: When the [Apply view settings to all spectra.] checkbox in the [Other]-[Options.] dialog box is selected, the pattern settings are applied to all Views.

Note 2: This operation can also be performed by clicking the tool button [Fonts].



3.4.7 [Gridlines...]

Select [View] - [Gridlines] to display the [Gridlines] dialog box, which is used to add/remove vertical and/or horizontal lines to the active View. (Fig. 3.21)

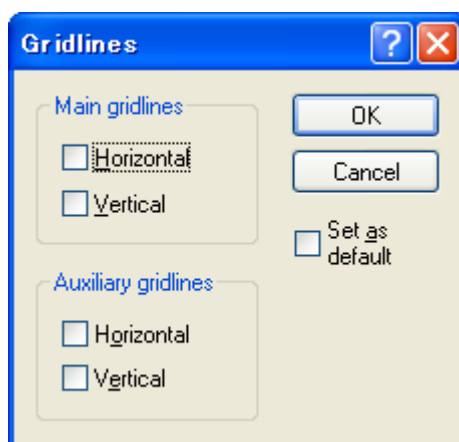


Figure 3.21 [Gridlines] dialog box

[Main gridlines] group	
[Horizontal]	Select the checkbox to display major horizontal axis lines.
[Vertical]	Select the checkbox to display major vertical axis lines.
[Auxiliary gridlines] group	
[Horizontal]	Select the checkbox to display minor horizontal axis lines.
[Vertical]	Select the checkbox to display minor vertical axis lines.
[Set as default]	Select this checkbox to use the current gridlines for all subsequently opened Views.
<OK>	Applies the changes to the [Gridlines] settings and closes the dialog box.
<Cancel>	Closes the dialog box without applying any changes to the [Gridlines] dialog box.

Note 1: When the [Apply view settings to all spectra.] checkbox in the [Other]-[Options.] dialog box is checked, the pattern settings are applied to all Views.

Note 2: This operation can also be performed by clicking the tool button



[Gridlines].

3.4.8 [Styles...]

Select [View] - [Styles...] to display the [Styles] dialog box used to change the scale settings for the active View. (Fig. 3.22)

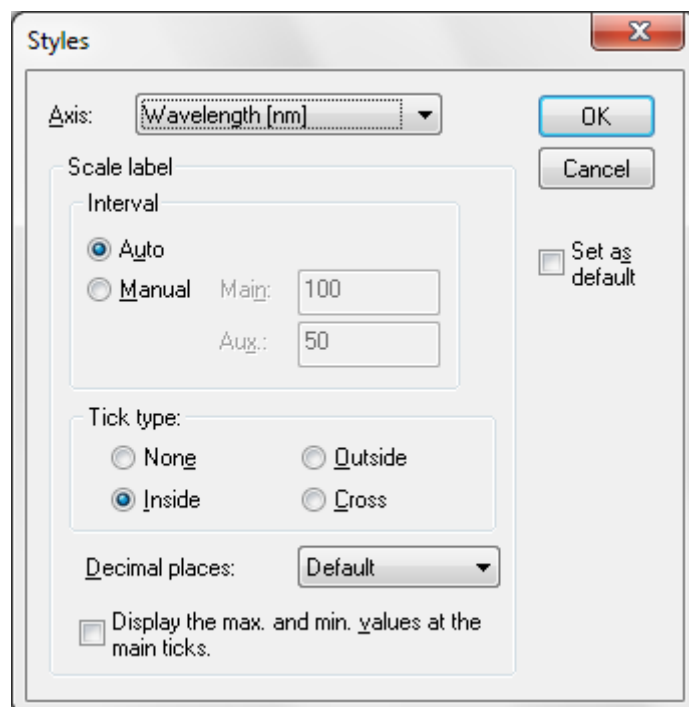


Figure 3.22 [Styles] dialog box

[Axis]	Selects an item to change its display style.
Interval	
[Auto]	Automatically selects the gridline format.
[Manual]	Manually specifies the gridline format of the main and auxiliary scales.
Tick type	
[None]	Displays no ticks on the axis
[Outside]	Displays ticks on the outside of the axis.
[Inside]	Displays ticks on the inside of the axis.
[Cross]	Displays ticks on both sides of the axis.
[Decimal places]	Selects the number of decimal places for the display values of horizontal and vertical axes.
[Display the max. and min. values at the main ticks]	Displays maximum and minimum values at the main ticks. Values at the auxiliary ticks are not displayed.
[Set as default]	Applies the specified styles to all subsequently opened Views.
<OK>	Applies the changes to the [Styles] settings and closes the dialog box.
<Cancel>	Closes the dialog box without applying any changes to the [Styles] settings.

Note: This operation can also be performed by clicking the tool button  [Styles].

3.4.9 [Horizontal Axis Units...]

Select [View] - [Horizontal Axis Units...] to display the [Horizontal Axis Units] dialog box used to change the horizontal axis (Fig. 3.23).

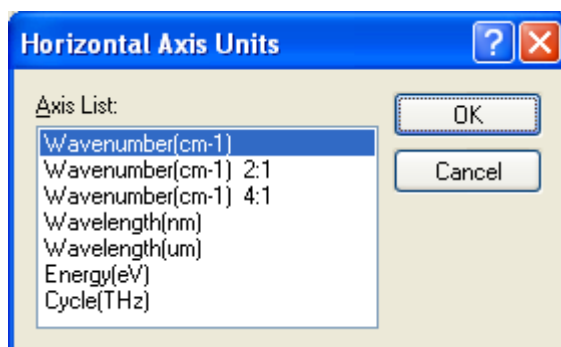


Figure 3.23 [Horizontal Axis Units] dialog box

[Axis List]	Lists the available formats for the horizontal axis.
<OK>	Accepts the changes to [Horizontal Axis Units] and closes the dialog box.
<Cancel>	Closes the dialog box without applying any change to the [Horizontal Axis Units] settings.

3.4.10 [Information...]

Displays spectrum information for the currently selected spectrum in the active View. (Fig.3.24)

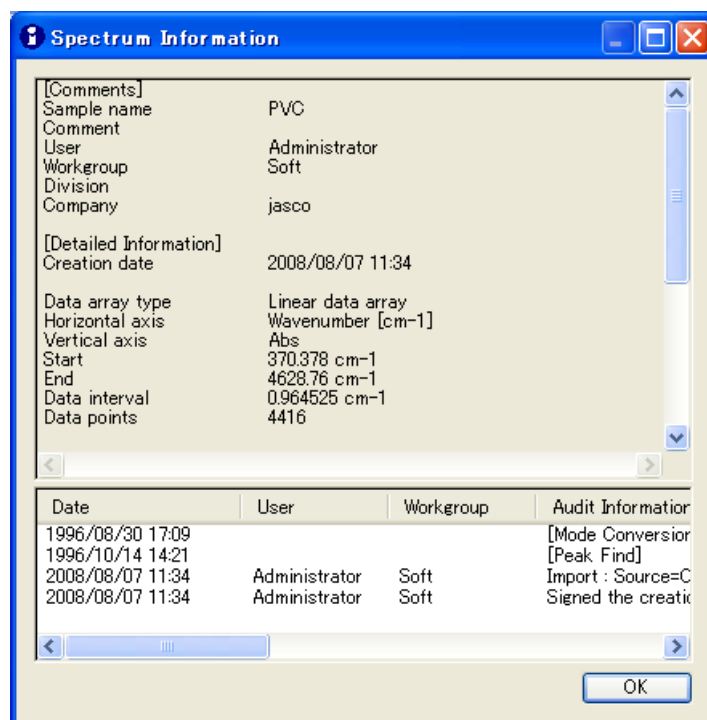


Figure 3.24 [Spectrum Information] dialog box

3.4.11 [Normalize]

Automatically normalizes the scale of each spectrum displayed in the active View. The purpose of this function is so that spectra having different vertical axis scales can be viewed simultaneously in a single View window. When the [Normalize] function is active, a box appears around the vertical axis label in the active View.

For example, for two spectra, displayed in the %T and Abs modes, respectively, the spectrum displayed in Abs mode is not visible, as shown in Fig. 3.25 (a).

By switching to [Normalize], both spectra are visible, as shown in Fig. 3.25 (b).

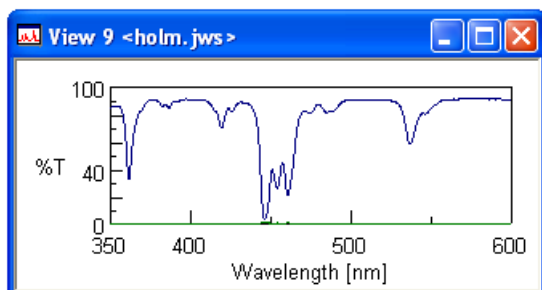


Figure 3.25 (a) Overlaid view

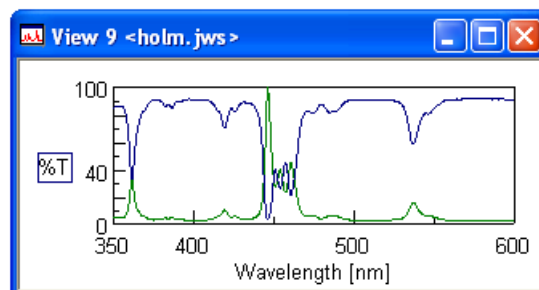


Figure 3.25 (b) Normalized view

Note 1: This function can also be executed by clicking . When this button is selected (shown as depressed), Normalize-View is displayed. A second click on the button restores the View to the original state, returning control to the normal View.

Note 2: Scale change in Normalize-View varies slightly from that in the Overlay-view. Only the scale of an active spectrum is changed. The [Scales.] dialog box for Normalized view is shown below. (Fig. 3.26) Change the scale after selecting a spectrum from the legend drop-down list at the top of the dialog box.

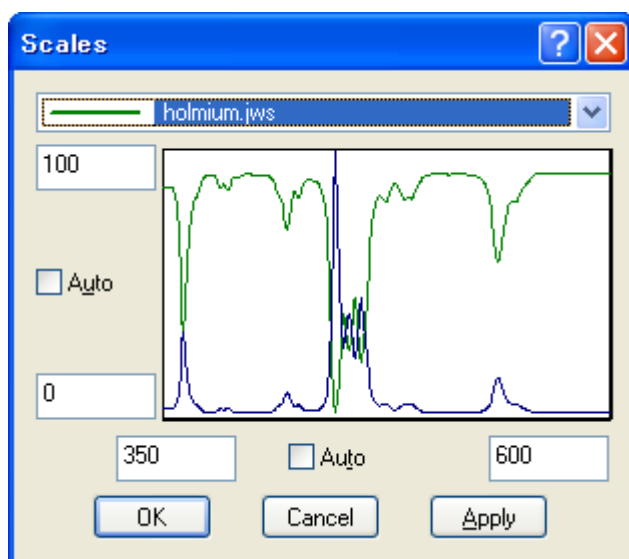


Figure 3.26 [Scales] dialog box for normalized view

3.4.12 [Peak Indicators...]

Click [View] - [Peak Indicators] to display a submenu in which you can designate how the peak value or peak position of the active spectrum should be displayed when the [Peak Find...] command is executed.

[None]	Peak position not displayed.
[Bar]	Denotes peak position using a vertical bar.
[Bar, Data No.]	Denotes peak position using a vertical bar, and displays the peak number.
[Bar, X]	Denotes peak position using a vertical bar, and displays the horizontal axis value.
[Bar, Y]	Denotes the peak position using a vertical bar, and displays the vertical axis value.
[Bar, X, Y]	Denotes the peak position using a vertical bar and displays both the horizontal and vertical axes values.

<i>Note: Specify the orientation of the labels to display x- and y-axes values in Fig. 3.20 of Section 3.4.6. It can be changed as desired.</i>
--

3.5 [Processing] Menu

This section describes the commands in the [Processing] menu.

3.5.1 [Corrections] submenu

3.5.1.1 [Baseline...]

Corrects the active spectrum using a free baseline (Fig. 3.27). A baseline having a maximum of 32 variable points (including both ends) can be created. The variable points can be connected by a straight line or by a spline curve.

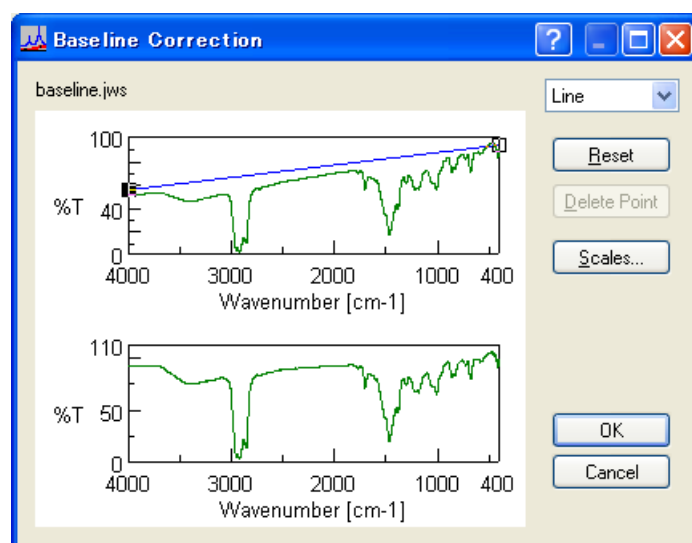


Figure 3.27 [Baseline Correction] dialog box

[Line]	Select to connect adjacent variable points by a straight line.
[Spline]	Select to connect adjacent variable points by a spline curve.
<Reset>	Click to restore the baseline to its original shape (i.e. two variable points connected by a straight line).
<Delete Point>	Click to erase a single variable point. Darkened points or points having thick square frames can be erased. If a variable point on the left or right end is erased, the next point becomes the end point.
<Scales...>	Sets the vertical and horizontal axis scales for both spectra views.
Spectrum display area	Set the spectrum baseline in the upper spectrum to display the results below it. Move the small squares on the baseline (initially only at the ends) to change the baseline shape. When the pointer approaches a square, the shape of the pointer changes to a set of tweezers that can be used to drag the square to another location. When the pointer approaches the baseline, the shape of the pointer changes to a syringe. Click the left mouse button to add variable points.
<OK>	Applies the baseline settings and displays a new View with the calculated baseline correction after closing the dialog box.
<Cancel>	Closes the dialog box without applying any setting of the [Baseline Correction] parameters.

3.5.1.2 [Smoothing...]

Smooths the active spectrum and reduce the apparent noise in the spectrum. (Fig.3.28) The following four smoothing methods are available: Means-Movement, Savitzky-Golay, Adaptive-Smoothing and Binomial.

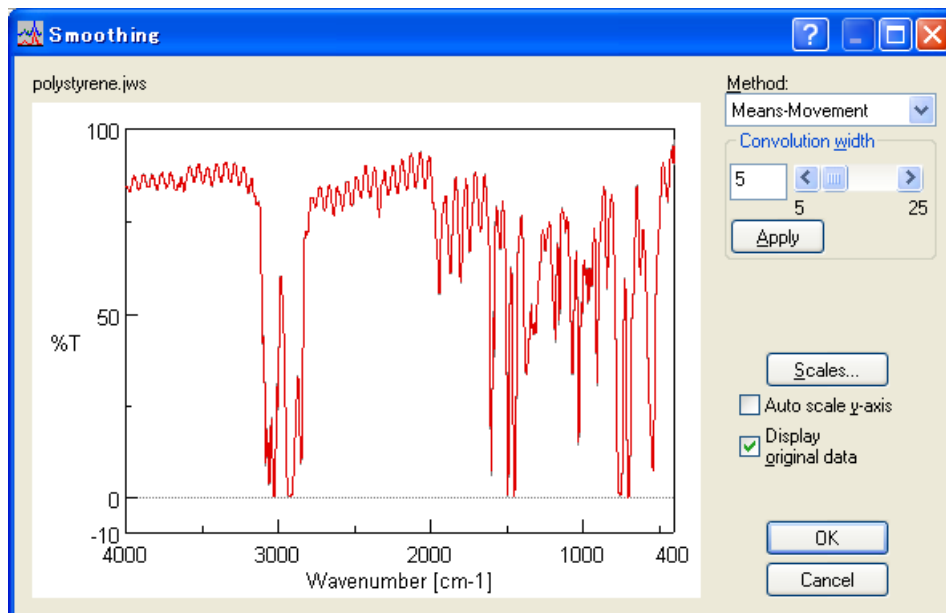


Figure 3.28 [Smoothing] dialog box

[Method] drop-down list	Lists the available smoothing methods.
[Convolution width]	Sets the number of data points to be bundled on both sides for each data point. Only odd numbers can be set. Values are immediately recalculated after each change.
<Apply>	Applies the set convolution width value, and then performs the calculation.
Spectrum display area	The original spectrum is displayed in gray (if the [Display original data] box is checked) while the smoothed spectrum is always displayed in red.
<Scales...>	Click to set the spectrum view range.
[Auto scale y-axis]	Select the checkbox to view the results using the auto scale function. Clear to view the changes resulting from the smoothing calculation.
[Display original data]	Overlays a spectrum before smoothing. The original spectrum is displayed in gray and the processed spectrum is displayed in red.
<OK>	Applies the smoothing parameters and displays a new View with the smoothed spectrum after closing the dialog box.
<Cancel>	Closes the dialog box without applying any setting of the [Smoothing] parameters.

Note: Refer to “Smoothing Method” in the Appendix for details on each method.

3.5.1.3 [Noise Elimination...]

Eliminates false peaks due to noise of known origin. (Fig.3.29) The peak is eliminated by connecting the designated horizontal axis ranges by a straight line. The horizontal axis range can also be changed.

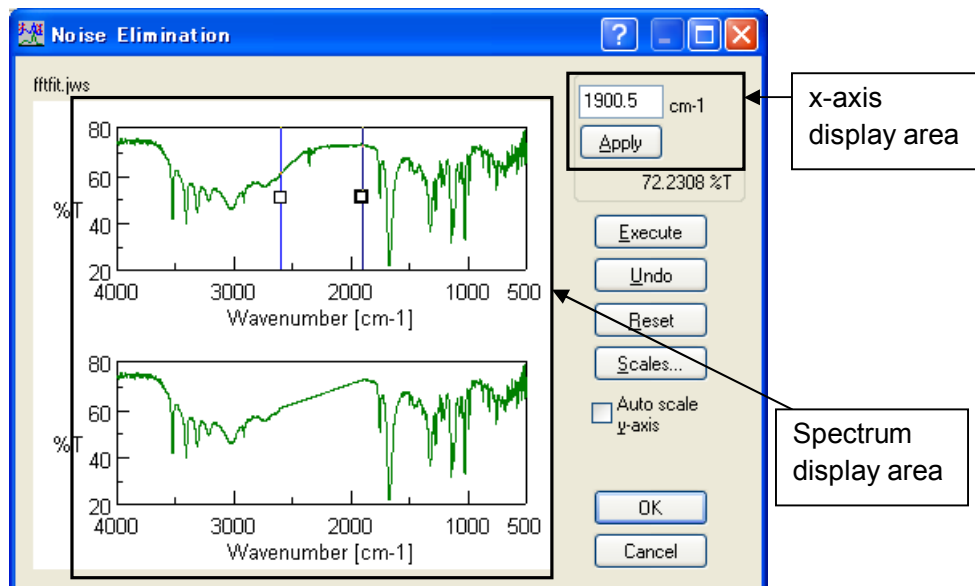


Figure 3.29 [Noise Elimination] dialog box

Spectrum display area	The top and bottom spectra are the same when the dialog box is opened. Specify the range for eliminating peaks by dragging the two vertical bars in the top spectrum and click the <Execute> button to display the resulting spectrum. Desired peaks can be eliminated by repeating this procedure.
x-axis display area	Displays the position of the active x-axis cursor. Input an x-axis value, and then click the <Apply> button to shift the active x-axis cursor to the designated position.
<Apply>	Moves the active x-axis cursor to the designated position.
<Execute>	Executes noise elimination.
<Undo>	Undoes the most recently executed noise elimination.
<Reset>	Resets the spectrum to its original condition before noise elimination.
<Scales...>	Click to set the spectrum range.
[Auto scale y-axis]	Select the checkbox to view the results using the auto scale function. Clear when viewing the change in size of the differential spectrum.
<OK>	Applies the [Noise Elimination] settings and displays a new View with the calculated spectrum after closing the dialog box.
<Cancel>	Closes the dialog box without applying any of the [Noise Elimination] settings.

3.5.1.4 [Deconvolution...]

This function analyzes bands that may contain overlapping Lorentzian curves having the same full width at half maximum value and accurately distinguishes the peak positions for each band. (Fig.3.30)

Note: *The region outside the wavelength range of interest will also be influenced by the deconvolution calculation, and the spectral profile will change as a result of the deconvolution calculation.*

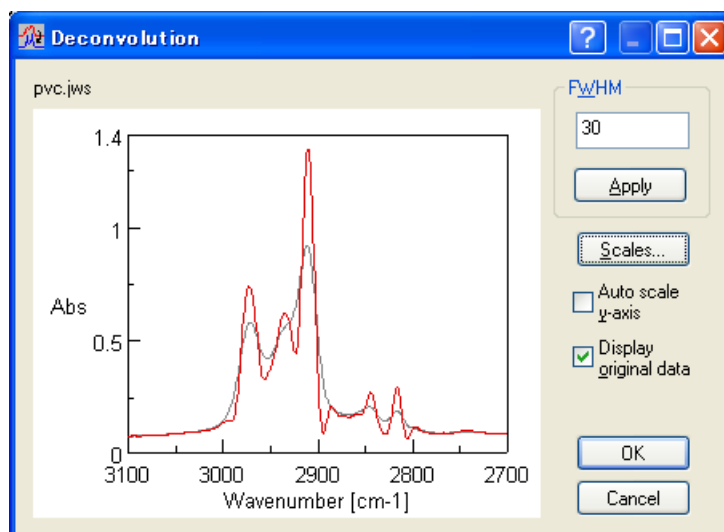


Figure 3.30 [Deconvolution] dialog box

[FWHM]	Specify the full width at half maximum value of the expected Lorentzian waveform.
<Apply>	Click to perform deconvolution using the input value.
<Scales...>	Click to set the spectrum range.
[Auto scale y-axis]	Select this checkbox to view the results using the auto scale function. Clear to view the change in size of the calculated spectrum.
[Display original data]	Overlays the original spectrum before deconvolution. The original spectrum is displayed in gray and the calculated spectrum is displayed in red after processing.
<OK>	Applies the final [Deconvolution] settings and opens a new View with the calculated spectrum after closing the dialog box.
<Cancel>	Closes the dialog box without applying any of the [Deconvolution] parameters.

Note: *Before clicking <OK> and confirming the changes, the deconvolution calculation can be executed with different full width at half maximum values any number of times. For more details on deconvolution refer to the Appendix 4.3.*

3.5.1.5 [FFT Filter...]

Separates the spectrum into frequency components using a Fast Fourier transformation and eliminates noise having a particular periodicity. (Fig.3.31) A filter containing a maximum of 32 variable points (including both ends) can be created. However, if the heights of the peaks and the noise are approximately equal, the noise cannot be removed.

Note: *[FFT Filter] processing is not compatible with data having uneven data point intervals (pitches). In order to conduct [FFT Filter] processing, it is necessary to interpolate the data into data with an even data interval by conducting [Truncate Data] in advance.*

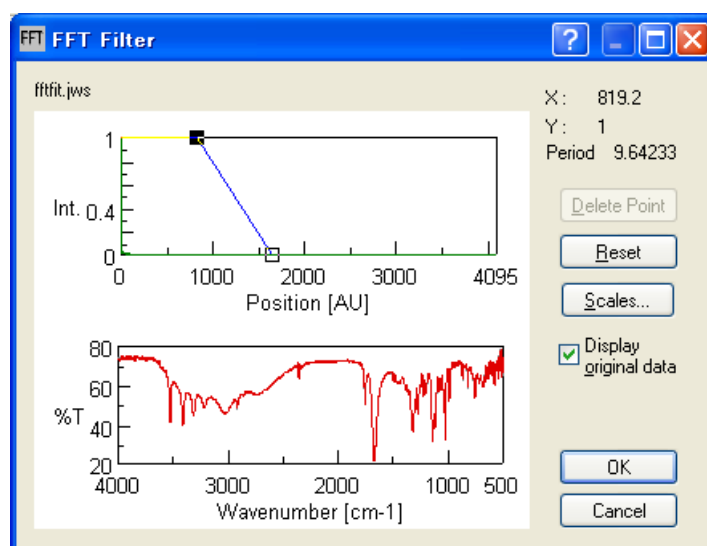


Figure 3.31 [FFT Filter] dialog box

Spectrum display area	Set the shape of the FFT filter in the upper spectrum and the results are displayed in the view below it. Move the small squares on the filter (initially only at the ends) to change the filter shape. When the pointer approaches a square, the shape of the pointer changes to a set of tweezers that can be used to drag the square to another location. When the pointer approaches the filter, the shape of the pointer changes to a syringe. Click the left mouse button to add additional filter points. The X and Y values at the left correspond to the coordinates of the currently selected filter point and the period to the period of noise (cm^{-1}).
<Delete Point>	Click to erase a single variable point. Darkened points or points having thick square frames can be erased. If a variable point on the left or right end is erased, the next successive point will become the end point. The minimum number of variable points is two. If only two points exist, neither can be erased.
<Reset>	Returns the filter to its original shape.
<Scales...>	Click to set the spectrum range.
[Display original data]	Overlays a spectrum before applying the FFT filter. The original spectrum is displayed in gray and the processed spectrum is displayed in red.

<OK>	Applies the parameter settings and displays the calculated spectrum in a new View after closing the dialog box.
<Cancel>	Closes the dialog box without applying the [FFT Filter] parameters.

[Example of application of FFT filter]

The FFT filter can only selectively eliminate noise having a specific period. As an example, the noise is eliminated using the supplied sample data file “fftfilt.jws”.

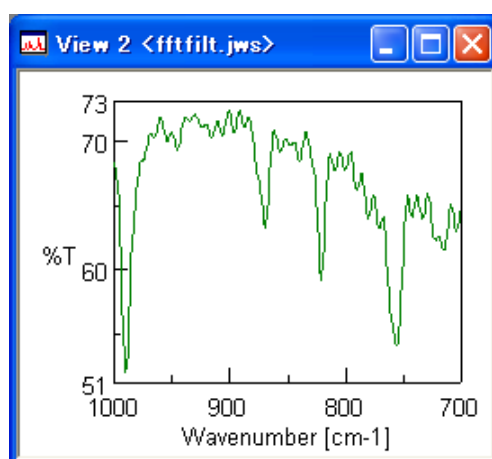


Figure 3.32 fftfilt.jws

First, open the spectrum fftfilt.jws, change the scale and display the range as shown in the figure above (Fig. 3.32) (accuracy is not necessary). This is a spectrum having high noise. Determine the period of the noise by using a simple trace. Determine the start and end wavenumbers of the period of noise by referring to the figures below. (Fig. 3.33) In this example, one period is approximately 10 cm^{-1} . This value should be noted for later use.

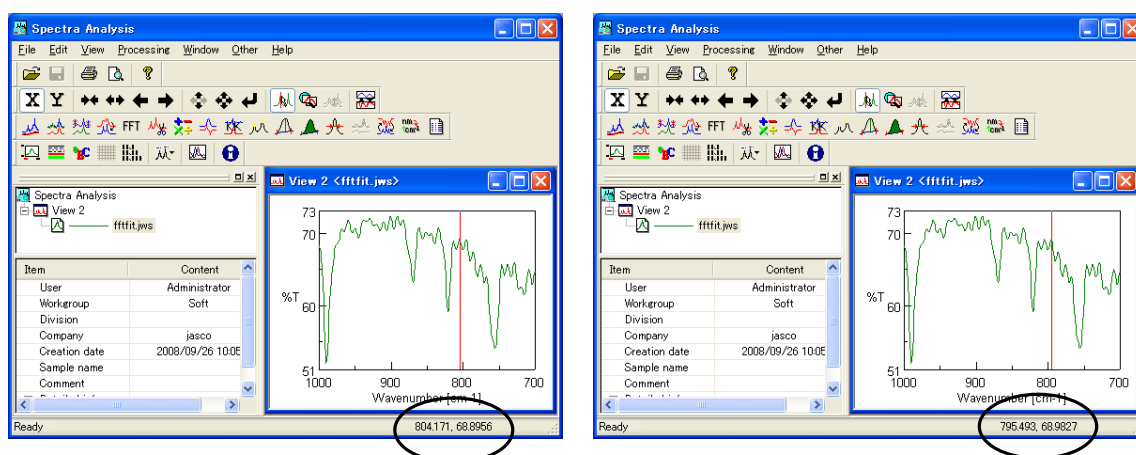


Figure 3.33 Determination of the noise period

When the period of the noise is known, execute FFT filter. The following dialog box appears. (Fig. 3.34)

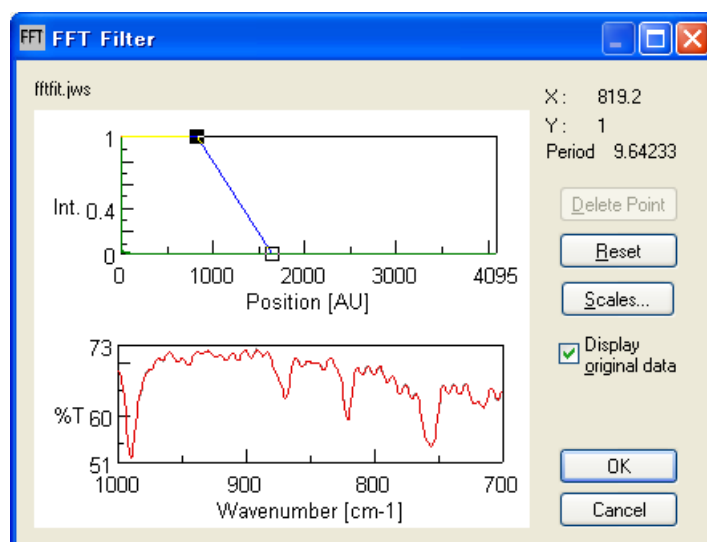


Figure 3.34 [FFT Filter] dialog box

Referring to the period of the noise displayed at the upper right of the window in Fig. 3.34, move the two small squares to the neighborhood of 10 cm^{-1} to indicate that the noise having a period of less than 10 cm^{-1} will be eliminated. Each time the filter shape is changed, the spectrum of the calculated result will be rewritten. Checking [Display original data] overlays the data before calculation, allowing the filter shape to be changed while comparing the resulting spectrum to the original data. (Fig. 3.35)

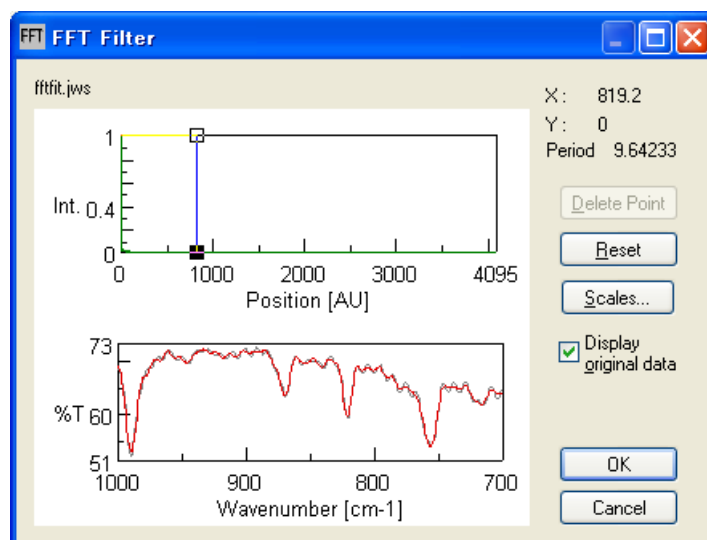


Figure 3.35 [FFT Filter] dialog box

A complex filter may also be set by increasing the number of rectangles. However, if only noise having a short period is to be eliminated, a simple shape such as that shown above will suffice.

3.5.1.6 [Truncate Data...]

Creates a spectrum having a defined data interval with a specified x-axis range. Data in any region of the x-axis can also be cut without changing the data point interval. (Fig. 3.36)

This function is used, for example, when converting data having an irregular data point interval (pitch) into data having an evenly spaced data point interval (pitch) in order to conduct [Derivatives], [Curve Fitting] or [FFT Filter] processing.

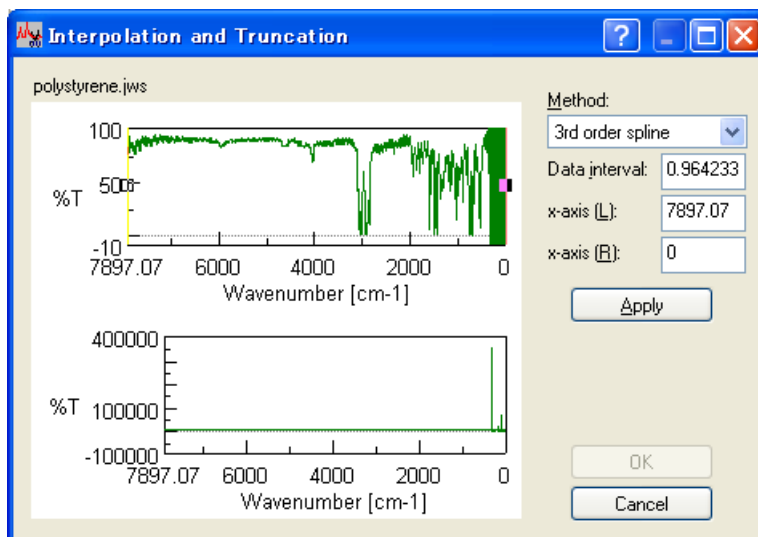


Figure 3.36 [Interpolation and Truncation] dialog box

Note: Figure 3.36 shows spectra before the execution of data interpolation and truncation calculations. Therefore, the upper and lower spectra in the spectrum display area are the same. After execution, the resulting spectrum is displayed in the lower half of the spectrum display area.

Spectrum display area	The original spectrum is displayed at the top, and the calculated spectrum with the data truncation and interpolation (if applied) is displayed below it.
[Method] drop-down list	Select the method of data interpolation. The following options are available (see reference section below): [Lagrange] Lagrange interpolation [3 rd order spline] Tertiary spline interpolation [No correction] Select this to simply truncate data.
[Data interval]	Specify the pitch (data point interval) of the data to be created.
[x-axis (L)]	Specify the upper x-axis value of the data to be created.
[x-axis (R)]	Specify the lower x-axis value of the data to be created.
<Apply>	Click to execute data interpolation and truncate using the entered values.
<OK>	Applies the parameter settings and displays a new View with the calculated spectrum after closing the dialog box.
<Cancel>	Closes the dialog box without applying any of the [Truncate Data] parameters.

Reference: Methods of data interpolation

Data interpolation determines the interpolation function, F that satisfies $Y_i = F(X_i)$; $i=1,2,3,\dots,N$ from the data points $\{(X_i, Y_i); i=1,2,3,\dots,N\}$ obtained by measurement in order to estimate Y with respect to X (except for known X_i).

“Lagrange interpolation” and “Tertiary spline interpolation” are representative methods of data interpolation.

(1) Lagrange interpolation

This method determines a function F that passes all data points in a specified range for interpolation. For N data points, the interpolation function F is basically a complicated $(N - 1)^{\text{th}}$ -order function. However, in practice, calculation is performed through special operations.

(2) Tertiary spline interpolation

This method calculates the values between N data points within a specified range using $(N - 1)$ items of different tertiary functions F_i . The interpolation functions F_i and F_{i+1} of adjacent sections are determined such that the value of the function and the primary and secondary derivative factors at the boundary point agree.

3.5.2 [Operations] submenu

3.5.2.1 [Arithmetic...]

Performs an arithmetic operation between spectra or between a spectrum and a constant(s). (Fig. 3.38) When only one spectrum is displayed in the active View, the arithmetic operation is performed using a constant(s). When two spectra are displayed, the arithmetic operation is performed between the two spectra. When three or more spectra are displayed, the second spectrum to be used in the arithmetic operation must be designated from the [Select Spectrum] dialog box (Fig. 3.37), which opens before the [Arithmetic] dialog box. (Fig. 3.38) The active spectrum in the View is always selected as the primary spectrum for the Arithmetic operation.

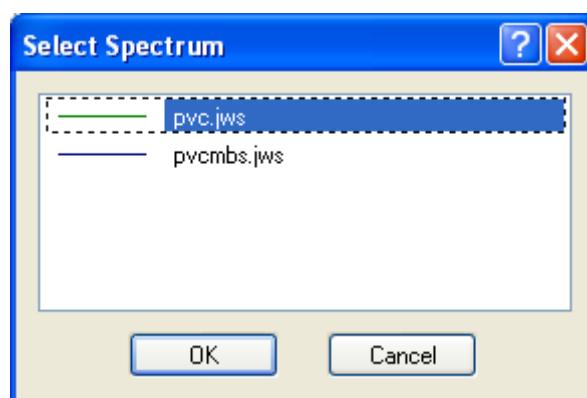


Figure 3.37 [Select Spectrum] dialog box

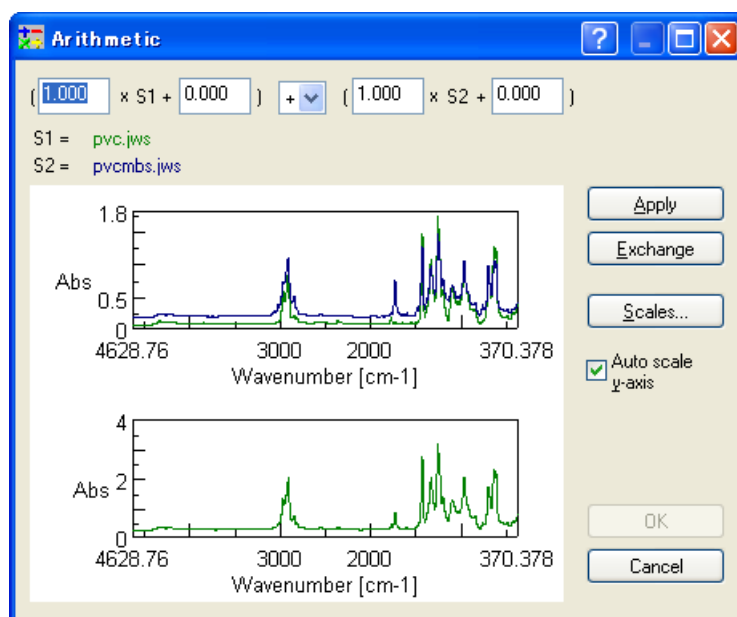


Figure 3.38 [Arithmetic] dialog box (when two spectra are displayed)

The operation to be performed is shown at the top of the dialog box. (Fig. 3.38) S1 and S2 represent the selected spectra and are defined immediately below the Arithmetic expression to be calculated. Enter constants into the appropriate text boxes and select the desired arithmetic operator from the drop-down list to generate an arithmetic expression.

<Apply>	Click to execute the calculation using the designated parameters.
<Exchange>	Reverses the order of spectra S1 and S2.
Spectrum display area	The original spectrum or spectra is (are) displayed at the top, the results of the arithmetic operation displayed below it.
<Scales...>	Click to set the spectrum display range.
[Auto scale y-axis]	Select the checkbox to view the results using the auto scale function. Deselect to view the change in the size of the calculated spectrum.
<OK>	Applies the settings for the Arithmetic expression and displays the calculated spectrum in a new View after closing the [Arithmetic] dialog box.
<Cancel>	Closes the dialog box without applying any setting of the [Arithmetic] parameters.

Note 1: To perform an arithmetic operation between one spectrum and a constant in a View in which multiple spectra are displayed, open the dialog box via the same procedure as when performing an arithmetic operation between two spectra. Input 0 for the factor for the second spectrum and a constant not related to the desired operation.

Note 2: The result of dividing between single beams (SB) is converted to %T. When adding or subtracting %T (%R), the values are added or subtracted using the same units. (Addition of 100% and 100% yields 200% and subtraction of 100% and 100% yields 0%. (This is in contrast to the difference spectrum obtained using [Subtraction...]).

3.5.2.2 [Derivatives...]

Determines the derivative of a spectrum. (Fig.3.39) First-, second-, third-, and fourth-order derivative calculations can be performed using a [Subtract] or [Savitzky-Golay] algorithm.

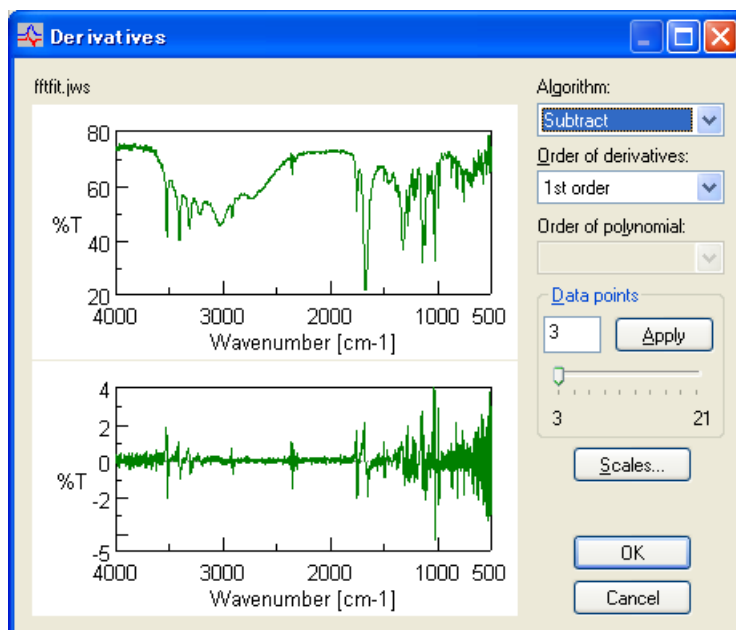


Figure 3.39 [Derivatives] dialog box

Spectrum display area	The original spectrum is displayed at the top, and the calculated spectrum is displayed below it.
[Algorithm]	Selects the derivative algorithm from the drop-down list.
[Order of derivatives]	Select the order of the derivative. Choose a 1 st , 2 nd , 3 rd , or 4 th order derivative function from the drop-down menu.
[Order of polynomial]	Selects the order of a smoothing polynomial for the [Savitzky-Golay] algorithm. Choose a 1 st , 2 nd , 3 rd , 4 th or 5 th order polynomial from the drop-down menu.
[Data Points]	Selects the interval between data points, in effect, a smoothing function.
<Apply>	Performs the smoothing interval selected in [Data Points].
<Scales...>	Selects an x-axis range for the spectra displayed in the dialog box.
<OK>	Applies the parameter settings and displays a new View after closing the dialog box.
<Cancel>	Closes the dialog box without applying any of the [Derivatives] parameters.

Note: For the derivative calculation of a spectrum containing high noise, it is recommended to conduct smoothing first or use a greater number of [Data Points].

3.5.2.3 [K-K Transform...]

Determines the absorbance from a specular reflection spectrum (Kramers-Kronig transformation). (Fig.3.40) Only available for spectra for which the vertical axis is absorbance (Abs), %T or %R. Since the specular reflection spectrum is affected by the refractive index and absorption of the sample, the measured spectrum is results in data with a derivative profile. The derivative type spectrum can be divided into two components: an absorption index (k) and a refractive index (n), by subjecting the reflectance spectrum to Kramers-Kronig (K-K) transformation. This processing can only be used effectively for a spectrum with a derivative profile and for which both ends are flat (containing no peaks). In addition, when applying Kramers-Kronig transformation to only a specific region, select a region for which both ends are flat.

Note: Only the display area is calculated, the data in other areas are truncated from the original spectrum.

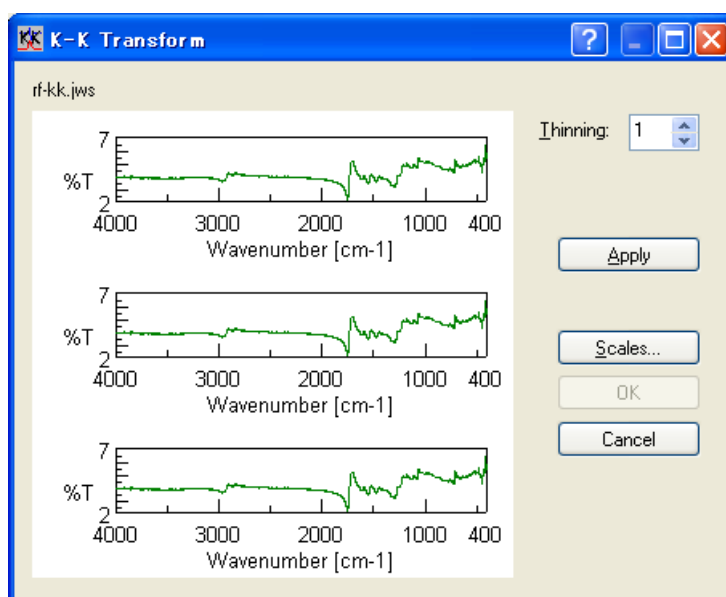


Figure 3.40 [K-K Transform] dialog box

Note: Figure 3.40 shows spectra before execution of K-K transform. All three spectra in the spectrum display area are the same. After execution, the top spectrum is the original spectrum, the middle spectrum is the K-K-transformed spectrum and the bottom spectrum is the refractive index spectrum obtained as a result of the calculation. (Fig. 3.41)

Spectrum display area	The top spectrum is the original spectrum, the middle spectrum is the K-K-transformed spectrum and the bottom spectrum is the refractive index spectrum obtained as a result of the calculation. (Fig. 3.41)
[Thinning]	Designates the interval between data points to be used for calculation. Calculation is faster when this value is large.
<Apply>	Click to execute calculation using the set parameters. The K-K-transformed spectrum is in Abs mode.
<Scales...>	Used to select the x-axis range of the calculation to be performed
<OK>	Click to close the dialog box. Two spectra containing the absorption index (k) spectrum and refractive index (n) spectrum are displayed in two separate Views.
<Cancel>	Closes the dialog box without applying any setting of the [K-K Transform] parameters.

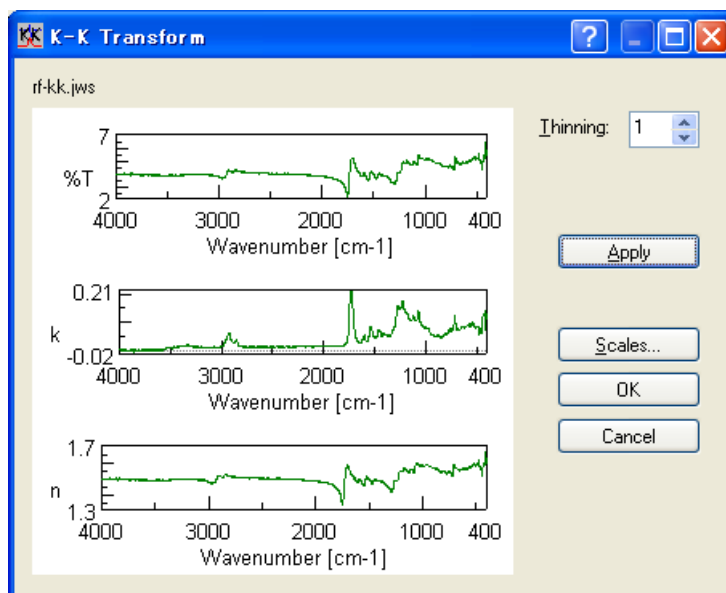


Figure 3.41 [K-K Transform] result dialog box

3.5.3 [Peak Processing] submenu

3.5.3.1 [Peak Find...]

Detects the peaks and/or valleys in the spectrum. (Fig.3.42) The Peak Table and marker display on the spectrum can be updated by changing the peak find parameters in the Parameters box and clicking the <Apply> button.

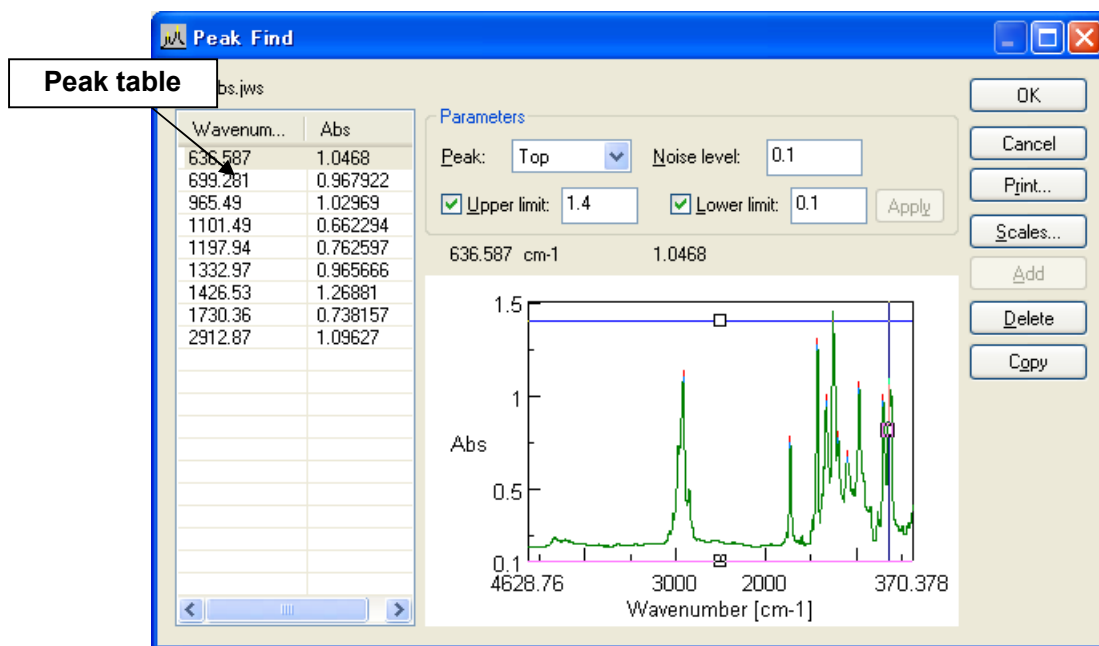


Figure 3.42 [Peak Find] dialog box

Peak table	Displays the peak find results for the active spectrum. Clicking on a value causes a vertical cursor to appear at the selected top/bottom position in the spectrum display area below. The top/bottom position and the intensity are displayed as x- and y-axis values, respectively, in the peak table on the left. [Copying Peak table]: The result of [Peak Find] can be copied to the <i>Windows</i> clipboard by selecting Peak table and pressing Ctrl + C.
[Peak] drop-down list	Lists the modes available for detecting the peaks or valleys in the spectrum. Top: Detects peaks. Bottom: Detects valleys. Both: Detects both peaks and valleys.
[Noise level]	Enter a threshold value used to recognize the top/bottom of the peak. When the difference between the start of the top/bottom of a peak to its apex (valley to peak to valley) does not meet or exceed this value, the peak will not be recognized. The units are the same as those used for the active spectrum.
[Upper limit], [Lower limit]	When these checkboxes are checked, the [Peak Find] function searches the portion of the spectrum currently specified by the upper and/or lower horizontal cursor(s). Therefore, by changing the position of the horizontal cursor(s) it is possible to select those peaks or valleys within a specific y-axis range of the spectrum. When this checkbox is not checked, the peak find function searches the portion of the spectrum contained within the limits of the current x-axis.

	Placing a check in the checkbox activates the corresponding input field. Only the peaks/valleys within a certain range will be found by entering the upper and lower limit values and clicking the <Apply> button. In addition, checking a single checkbox will display the corresponding upper or lower limit horizontal cursor in the spectrum display field. The upper and lower limit can also be set by dragging the cursors using the square bar handles (■).
<Apply>	Executes peak find using the declared [Peak Find] parameters.
Spectrum display area	By clicking on the short bar that indicates a peak position, the vertical cursor will move to that position and the selected value in the peak list becomes highlighted. This operation is identical to clicking on a value in the peak list. The vertical cursor can also be moved to locations other than the selected top/bottom positions by left-clicking the desired position within the spectrum. The wavelength (wavenumber) and y-axis value for the selected position, as indicated by the x-axis and y-axis values, respectively, are displayed above the spectrum display area. To add a peak or valley, click on the desired peak/valley position, and then click the <Add> button.
<Print...>	Click to open the [Print Preview] window. For details refer to Section 3.2.11 [Print Preview].
<Scales...>	Click to set the spectrum range. Peak find is executed only within the x-axis range set here. Clicking the <Apply> button after changing the scale updates the peak table.
<Add>	Click to add a peak/valley. Click the <Add> button after clicking on the desired position within the spectrum.
<Delete>	Click to delete a peak/valley. Select the peak(s) to be deleted in the peak list or in the spectrum display and then click the <Delete> button.
<Copy>	Copies the peak table to the Clipboard.
<OK>	Accepts the [Peak Find] parameters within the dialog box and displays the calculated Peak labels in the original View after closing the dialog box. The Peak labels can be pre-selected prior to execution of [Peak Find] by choosing a Peak Mark setting from the [View] - [Peak Indicators] submenu (Section 3.4.12).
<Cancel>	Closes the dialog box without applying any of the [Peak Find] parameters.

Note: *Easy detection of peak top (bottom)*

*This is a convenient function that can be used when a peak that is smaller than the noise level is to be added. Dragging the mouse while holding down the **Ctrl** key moves the cursor to the position at which the vertical axis value is the largest (or smallest, in the case of a valley) in the range over which the mouse is dragged. Clicking the <Add> button adds the selected peak (or valley). Dragging the mouse while holding down the **Ctrl** key displays a red line on the spectrum display that indicates the range to be examined for the maximum or minimum value (peak or valley).*

3.5.3.2 [Peak Height...]

Determines the peak height and peak height ratio for a selected peak(s). (Fig. 3.44)

[How to draw the Baseline]

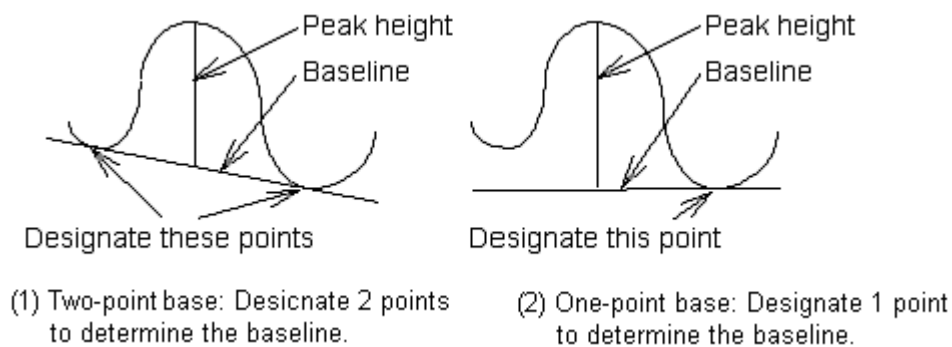


Figure 3.43 Methods for determining peak height

Note 1: It is always best and more accurate to calculate Peak Height by using a spectrum in the Absorbance format.

Note 2: The [Peak Height] function includes variety of parameters to be set. Follow the procedures below to learn about general ideas of [Peak Height]

[Operation Procedures]

These procedures explain how to calculate peak height of a spectrum by using [Two-point base (auto)].

(1) Select [Peak Processing] – [Peak Height] to display [Peak Height] dialog box.

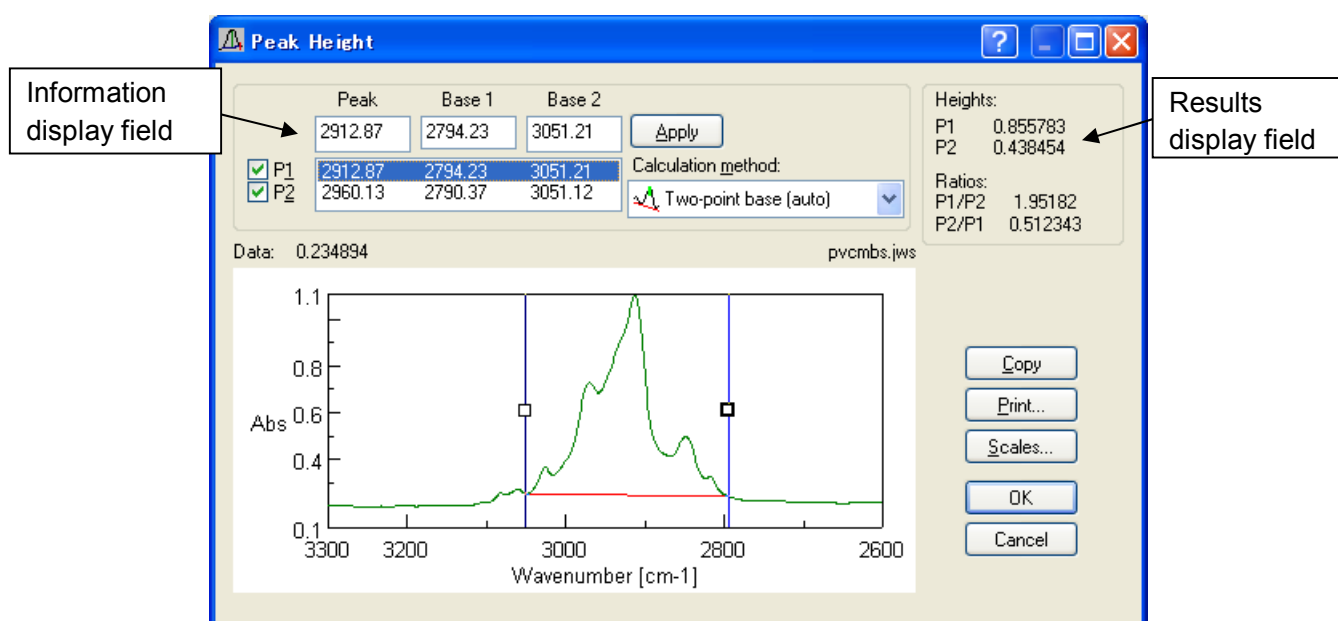


Figure 3.44 [Peak Height] dialog box

(2) Check in the [P1] check box to calculate the peak height.

Note: Check in the [P1] and [P2] check boxes to calculate the peak height ratio.

- (3) Select [Two-point base (auto)] from the [Calculation method] drop-down list.
- (4) Select Base 1 and Base 2 to set base range of horizontal axis by using vertical cursors. Base range of the horizontal axis can also be set by typing the value in the both text boxes and click the <Apply> button. Peak height of the spectrum is calculated and the results are displayed on the results display field.

Note: To calculate peak height ratio, follow the steps (3) and (4) after selecting a peak to be set in the information display field.

- (5) Click the <Print...> button to display [Print Preview] dialog box. Clicking the <Print> button in the dialog box prints the results.

[Peak], [Base 1], [Base 2]	Horizontal axis fields for Peak, Base 1 and Base 2. Enter the values using the numeric keypad and click the <Apply> button. Alternatively, the displayed horizontal cursors can be dragged to the desired positions for the Peak, Base 1 and Base 2 fields. Items to be set vary with [Calculation method].
[P1], [P2] checkboxes	P1 and P2 represent Peak 1 and Peak 2, respectively. Items having a check in the checkbox are ascertained and calculation results are displayed in the results display field.
[Calculation method]	Select how to draw the baseline for peak height calculation and how to set the peak position (manually or automatically). When (Auto) is set for [Two-point base], the maximum absorption position is recognized as a peak within the range specified by Base 1 and Base 2. (Auto) cannot be set in the case of a [One-point base]. Click the drop-down menu for [Calculation method] and select the desired calculation method from the drop-down list. • One-point base • Two-point base (manual) • Two-point base (auto) The method used to draw the baseline is described in Fig. 3.43.
Information display field:	Displays the peak and base positions of peaks ([P1], [P2]) used to calculate the peak height/height ratio. The horizontal range can be changed using the vertical cursors or by entering the desired values into the peak and base fields and clicking the <Apply> button.
Spectrum display area	The peak and base positions can be set by dragging the vertical cursors on the spectrum display. Peak height and peak height ratio are calculated simultaneously upon setting the x-axis values. The y-axis value for the active cursor is displayed in the [Data] field.
Results display field:	Displays the calculation result of peak height/height ratio. The results are calculated and re-displayed each time the peak and base position(s) are changed.
<Apply>	Clicking this button after the values for Peak, Base 1 and Base 2

	have been entered using the keyboard moves the appropriate vertical cursors to the specified position and executes the peak height calculation.
<C <u>o</u> py>	Copies the [Peak Height] calculation results and its information to the Clipboard.
<P <u>r</u> int...>	Opens the [Print Preview] window. Refer to 3.2.11 [Print Preview] for more information.
<S <u>c</u> ales...>	Sets the display range of the spectrum display field.
<O <u>K</u> >	Executes the peak height/peak height ratio calculation using the defined parameters and closes the dialog box. Results are attached to the original spectrum file as metadata and the file must be saved for the results to be retained.
<C <u>a</u> ncel>	Closes the dialog box without applying any of the [Peak Height] parameters.

3.5.3.3 [Peak Area...]

Determines the peak area and peak area ratio for a selected peak(s). (Fig. 3.47)

[How to draw the Baseline]

- (1) [Two-point base]: Determines the total area within the baseline range
- (2) [No base]: Assumes 0 Absorbance or 100%T as the baseline
- (3) [Independent base]: Sets the baseline and peak ranges separately

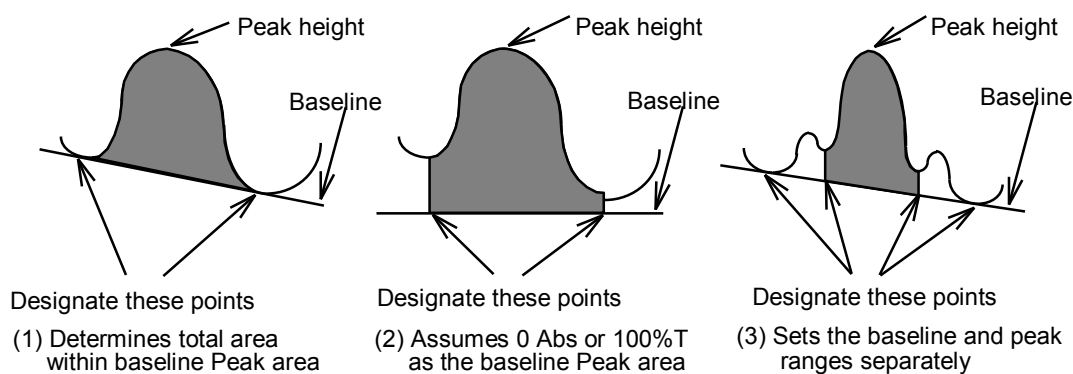


Figure 3.45 Peak Area determination methods

[Calculation property]

- (1) Ignore under baseline
- (2) Subtract under baseline
- (3) Add under baseline

To calculate peak area $aa' + bb'$ in Fig. 3.46 (a), designate a [Two-point base] from Point A to Point B and select [Add under baseline]. To calculate peak area aa' , designate a [Two-point base] from Point A to Point B and select [Ignore under baseline]. To calculate peak area $aa' - bb'$, designate a [Two-point base] from Point A to Point B and select [Subtract under baseline].

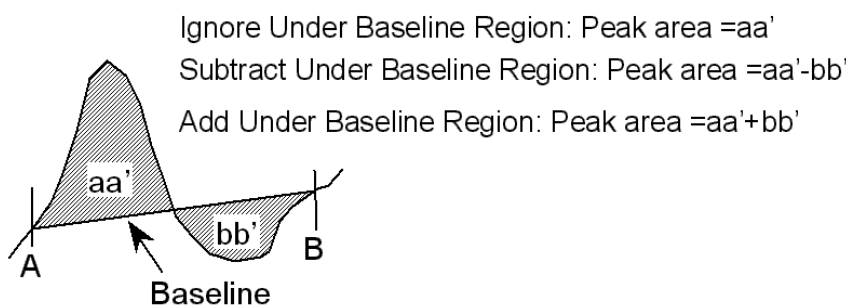


Figure 3.46 (a) Example 1

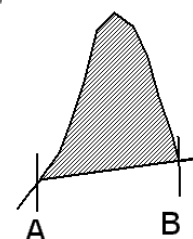


Figure 3.46 (b) Example 2

In Fig. 3.46 (b), when a baseline is designated from point A to B, all three methods give the same result.

Note1: It is always better and more accurate to calculate Peak Area using a spectrum in the Absorbance format.

Note2: The [Peak Area] function includes variety of parameters to be set. Follow the procedures below to learn about general ideas of [Peak Area]

Operation Procedures

These procedures explain how to calculate peak area of a spectrum by using [Independent base (manual)].

- (1) Select [Peak Processing] – [Peak Area] to display [Peak Area] dialog box.

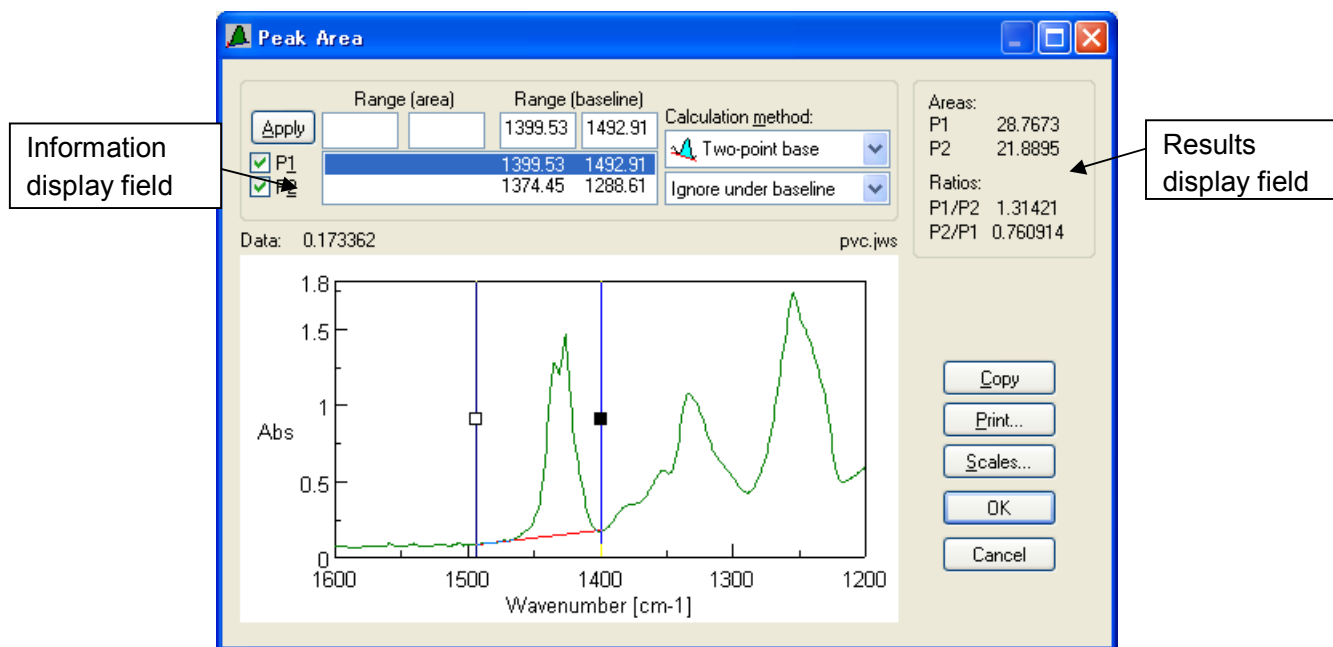


Figure 3.47 [Peak Area] dialog box

- (2) Check in the [P1] check box to calculate the peak area.

Note: Select the [P1] and [P2] check boxes to calculate the peak area ratio.

- (3) Select [Two-point base] and [Ignore under baseline] from the [Calculation method] drop-down list.
- (4) Select baseline range of Peak 1 by using vertical cursors. Baseline range can also be set by typing the value in the both text boxes and click the <Apply> button. Peak area of the spectrum is calculated and the results are displayed on the results display field.

Note: To calculate peak area ratio, follow the steps (3) and (4) after selecting a peak to be set in the information display field.

- (5) Click the <Print...> button to display [Print Preview] dialog box. Clicking the <Print> button in the dialog box prints the results.

[Range (area)], [Range (baseline)]	Fields for setting the peak area calculation range and the peak baseline range. Enter the values using the numeric keypad and click the <Apply> button to set the area and baseline values. Alternatively, drag the vertical cursors to the desired positions for the area and baseline fields. Items to be set vary with [Calculation method].
[P1], [P2] checkboxes	P1 represents peak area 1 and P2 represents peak area 2. Items having a check in the checkbox are ascertained and calculation results are displayed in the results display field.
[Calculation method]	Declares how to draw the baseline for peak area calculation and the method of peak area calculation. Select P1 or P2 and click the drop-down menu for [Calculation method] to select the calculation method from the drop-down list. [Baseline]: • Two-point base • No base • Independent Base [Method]: • Ignore under baseline • Subtract under baseline • Add under baseline The method used to draw the baseline and the calculation properties are described in Figs. 3.45 – 3.46.
Information display field	Displays the area calculation range of peaks ([P1], [P2]) for calculating peak area/area ratio and the horizontal axis value of the baseline calculation range. The horizontal range can be changed using the vertical cursors or by entering the desired values into the base fields and clicking the <Apply> button.
Spectrum field:	The area calculation range and base calculation range can be set by dragging the vertical cursors on the spectrum. Peak area and peak ratio are calculated simultaneously upon setting of the cursors. The y-axis value for the active cursor is displayed in the [Data] field.
Results display field:	Displays the calculation results for peak area/area ratio. The results are calculated and re-displayed each time the area and baseline range are changed.
<Apply>	Clicking this button when the values of the [Range (area)] and [Range (baseline)] are entered by the keyboard moves the vertical cursor to the specified position and executes the area calculation.
<Copy>	Copies the [Peak Area] calculation results and its information to the Clipboard.
<Print...>	Opens the [Print Preview] dialog box. Refer to 3.2.11 [Print Preview] for more information.
<Scales...>	Sets the display range of the spectrum display field.
<OK>	Executes the peak area/peak area ratio calculation using the defined parameters and closes the dialog box. Results are attached to the original spectrum file as metadata and the file must be saved for the results to be retained.
<Cancel>	Closes the dialog box without applying any of the [Peak Area] parameters.

3.5.3.4 [FWHM...]

Calculates the full width at half maximum. (Fig. 3.48)

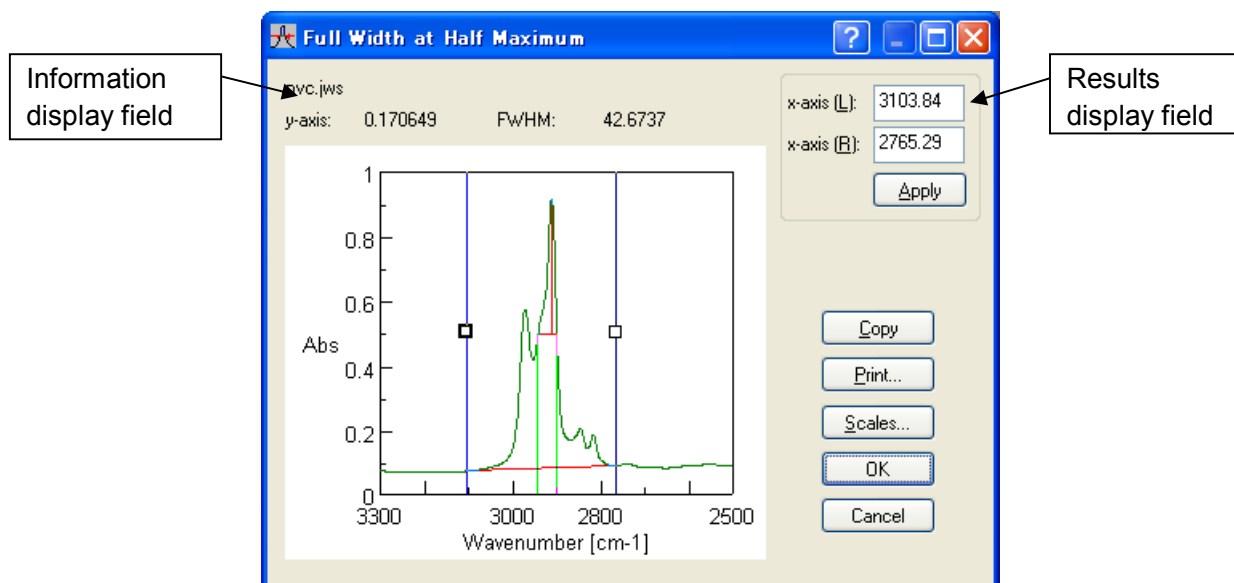


Figure 3.48 [FWHM] dialog box

Spectrum area:	Set the full width at half maximum calculation range by dragging the vertical cursors on the spectrum. The full width at half maximum value is calculated simultaneously. Alternatively, the full width at half maximum calculation range can be entered directly in [x-axis(L)] and [x-axis(R)] using the keyboard and then clicking the <Apply> button.
[x-axis(L)]	Enter the lower x-axis value for the full width at half maximum calculation range.
[x-axis(R)]	Enter the upper x-axis value for the full width at half maximum calculation range.
[y-axis]	Displays the y-axis value intercepting the bar of the active vertical cursor.
[FWHM]	Displays the result of the full width at half maximum calculation.
<Apply>	Clicking this button when the values for [x-axis(L)] and [x-axis(R)] are entered by the keyboard moves the vertical cursors to the specified position and executes the full width at half maximum calculation.
<Copy>	Copies the [FWHM] calculation results and its information to the clipboard.
<Print...>	Click to open the [Print Preview] window. For details refer to 3.2.11 [Print Preview].
<Scales...>	Selects the range of the spectrum to be displayed within the dialog box.
<OK>	Executes the full width at half maximum calculation using the defined parameters and closes the dialog box. Results are attached to the original spectrum file as metadata and the file must be saved for the results to be retained.
<Cancel>	Closes the dialog box without applying any of the [FWHM] parameters.

3.5.4 [Subtraction...]

Determines a difference spectrum from two spectra. (Fig.3.49)

One method of identifying an unknown sample is to compare the spectrum of the sample with other known spectra. To identify the components of an unknown mixture, the spectrum of the known component(s) can be removed, leaving only the spectrum containing the unknown component(s) of the mixture. Thus, if the spectrum of the mixture is assumed to be A, and the spectrum of a known component is assumed to be B, the difference spectrum can be determined using the formula $((A \times \text{coefficient}) - B)$ or the formula $(A - (B \times \text{coefficient}))$. The coefficient is then selected to cancel the intensity of a common absorption peak (key band) present in the two spectra.

The coefficient can be applied using one of two methods. The first method involves continuously changing the coefficient to remove the key band by trial and error while viewing the spectrum. The second method involves calculating a coefficient from an observed peak height or area in order to remove the key band.

The number of spectra overlaid in the View determines which dialog box opens first when [Subtraction] is executed.

- When two spectra are displayed, the [Subtraction] dialog box appears. (Fig. 3.49)
- When three or more spectra are displayed, select the unknown spectrum from which you wish to subtract the known component(s). When [Subtraction] is selected, the [Select Spectrum] dialog box is displayed. (Fig. 3.37) Select the spectrum that contains the component(s) to be subtracted from the unknown.

Note: *[Subtraction] is not valid when only one spectrum is displayed in the View. Execute [Subtraction] after overlaying at least one additional spectrum containing a known component of the mixture.*

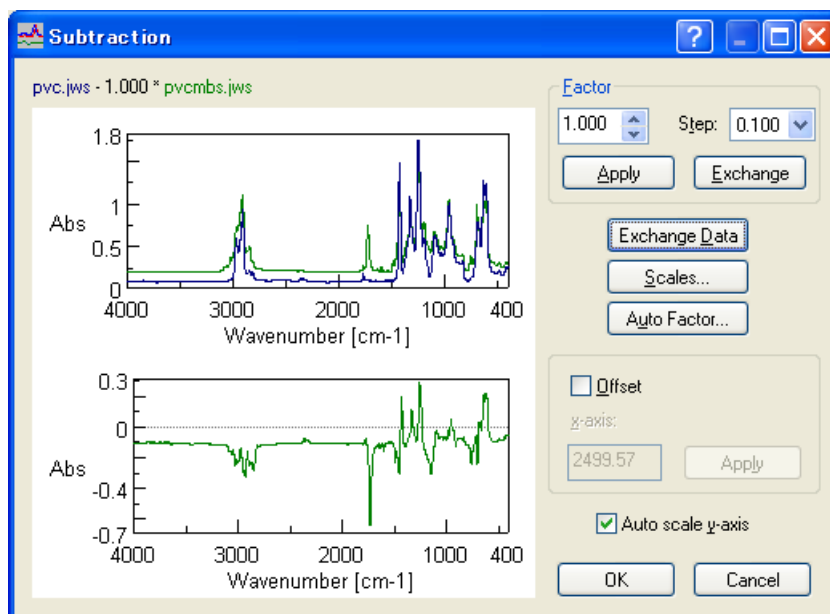


Figure 3.49 [Subtraction] dialog box

The calculated result is displayed in the spectrum field at the bottom.

[Factor]	Used to input a multiplication (coefficient) factor. The factor can also be changed using the arrow keys   . Each click on an arrow key   increases or decreases the factor by the amount designated in [Step]. The difference spectrum is recalculated each time the factor is changed.
[Step]	Designates the increment or decrement by which [Factor] increases or decreases each time an arrow key   is clicked.
<Exchange>	Exchanges the spectrum to be multiplied by the factor. In the above figure, (Fig. 3.49) clicking the <Exchange> button will change the calculation from $((\text{Factor} \times \text{pvc.jws}) - \text{pvcmb.jws})$ to $(\text{pvc.jws} - (\text{Factor} \times \text{pvcmb.jws}))$.
<Exchange Data>	Click to exchange the spectra. In Fig. 3.46, clicking the <Exchange Data> button will change the calculation from $((\text{Factor} \times \text{pvc.jws}) - \text{pvcmb.jws})$ to $((\text{Factor} \times \text{pvcmb.jws}) - \text{pvc.jws})$.
<Scales...>	Click to set the spectrum range displayed in the spectrum fields.
<Apply>	Clicking this button when the value for the [Factor] has been entered by the keyboard executes the subtraction calculation and shows the subtraction result in the lower spectrum display field. <i>Note: When [Offset] is selected, the <Apply> button in this area will move the vertical cursor in the upper spectrum display field to the position entered by the keyboard and will shift the spectra in the upper and lower spectrum display fields to correspond to the [Offset] entered in the [Offset] field.</i>
<Auto Factor>	Click to display the [Subtraction – Auto Factor] dialog box used to calculate the factors that cancel a particular peak. These factors can be calculated using a peak height or peak area. (The contents of the [Subtraction – Auto Factor] dialog box are described below.(Fig. 3.50)
<Offset>	Toggling offset calculates the overall spectrum, where the difference at an arbitrary x-axis value for two spectra is given as zero. Move the vertical cursor to any position or enter the x-axis value using the keyboard and click the <Apply> button.
[Auto scale y-axis]	Select the checkbox to view the results using the auto scale function. Clear the checkbox to view the change in the size of the resulting difference spectrum.
<OK>	Applies the [Subtraction] parameters and displays the subtraction result in a new View after closing the dialog box.
<Cancel>	Closes the dialog box without applying any setting of the [Subtraction] parameters.

3.5.4.1 Automatic calculation of factor

Calculates a factor that cancels a peak of interest by subtraction. The factor can be calculated by either the peak height or peak area.

The factor of the subtraction equation displayed at the upper left of the spectrum changes in real time.

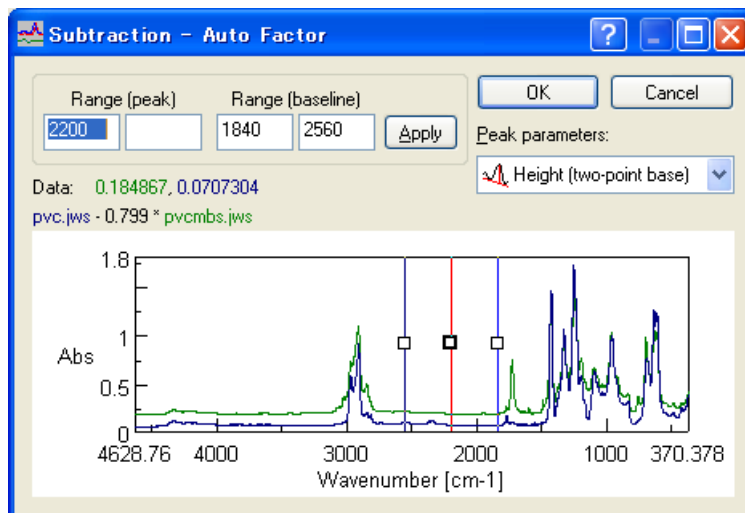


Figure 3.50 [Subtraction – Auto Factor] dialog box

[Peak Parameters]	Sets the method of calculating the subtraction factor (either peak height or peak area) and how the baseline is defined. Height (two-point base) Height (one-point base) Area (two-point base) Area (one-point base) For more information about how to define the baseline and the calculation range, refer to 3.5.3.2 [Peak Height] and 3.5.3.3 [Peak Area].
[Range (peak)] [Range (baseline)]	Items to set differ depending on the calculation method selected and how the baseline is to be defined. Select a specific x-axis position using the vertical cursor or enter the horizontal axis value with the keyboard and click the <Apply> button.
<Apply>	Clicking this button when the values of the [Range (peak)] and [Range (baseline)] are entered by the keyboard moves the vertical cursor(s) to the specified position and executes the subtraction calculation.
[Data]	Displays the current x-axis value for the active vertical cursor.
<OK>	Applies the entered parameters and displays the result within the [Subtraction] dialog box after closing the [Subtraction – Auto Factor] dialog box.
<Cancel>	Closes the dialog box without applying any setting of the [Auto Factor] parameters.

3.5.5 [X Unit Conversion...]

Converts the X-unit units of a spectrum and open new view for the converted spectrum. (Fig. 3.51)

Note : The units for the x-axis can be changed by executing [View]-[Horizontal Axis Units...], but this only changes the apparent unit of the View window while the unit of the data in memory remains the same.

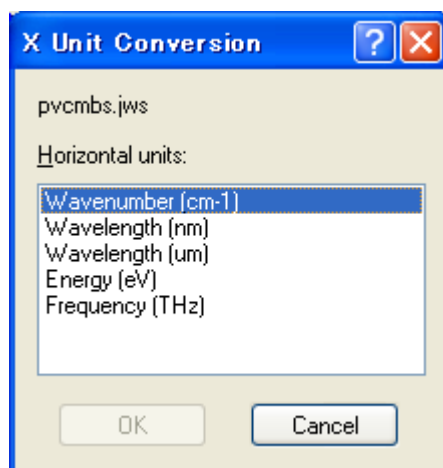


Figure 3.51 [X-Unit Conversion] dialog box

3.5.6 [Y Unit Conversion...]

Converts the y-axis units of the spectrum. (Fig. 3.52)

This function is applicable for spectra for which the vertical axis is absorbance (Abs), %T or %R.

Note: When a spectrum is selected for which the vertical axis is intensity, a [Data mode error] message will appear.

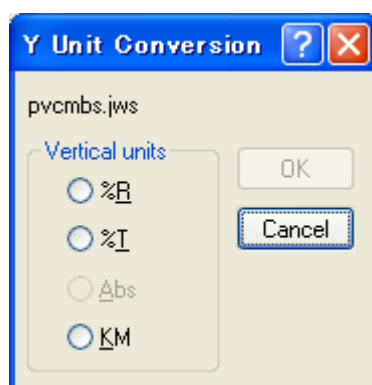


Figure 3.52 [Y Unit Conversion] dialog box

3.5.7 [Common Options]

The contents of [Common Options] vary depending on the instrument.

3.5.7.1 [Curve Fitting...]

Selecting [Processing] – [Common Option] – [Curve Fitting] runs the [Curve Fitting] program. The active spectrum is automatically transferred to the program. Refer to the [Curve Fitting] program manual for details.

3.5.7.2 [Data Dump...]

Displays data values of a spectrum. The values can be copied to the clipboard.

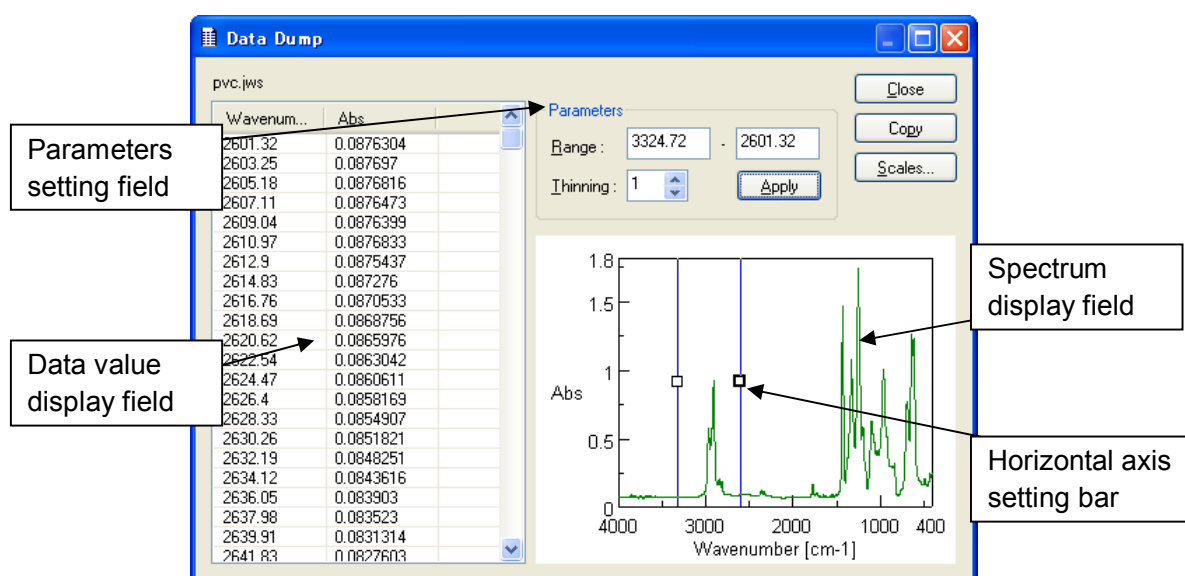


Figure 3.53 [Data Dump] dialog box

[Range]	Enter the values of range using numeric keyboard to get data value and click the <Apply> button. Range can also be input by moving horizontal axis setting bars.
[Thinning]	Designates the interval between data points to be used for data value display.
<Apply>	Clicking this button when the values of the [Range] are entered by the keyboard moves the horizontal axis setting bars to the specified position and executes the data dump processing.
<Close>	Closes the dialog box.
<Copy>	Copies the data values to the Clipboard.
<Scales...>	Click to set the spectrum range displayed in the spectrum display field.

3.5.7.3 [Data Accumulation...]

Obtains the averaged spectrum of spectra overlaid in the same view. This function cannot be performed on spectra that have different data modes, data intervals, or measurement ranges.

3.5.7.4 [Arithmetic between Data...]

Performs an arithmetic operation between spectra overlaid in the same view. This function cannot be performed on spectra that have different data modes or data intervals. An arithmetic operation is performed only on the overlapping ranges of the spectra. When only one spectrum is displayed in the active View the arithmetic operation is performed using a constant(s).

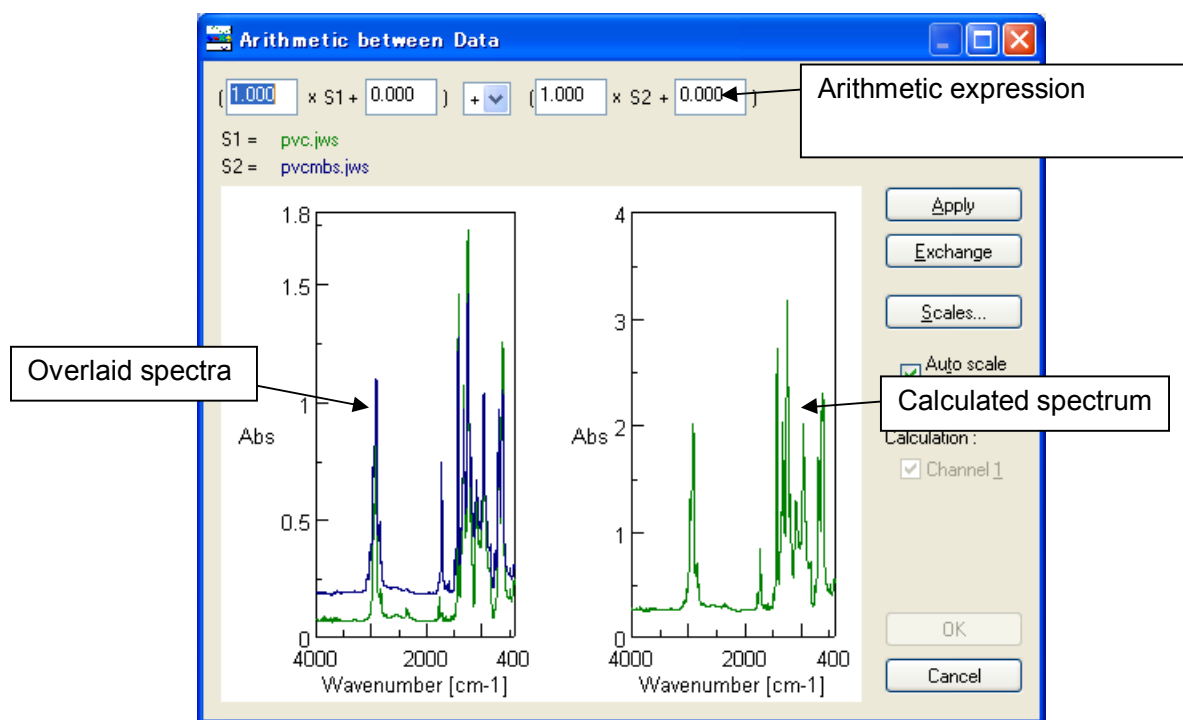


Figure 3.54 [Arithmetic between Data] dialog box

The operation to be performed is shown at the top of the dialog box (Fig. 3.54). S1 and S2 represent the selected spectra and are defined immediately below the Arithmetic expression to be calculated. Enter constants into the appropriate text boxes and select the desired arithmetic operator from the drop-down list to generate the arithmetic operation.

Spectrum display field	Displays overlaid raw spectra on the left side and calculated spectra on the right side.
<Apply>	Executes the calculation using the designated parameters.
<Exchange>	Reverses the order of spectra S1 and S2.
<Scales...>	Sets the spectrum display range.
[Auto scale y-axis]	Select the checkbox to view the results using the auto scale function. Clear to view the change in the size of the calculated spectrum.
<OK>	Applies the settings for the Arithmetic expression and displays the calculated spectrum in a new View after closing the [Arithmetic between Data] dialog box.
<Cancel>	Closes the dialog box without applying any setting of the [Arithmetic between Data] parameters.

Note 1: To perform an arithmetic operation between one spectrum and a constant in a View in which multiple spectra are displayed, open the dialog box via the same procedure as when performing an arithmetic operation between data. Input 0 for the factor of the second spectrum and a constant not related to the desired operation.

Note 2: Following dialog box is displayed when performing this function on a spectra. In the dialog box, an arithmetic operation between one spectrum and a constant can be performed.

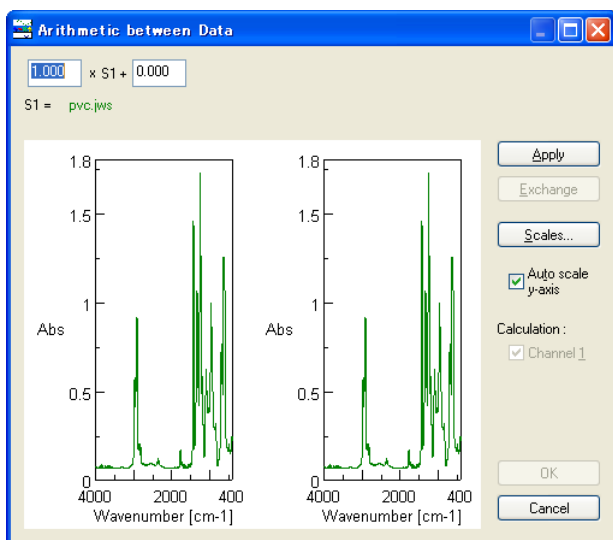


Figure 3.55 [Arithmetic between Data] dialog box

3.5.7.5 [Channel Conversion...]

Converts channels of data in a View and picks out data having a desired channel from overlaid spectra or multiple channels in a View. This function is usable for the following cases: performing data processing on data with channel Nos. 2 to 4. Such kind of processing requires converting channel Nos. 2 to 4 to channel No. 1 or selecting desired channel data, and then opening it in a new view.

Note1: Data processing is performed on channel No. 1 data.

Note2: Channels are not included in measurement data of FT-IR.

Note3: The data processing result is displayed with the specified spectra arranged lengthwise in one column.

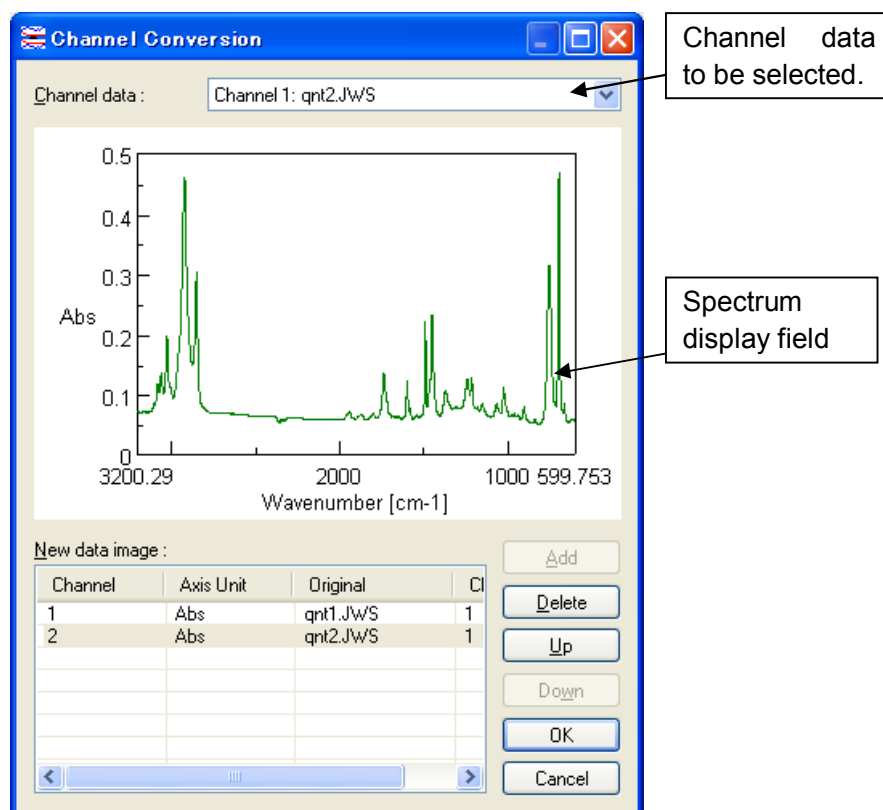


Figure 3.56 [Channel Conversion] dialog box

[Channel data]	Select a channel data from the drop-down list. The data is displayed on the spectrum display field. Click the <Add> button to register the data to [New data image]. Data is numbered according to the order it is added. Repeat this procedure to convert all channels. Then, select the required channel to display the desired data.
[New data image]	Registered data are displayed in the list. The maximum number of data that can be added is 4.
<Add>	Adds the selected data in the [Channel data] drop-down list to the [New data image] list.
<Delete>	Deletes the selected data in the [New data image] list.
<Up>	Moves the selected spectrum to a lower channel number.
<Down>	Moves the selected spectrum to a higher channel number.
<OK>	Displays registered data in a new View after closing the [Channel Conversion] dialog box..
<Cancel>	Closes the dialog box without applying any setting of the [Channel Conversion] parameters.

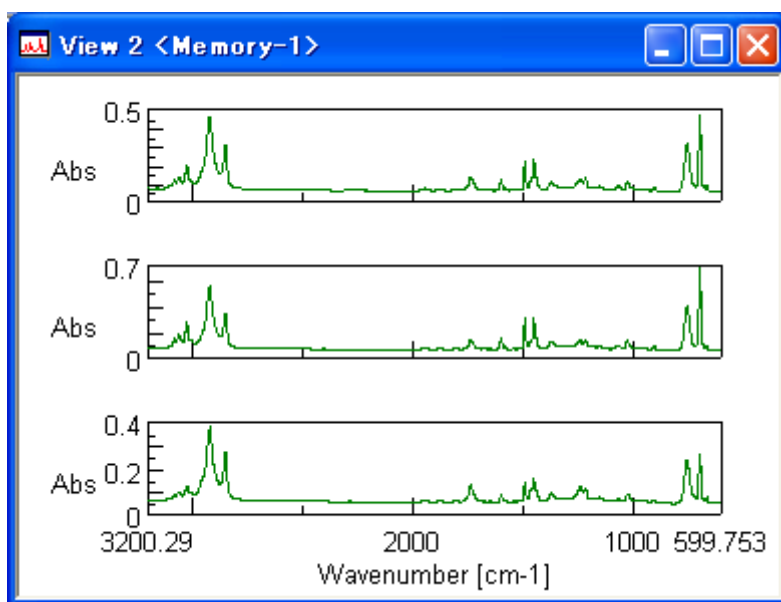


Figure 3.57 [View after Channel Conversion] dialog box

3.5.7.6 [Arithmetic between Channels...]

Performs an arithmetic operation between channels of data displayed in the same view. The [Arithmetic between data] function performs an arithmetic operation between overlaid spectra. This function performs arithmetic operation by selecting two data sets from multiple channel data displayed in the same View. The attributes of the vertical axis can also be selected. This function is performed on the horizontal axis range covered by both data sets.

Note: Channels are not included in measurement data of FT-IR and channels are not displayed for FT-IR data. Also, the operation is the same as the [Arithmetic] function.

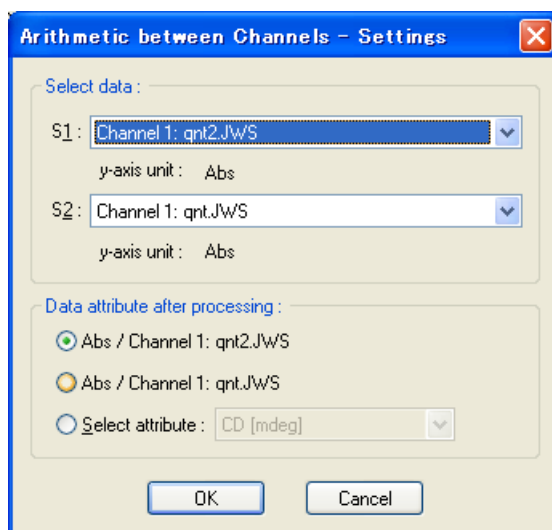


Figure 3.58 [Arithmetic between Channels - Settings] dialog box

[Select data]	Channels and file names are displayed in the [S1] and [S2] drop-down lists. Click on the  button to display file names of data overlaid in the same View. Select a data set and channel.
[Data attribute after processing]	Selects an attribute after processing. When the data attribute changes, select the [Select attribute] check box and select an attribute from its drop-down box. For example, attribute after dividing a single beam by another single beam, it becomes %T.

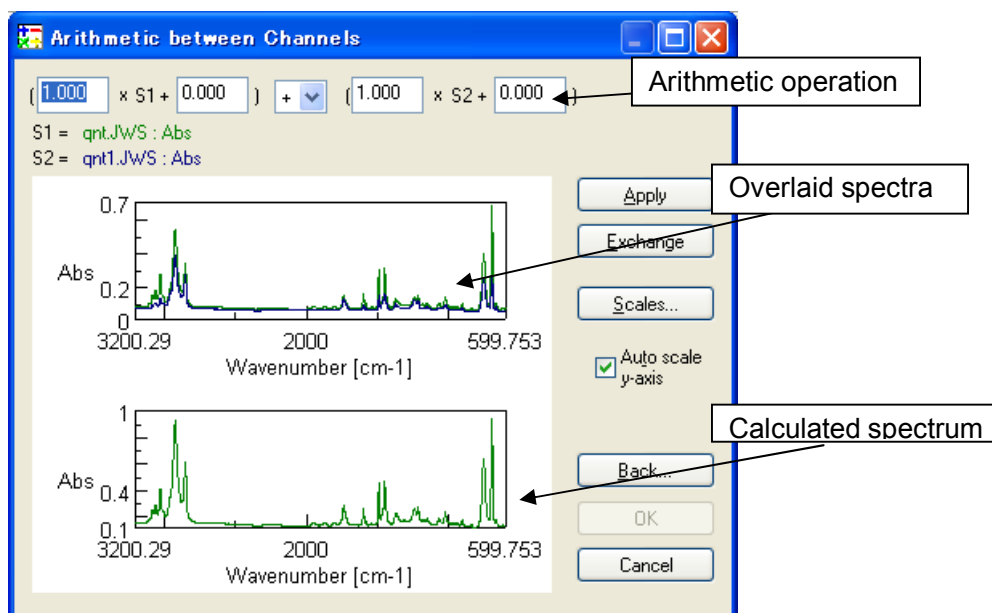


Figure 3.59 [Arithmetic between Channels] dialog box

[Spectrum display field]	S1 and S2 represent the selected spectra and are defined immediately below the Arithmetic operation. Enter constants into the appropriate text boxes and select the desired arithmetic operator from the drop-down list to generate an arithmetic expression. Refer to 3.5.7.4 "[Arithmetic between Data]" for details. Click the <Back> button to return to the dialog box shown in Fig. 3.59
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3.5.7.7 [X Unit Conversion – Photon...]

Converts the horizontal unit of a spectrum to the selected unit and displays the converted spectrum in a new View. In [X Unit conversion - Photon], the vertical axis value is divided by the conversion factor for the horizontal axis unit to maintain the peak area after conversion.

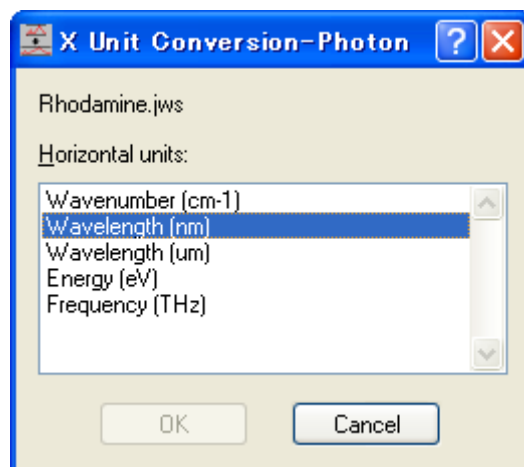


Figure 3.60 [X Unit Conversion - Photon] dialog box

<OK> Converts the horizontal axis of a spectrum and displays the converted spectrum in a new View.

Note: *This function cannot be performed on the converted data by using [X Unit Conversion].*

3.5.7.8 [Time Axis Offset...]

Sets the specified time of the time-course data to zero.

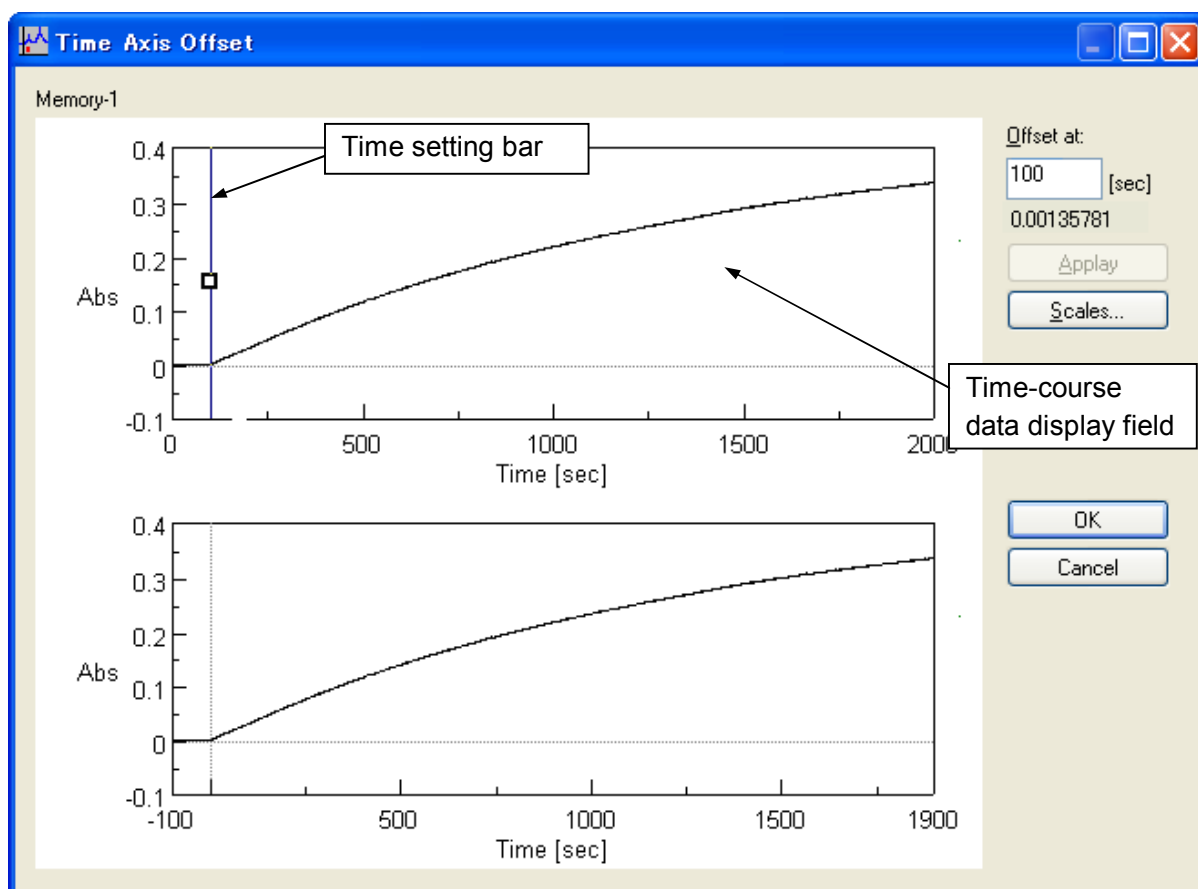


Figure 3.61 [Time Axis Offset] dialog box

Time-course data
display field

Displays the original and offset time-course data in the upper and lower graphs, respectively. When the dialog box is first opened, the data displayed in the lower graph is not offset. Drag the time setting bar in the upper graph to select the offset position.

[Offset at]
<Apply>

Specifies the time to offset the time-course data in the upper graph. Offsets the data with the time specified in the [Offset at] text box and displays the result in the lower graph.

<Scales...>

Specifies the display scale of the time-course data displayed in the upper graph.

3.5.7.9 [Peak Normalization...]

Normalizes the intensity of the specified peak position to an arbitrary intensity. [Peak Normalization] can also be used for overlaid spectra. Select [One-point base] for [Baseline] to normalize the intensity difference at peak and base wavelengths.

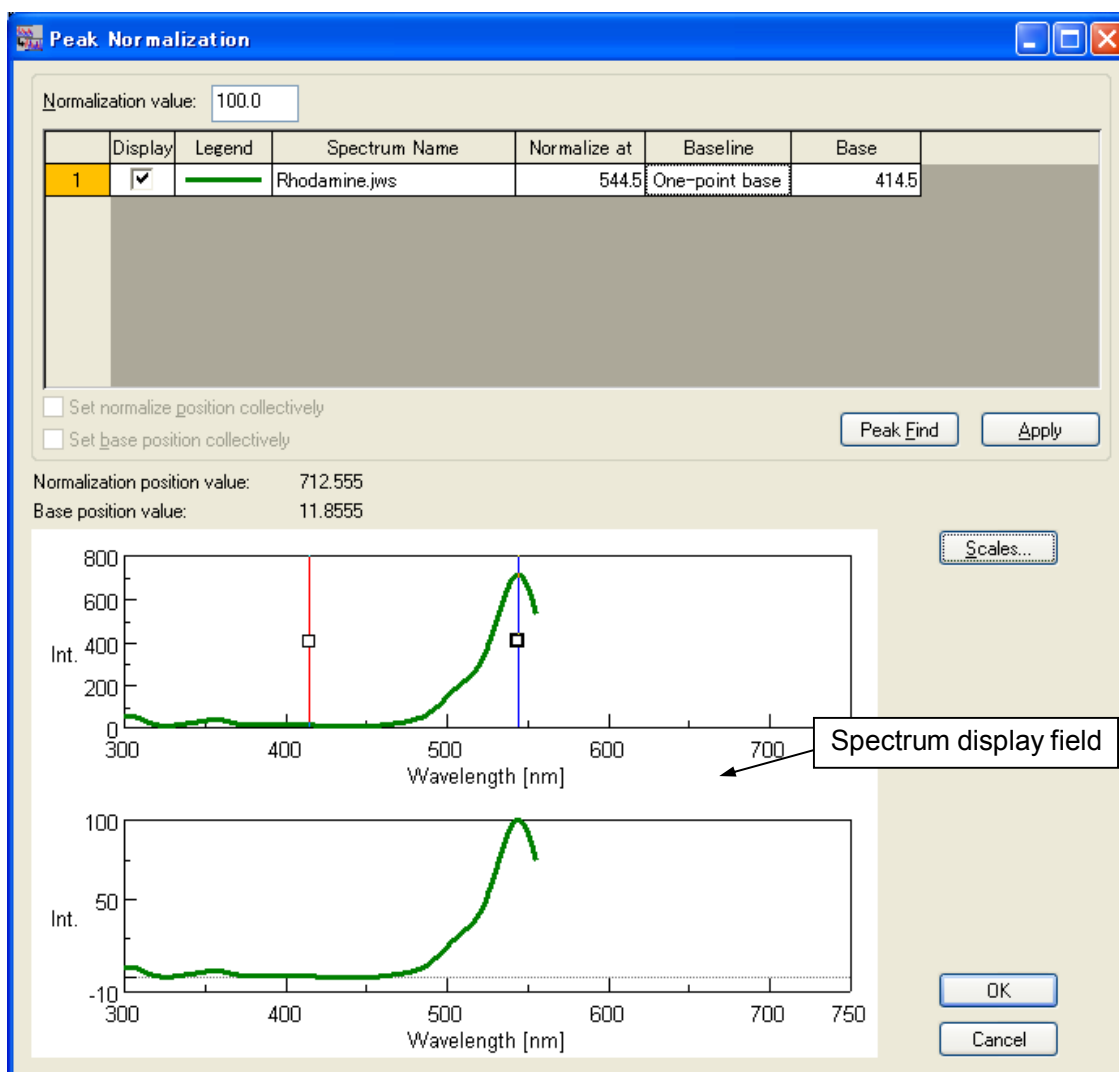


Figure 3.62 [Peak Normalization] dialog box

Spectrum display field	The upper and lower graphs show the original and normalized spectra, respectively.
[<u>N</u> ormalization value]	Specifies the intensity after normalization.
[D <u>i</u> splay]	Displays the spectrum in the spectrum display field.
[L <u>e</u> gend], [S <u>p</u> ectrum Name]	Displays the line style and name of the spectrum.
[N <u>o</u> rmalize at]	Specifies the wavelength for normalizing the spectrum.
[B <u>a</u> seline]	Selects the correction method for baseline. Double-click the text box to display a drop-down list with [No base] and [One-point base] options. Select [One-point base] to normalize the intensity difference at peak and base wavelengths.
[B <u>a</u> se]	Specifies the base wavelength. [Base] can be specified only when [One-point base] is selected for [Baseline].
[S <u>e</u> t normalize position collectively]	Applies the normalization wavelength for a spectrum to the other spectra. [Set normalize position collectively] can only be selected when two or more spectra are opened in [Peak normalization].
[S <u>e</u> t <u>b</u> ase position collectively]	Applies the base wavelength for a spectrum to the other spectra. [Set <u>b</u> ase position collectively] can only be selected when two or more spectra are opened in [Peak normalization].
<P <u>e</u> ak F <u>i</u> nd>	Locates the wavelength of the spectrum with the maximum intensity and specifies this wavelength as the normalization position.
<A <u>p</u> ply>	Normalizes the data with the specified parameters.
<S <u>c</u> ales...>	Specifies the display scale of the spectrum displayed in the upper graph.

3.5.7.10 [Run KnowItAll]

Transfers an active spectrum to the search software *KnowItAll*. The *KnowItAll* is the spectrum search program of *Bio-Rad Laboratories, Inc.*, and is a standard program of FT-IRseries. Refer to the *KnowItAll* manual for details.

Note 1: This function is not available to [Spectra Manager CFR] users.

Note 2 Installation of KnowItAll software on the PC is required.

Note 3: If Search Master 4.0.6.5 is installed on the PC, the search Master link program is required.

3.5.8 IR Options

Analysis functions for data collected by an FT-IR are displayed on the [IR Options] menu. Some instruments have standard menu items that are registered when the infrared spectrometer driver is installed.

3.5.8.1 [ATR Correction...]

Two types of ATR correction method are prepared. Select the ATR correction method from [Method].

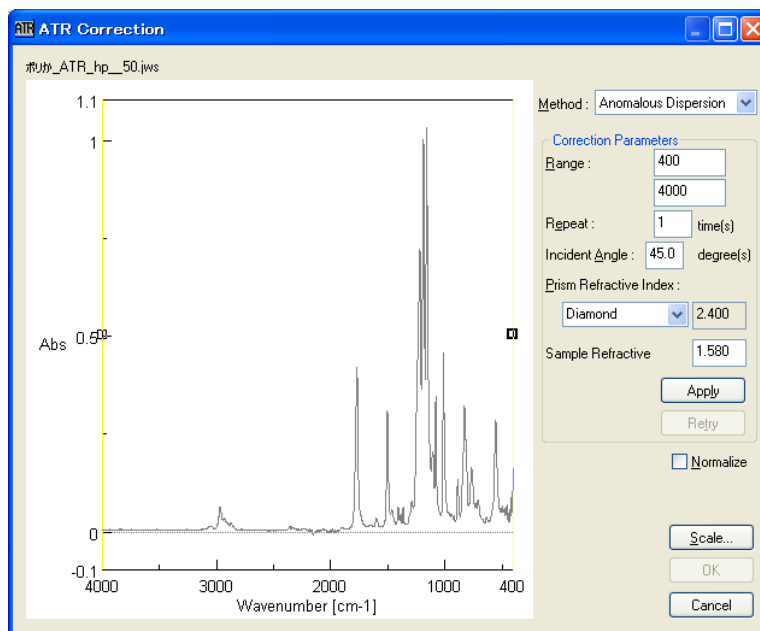


Figure 3.63 [ATR Correction] dialog box

[Method]: Penetration Depth

Penetration Depth Method corrects the relative intensity ratios for high wavenumber and low wavenumber data such that the ATR data more closely corresponds to spectra data collected using the transmission method. It is not necessary for Penetration Depth correction to input the correction parameter.

[Method]: Anomalous Dispersion

The peaks of ATR spectrum will shift due to Anomalous Dispersion compared with the peaks of transmission spectrum. Anomalous Dispersion correction makes this effect reduced. Please input wavenumber range, measurement parameter, and sample refractive index into the correction parameter.

[Correction parameter]

[Range] Input the correction range of spectrum.

[Repeat] Input the number of repetitions of correction.

[Incident Angle] Input the incident angle of ATR.

[Prism refractive Index] Select the type of prism used for measurement. [Diamond], [ZnSe], [Ge], [KRS5], [Si], [ZnS] are registered by default. If you use other ATR prism, you require register of that newly. The way of register is to input

	the prism name and prism refractive index. In addition, the way of delete is to erase prism name and prism refractive index.
[Sample refractive]	Input the sample refractive index. The refractive index of the general organic matter is approximately 1.5.
<Apply>	The correction is started.
<Retry>	After the correction, you can reset the correction parameter.
[Normalize]	The view of spectrum before and after correction is normalized.
<Scale>	Modifies the scale of the spectral data.

Note1: When you check [Normalize], you cannot change scale of spectrum.
Note2: The baseline of ATR spectrum had better be Abs >0.
Note3: The setting of register and delete of prism refractive index is available only for user who has the privileges of Manager or Analysts.

3.5.8.2 [Spectrum to Interferogram...]

This option converts a single-beam spectrum (Single) into an interferogram (IF). (Fig. 3.64) There are no parameters to be set for this dialog box.

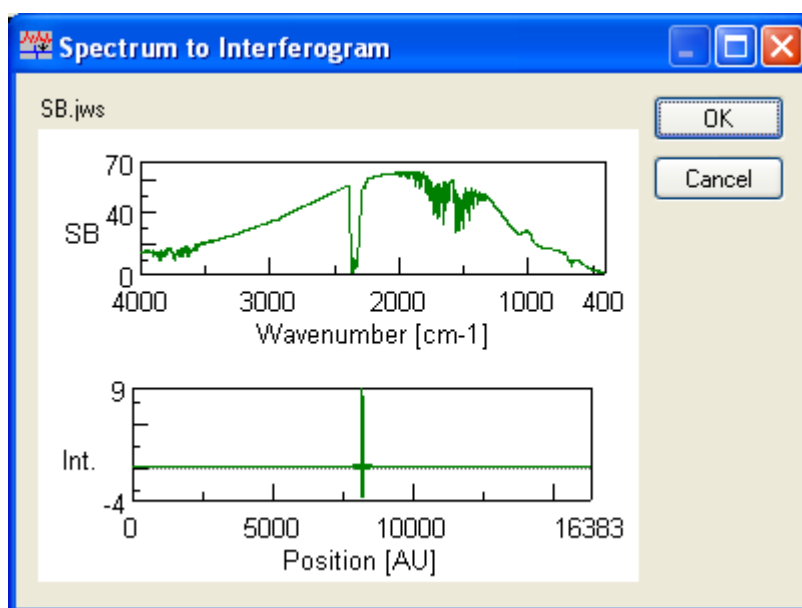


Figure 3.64 [Spectrum to Interferogram] dialog box

<OK>	Applies the reverse transform and closes the dialog box. The interferogram will be displayed in a new spectrum View.
<Cancel>	Closes the dialog box without performing the reverse FFT transform on the single-beam spectrum.

3.5.8.3 [Auto Baseline]

Automatically corrects a spectrum baseline. Entering a fitting range or setting the range with the vertical cursors and clicking the <Apply> button executes the baseline correction.

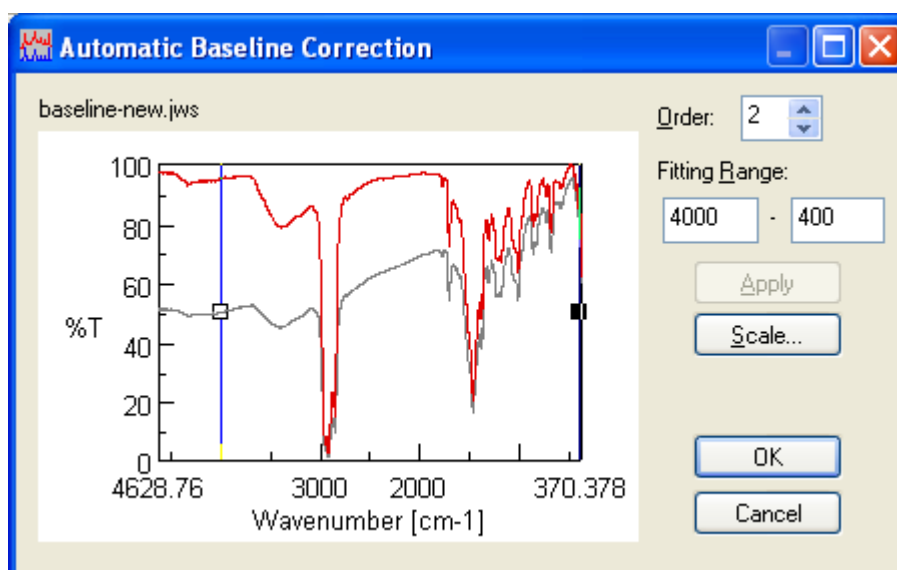


Figure 3.65 [Automatic Baseline Correction] dialog box

[Order]	Sets the order of the baseline correction. Setting an order and clicking the <Apply> button performs the correction and re-displays the corrected spectrum.
[Fitting Range]	Sets a correcting range. Setting a range and clicking the <Apply> button performs the correction and re-displays the corrected spectrum.
<Scale...>	Modifies the scale of the spectral data.
<Apply>	Performs the baseline correction according to the set conditions and displays the corrected spectrum.
<OK>	Applies the corrected spectrum and displays the corrected spectrum in a new spectrum View after closing the dialog box.
<Cancel>	Closes the dialog box without applying the corrected spectrum.

3.5.8.4 [Interferogram to Spectrum...]

This option converts an interferogram into a single-beam spectrum (Single) (Fig. 3.66). Eleven types of apodization functions are available for the FFT transformation of the interferogram.

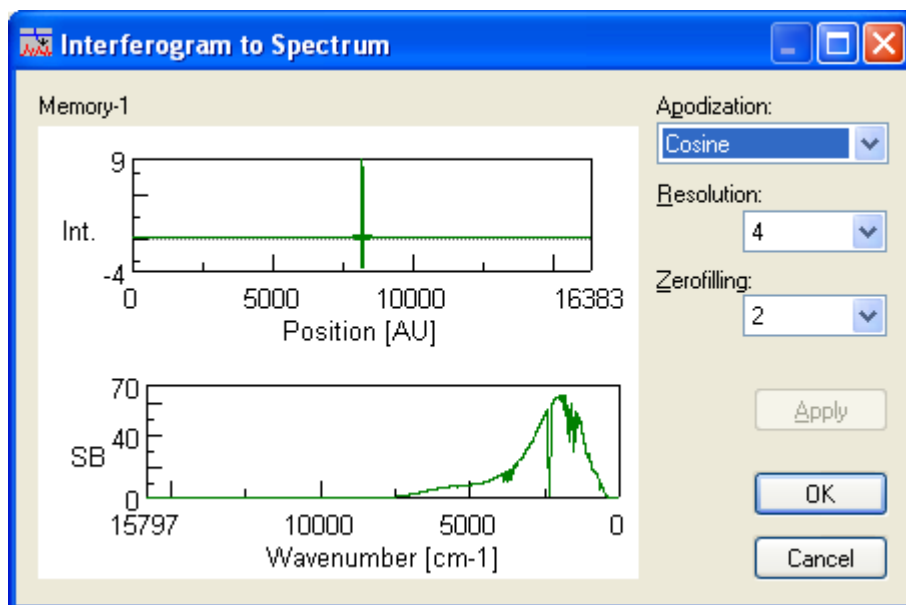


Figure 3.66 [Interferogram to Spectrum] dialog box

Spectrum display area	Displays the original spectrum in the upper spectrum display field, and the transformed single-beam spectrum in the lower spectrum display field.
[Apodization]	Sets the desired apodization function for the FFT transform.
[Resolution]	Sets the transformed spectrum resolution. Note that a resolution lower than that of the interferogram cannot be selected.
[Zerofilling]	Sets the amount of zero filling to be done.
<Apply>	Click this button to perform the calculation using the set parameters and display the transformed spectrum.
<OK>	Applies the dialog box parameters and closes the dialog box. The transformed spectrum will be displayed in a new spectrum View.
<Cancel>	Closes the dialog box without performing the FFT transform on the interferogram.

3.5.8.5 [H₂O Reduction]

Water vapor in the air may affect the spectrum being measured by obscuring some of the analyte peaks. This function eliminates the water vapor absorption in the infrared spectrum. (Fig.3.67)

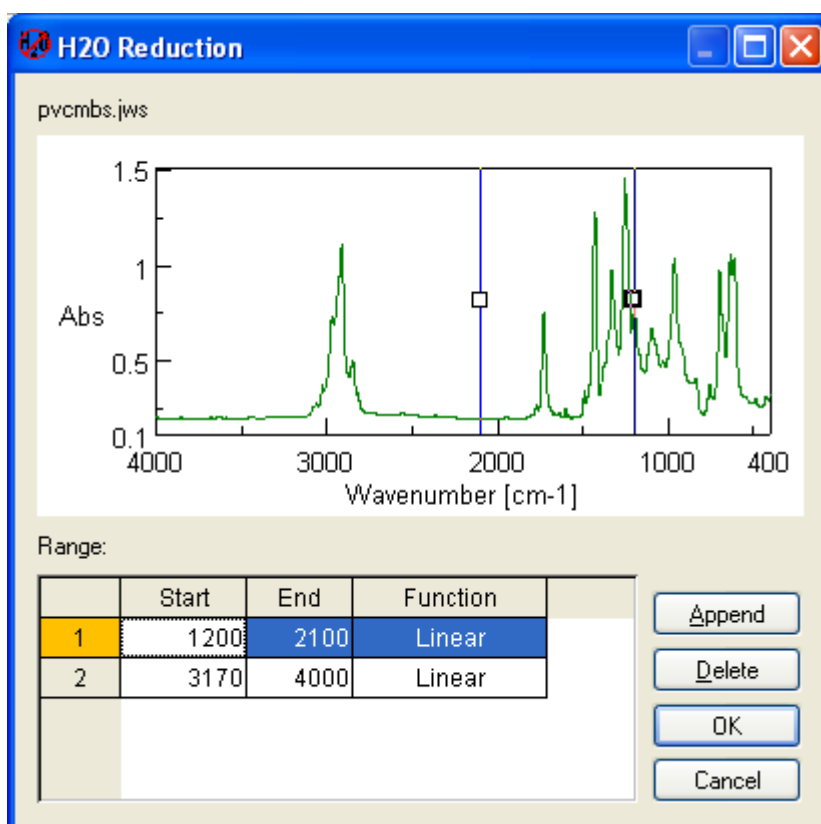


Figure 3.67 [H₂O Reduction] dialog box

[Start] [End]	Sets the range for conducting water vapor reduction. After entering a range in the text box, click the <Append> button. The range is registered in the display field at the bottom. Up to 16 spectral ranges can be set.
<Append>	Registers a new water vapor reduction range.
<Delete>	Deletes the currently selected range from the display field.
[Function]	Selects a calculation function from Constant (0-th order), Linear (primary) and Quadratic (secondary) functions.
<OK>	Applies the water vapor reduction parameters and closes the dialog box. The corrected spectrum is displayed in a new View.
<Cancel>	Closes the dialog box without applying the parameters or performing the water vapor reduction.

3.5.8.6 [CO₂ Reduction]

Carbon dioxide in the air may affect the spectrum being measured by obscuring some of the analyte peaks. This function eliminates the CO₂ absorption in the spectrum. (Fig. 3.68)

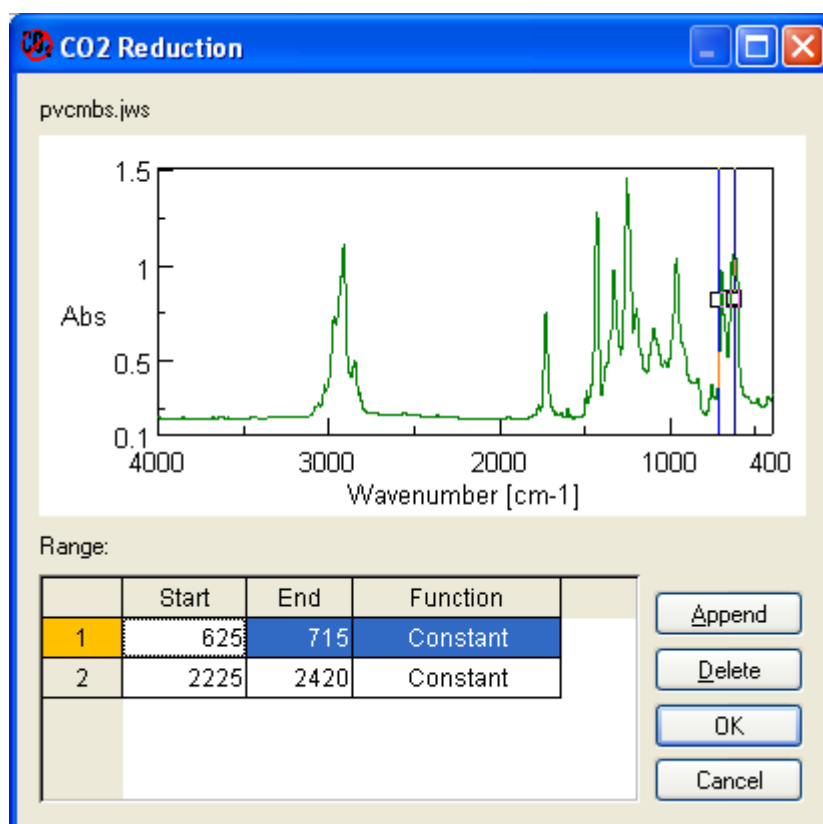


Figure 3.68 [CO₂ Reduction] dialog box

[Start] [End]	Sets the range for conducting CO ₂ reduction. Enter a spectral range for the reduction in the text box, and then click the <Append> button. The range is registered in the display field at the bottom. Up to 16 spectral ranges can be set.
<Append>	Registers a new CO ₂ reduction range.
<Delete>	Deletes the currently selected range.
[Function]	Selects a calculation function from Constant (0-th order), Linear (primary) and Quadratic (secondary) functions.
<OK>	Applies the CO ₂ reduction parameters and closes the dialog box. The corrected spectrum is displayed in a new View.
<Cancel>	Closes the dialog box without applying the parameters or performing the CO ₂ reduction.

3.5.8.7 [Data Comparison]

"Data Comparison" compares measured spectrum with reference spectrum to distinguish spectrum.

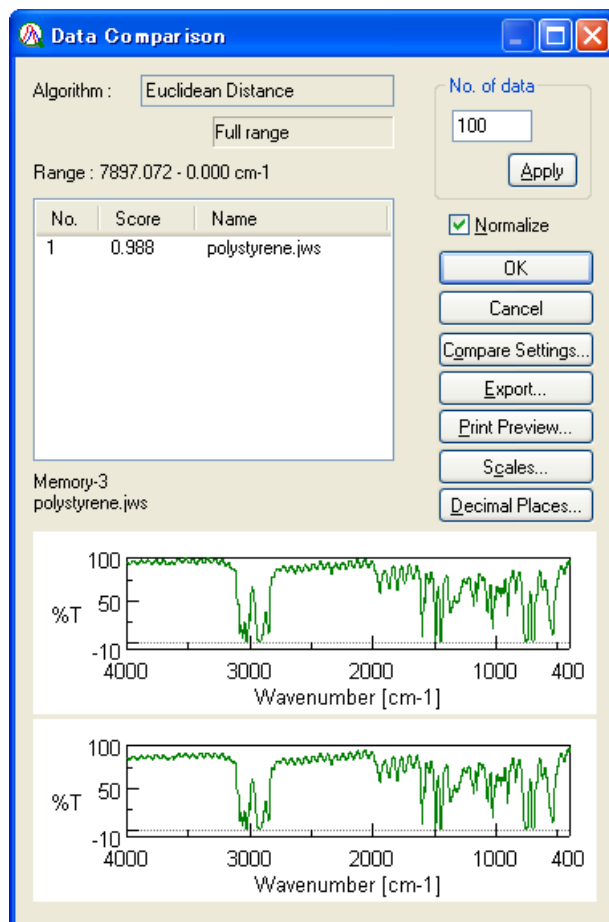


Figure 3.69 [Data Comparison] dialog box

[No. of data]

Input the number of representation of the comparison result.

[Normalize]

The score is normalized from 0-1 range. The height of score means agreement degree.

<Compare Settings...>

Click the <Compare Settings...> button, and [Data Comparison Settings] dialog box opens (Fig. 3.70).

<Export...>

The comparison results are saved as text format.

(1) The file name of spectrum comparing

(2) Compare Settings (Algorithm, Calculation range, Threshold)

(3) Comparison Results (No., Score, Judgment, File name)

<Print Preview...>

A comparison spectrum and the original spectrum you chose are printed.

<Scales...>

Modifies the scale of the spectral data.

<Decimal Places...>

Modifies the format of the spectral data, score, and threshold.

Click the <Compare Settings...> button in [Data Comparison] dialog box and [Data Comparison Settings] dialog box opens.

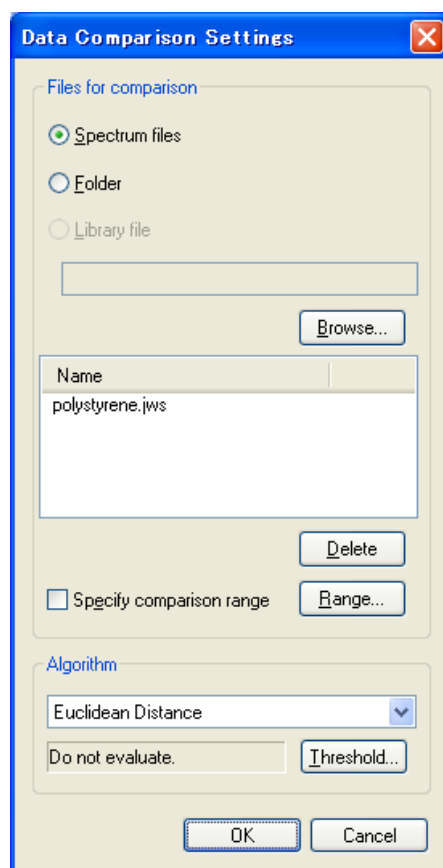


Figure 3.70 [Data Comparison Settings] dialog box

[Files for comparison]	Setting of the comparison file.
[Spectrum files]	Selected spectra are comparison objects.
[Folder]	All spectra in a chosen folder are comparison objects.
[Library file]	The spectra of library files made in [Spectra Search] program are comparison objects.
<Browse...>	Select comparison objects from spectrum file or folder or library file.
<Delete>	A selected spectrum from [Spectrum files] is deleted.
[Algorithm]	Selects algorithm comparing.
[Threshold...]	Sets the threshold of Data Comparison.

3.6 [Window] Menu

The commands contained in the [Window] menu are described below.

[C <u>a</u> scade]	Cascades all open Views.
[T <u>i</u> le H <u>o</u> rizontally]	Tiles all open Views horizontally.
[T <u>i</u> le V <u>e</u> rtically]	Tiles all open Views vertically.
[A <u>r</u> range <u>I</u> cons]	Arranges the minimized window icons.
[C <u>l</u> ose All]	Closes all Views. If any spectral data has been changed or if the data is new, a message is displayed which asks if you wish to save the data before closing. (Fig. 3.68) Click the <Y<u>e</u>s> button to save the data before closing. Click the <N<u>o</u>> button to close the View without saving. Click the <C<u>a</u>ncel> button to cancel closing the View.

Figure 3.71 Save message

3.7 [Other] Menu

3.7.1 [Customize Toolbar...]

Customizes the [Data Processing] toolbar in the tool button group, which is divided into four categories (Fig. 3.72).

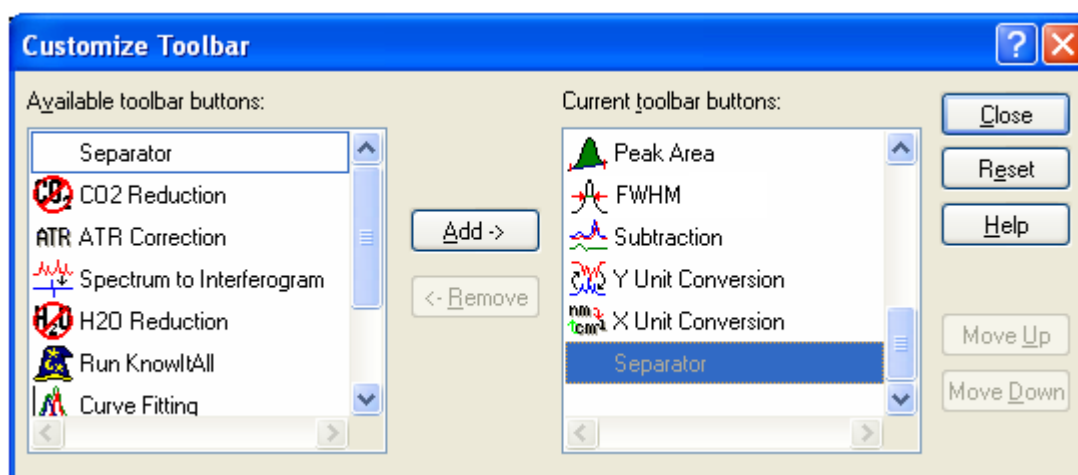


Figure 3.72 [Customize Toolbar] dialog box

[Available toolbar button]	Select a tool button and click the <Add> button to register the toolbar button and move it to the [Current toolbar button].
[Current toolbar button]	Displays the currently registered toolbar buttons. Selecting a tool button and clicking the <Remove> button removes the button from the toolbar and moves the toolbar button to the [Available toolbar button] list.
[Add->]	Registers a selected tool button from the [Available toolbar button] list.
[<- Remove]	Removes a tool button from the [Current toolbar button] list.
[Reset]	Returns the toolbar to the settings before the start of registering toolbar buttons.
[Move Up], [Move Down]	Changes the order of a selected button in the [Current toolbar button] list.
<Close>	Closes the dialog box.

3.7.2 [Options...]

Select [Other] - [Options...] to display the [Options] dialog box used to change the View and print style options (Fig. 3.73).

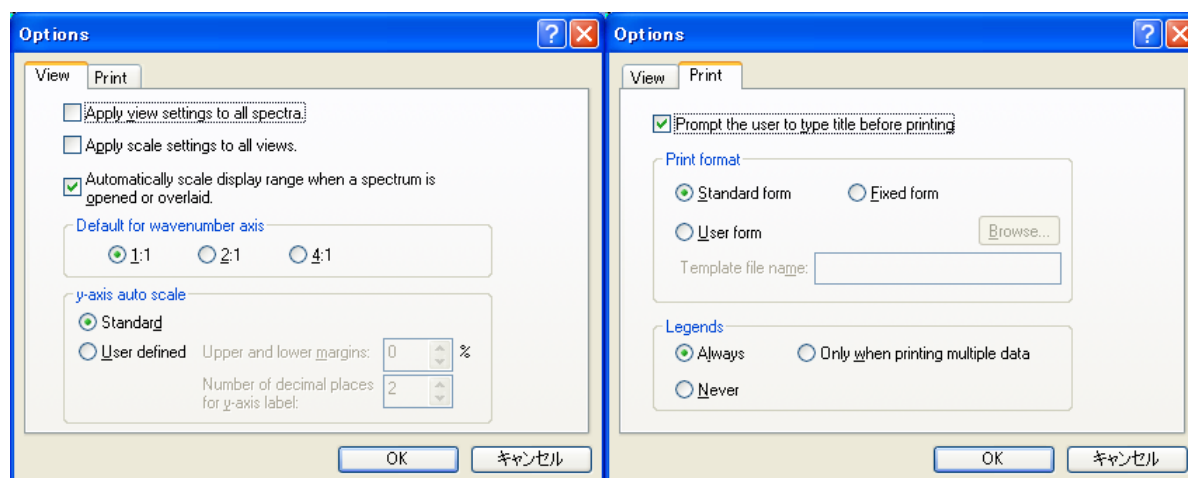


Figure 3.73 [Options] dialog box

[View] Tab

[Apply view settings to all spectra]	Place a check in the checkbox to apply the parameter settings of the [Patterns], [Fonts], and [Gridlines] dialog boxes from the [View] menu to all open Views.
[Apply scale settings to all views]	Place a check in the checkbox to apply the parameter settings of the [Scales] dialog box from the [View] menu to all open Views.
[Automatically scale display range when a spectrum is opened or overlaid]	Place a check in the checkbox to use the y-axis auto scale function when opening or overlaying spectra in a View.
[Default for wavenumber axis] group	Select the default setting for the wavenumber axis. Select: 1:1 for a constant wavenumber interval 2:1 for an expanded scale from 2000 cm ⁻¹ and below, values above 2000 cm ⁻¹ will be compressed 2:1 compared to data displayed below 2000 cm ⁻¹ 4:1 for an expanded scale from 2000 cm ⁻¹ and below, values above 2000 cm ⁻¹ will be compressed 4:1 compared to data displayed below 2000 cm ⁻¹
[y-axis auto scale] group	Allows customization of the settings used for the y-axis auto scale.
[Standard]	Fits the y-axis scale to the maximum and minimum values displayed in the View.
[User defined]	
Upper and lower margins	Automatically calculates the y-axis scale with a specified proportion of blank space above and below the maximum and minimum values displayed in the View. Settings range from 0 to 20%.
Number of decimal places for y-axis label	Specifies the number of significant figures to be used for the y-axis label. Settings range from 2 to 6 significant figures. The y-axis label is always rounded up.

[Print] Tab

[Prompt the user to type title before printing]	Place a check in the checkbox to be prompted to enter a title for the printout before printing.
[Print format] group	
[S <u>ta</u> ndard form]	Select to print using the standard format.
[F <u>i</u> xed form]	Select to print using the fixed format.
[U <u>s</u> er form]	Select to print using a template file created in JASCO Canvas. Click the <Browse...> button and designate the JASCO Canvas template file to be used.
[Legends] group	
[A <u>l</u> ways]	Select to always print a legend with the spectra.
[N <u>e</u> ver]	Select to never print a legend with spectra.
[Only <u>w</u> hen printing multiple data]	Select to print a legend with spectra only when multiple data are displayed.
<OK>	Click to apply the settings and close the dialog box.
<Cancel>	Click to close the dialog box without applying any changes to the settings.

3.7.3 [Decimal Places...]

Sets the display format for the vertical and horizontal axes. (Fig. 3.74) The format set here will be reflected in the numeric value field for each item in the status bar, the [Processing]-[Peak Processing] submenu and the [View Offset] dialog box (see Sections 3.5.3 and 3.1.4.6, respectively).

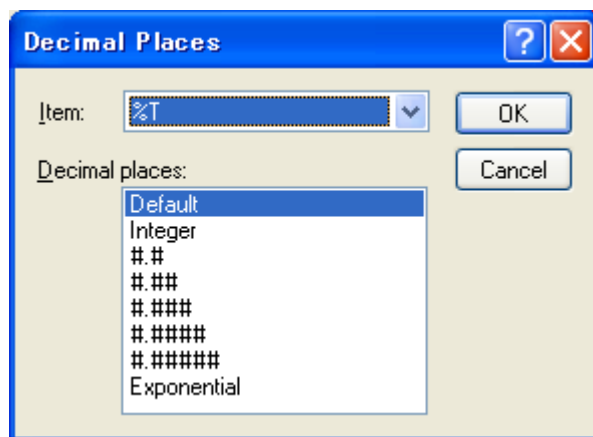


Figure 3.74 [Decimal Places] dialog box

[Item]	Selects the units of the vertical and horizontal axis for which the format is to be set.
[Decimal Places]	Selects the decimal point format for the unit selected with [Item].
<OK>	Applies the format settings and closes the dialog box.
<Cancel>	Closes the dialog box without applying any changes to the format settings.

Note: The View format is set with [View]-[Styles]. (Refer to 3.4.8.)

3.8 [Help] Menu

3.8.1 [Contents]

Displays the contents of the Help function for the [Spectra Analysis] program with the [Contents] tab.

3.8.2 [Search Topic]

Displays the contents of the Help function for the [Spectra Analysis] program with the [Index] tab

3.8.3 [About...]

Displays information about the version of the [Spectra Analysis] program and product license information.

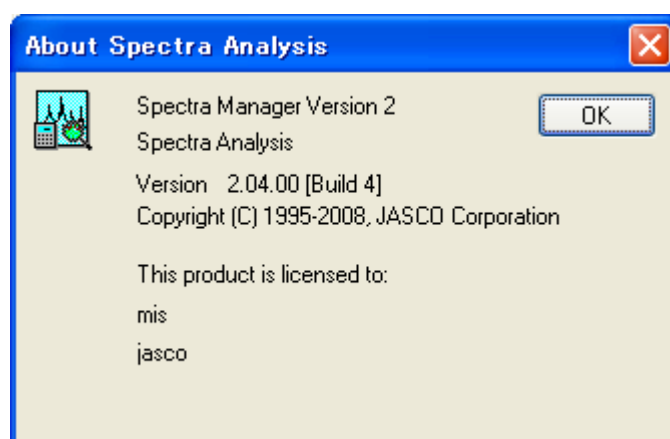


Figure 3.75 [About Spectra Manager] dialog box

4. Appendix

4.1 JCAMP-DX Format Files

The JCAMP-DX file format allows data conversion between various computers and media. This format was created by JCAMP (Joint Committee of Atomic and Molecular Physical Data) ¹⁾. The JCAMP-DX format offers the following advantages:

- Format file data can be easily transferred between large computers, multi-purpose laboratory computers, common data processing systems, workstations, and personal computers.
- Because JCAMP-DX files are text files, the data can be viewed or edited using various commercially available editors.
- In addition to spectra, peak tables and three-dimensional data can be handled.

The Spectra Analysis software can save measured and calculated data in the JCAMP-DX format. Data saved in this format can be manipulated using the Grams data processing program or the Sadtler reference program; among others. Although JCAMP-DX has several formats, only the FT-IR format is explained here. The following explanation requires some knowledge of ASCII code. A sample of JASCO's JCAMP-DX obtained using a JASCO FT-IR is given below as an example.

##TITLE= Sample name	JCAMP-DX files always start with the ##TITLE= line.
##JCAMP-DX= 4.24	JCAMP-DX version
##DATA TYPE= INFRARED SPECTRUM	Designation for IR spectrum.
##ORIGIN= JASCO	Always JASCO.
##OWNER= xxxxxx	Contents entered in [Organization] in [Comment Edit]
##DATE= 96/08/17	Measurement date
##TIME= 13:33:46	
##SPECTROMETER/DATA SYSTEM	
##RESOLUTION=	
##DELTA X= 1.92905045	Interval between data points (units in ##XUNITS=)
##XUNITS= 1/CM	Units of horizontal axis
##YUNITS= TRANSMITTANCE	Units of vertical axis (attribute)
##XFACTOR= 1.0000	Actual value is obtained by multiplying value in ##XYDATA= by this factor.
##YFACTOR= 0.00001	
##FIRSTX= 370.3777	First value for x-axis of ##XYDATA= (actual value)
##LASTX= 4627.7920	Last value for x-axis of ##XYDATA= (actual value)
##NPOINTS= 2208	Number of data points
##FIRSTY= 74.58664	Y-axis value corresponding to ##FIRSTX= (actual value)
##MAXY= 91.44050	Maximum value of y-axis (actual value)
##MINY= 0.46792	Minimum value of y-axis (actual value)
##XYDATA= (X++(Y..Y))	Data are written from this point on
370.3777 7458664 7164965 6797073 6463581 6262086 6216446 6313309 6520278 385.8191	
6759706 6993072 7203347 7370707 7518471 7794603 7953664	
----- (all data not shown)	
4608.5015 8470810 8478381 8484588 8489256 8495972 8500248 8504088 4623.9341 8505672	
85807172 8509182	
##END=	JCAMP-DX files always end with the ##END=.

4.1.1 JCAMP-DX file structure

4.1.1.1 Labeled-Data-Records (LDRs)

Labeled Data Records (LDRs) consist of a data label (described later) and data. LDRs start with the `##` data label flag and end with the next data label flag.

4.1.1.2 Lines

One line can contain up to 80 characters. The terminator for a line is CR-LF. LDRs can include several lines. Therefore, CR-LF does not necessarily designate the end of an LDR.

4.1.1.3 Data labels

A data label clarifies an LDR. Several labels are defined in the above example and further details are provided in Section 3.1.2. Data labels include the reserved labels defined by JCAMP as well as user defined labels that can record any information input by the user. The JASCO Spectra Analysis software does not employ user-defined labels.

4.1.1.4 Blocks

A block is a set of LDRs starting with `##TITLE=` and ending with `##END=`. A single JCAMP-DX file can contain multiple blocks (the Spectra Analysis software only responds to a one-block format).

4.1.2 Reserved labels

Only JCAMP-DX reserved labels that the Spectra Analysis software can use are described here. Reserved labels that start with `##`, and multiple labels cannot be placed on one line.

- (1) `##TITLE=`
All JCAMP-DX files begin with the LDR `##TITLE=`. Spectra Analysis data files use the contents of the "Sample name" field as the `##TITLE=`.
- (2) `##JCAMP-DX= 4.24`
This LDR always follows `##TITLE=`, and indicates the version of the file format.
- (3) `##DATA TYPE=`
Defines the data format of the block. The `##DATA TYPE=` includes ULTRAVIOLET SPECTRUM, FLUORESCENCE SPECTRUM, RAMAN SPECTRUM, INFRARED PEAK TABLE, INFRARED INTERFEROGRAM, and INFRARED TRANSFORMED SPECTRUM.
- (4) `##ORIGIN=`
Name and address of the file creator. This is always JASCO.
- (5) `##OWNER=`
Name of the owner of the spectrum. Here, the content entered is described in the "Organization" field of a Spectra Analysis data file.
- (6) `##DATE=, ##TIME=`
The date (yy/mm/dd) and time (hh:mm:ss) when the spectrum was measured.
- (7) `##SPECTROMETER/DATA SYSTEM=`
Describes the measurement system and data processing system. This reflects the instrument

used to record the spectral data and the Spectra Manager version.

(8) **##RESOLUTION=**

(9) **##DELTAX=**

Interval between data points. The unit is **##XUNITS=**. In the case of data subjected to zero filling, the unit differs based on the value of **##RESOLUTION=**, as mentioned earlier.

(10) **##XUNITS=**

Unit (attribute) of the horizontal axis. **##XUNITS=** is always 1/CM (cm^{-1}) because Spectra Manager cannot save an interferogram as a JCAMP-DX file. Other **##XUNITS=** include MICROMETERS, NANOMETERS and SECONDS.

(11) **##YUNITS=**

Unit of the vertical axis. Spectra Manager supports the following units (units enclosed by parentheses are as named in Spectra Manager). TRANSMITTANCE (%T), REFLECTANCE (%R), ABSORBANCE (ABS), KUBELKA-MUNK (KM) and ARBITRARY UNITS (other y-axis labels).

(12) **##XFACTOR=, ##YFACTOR=**

Multiplying the data of **##XYDATA=** by this factor gives the actual value.

(13) **##FIRSTX=, ##LASTX=**

The first value and the last value (actual value) of the horizontal axis of **##XYDATA=**.

(14) **##NPOINTS=**

Number of data points of **##XYDATA=**.

(15) **##FIRSTY=**

The value of the vertical axis (actual value) corresponding to **##FIRSTX**.

(16) **##MAXY=, ##MINY=**

The maximum and minimum values (actual values) of the vertical axis.

(17) **##XYDATA=**

Data starts here. Indicates that (X++(Y..Y)) is the data format and that the first datum in each row is the X-axis datum and that the remaining data on the row until the CR-LF are Y-axis data.

In the examples above, the data 370.3777, 385.8101,... represent wavenumbers. Here, 7458664 indicates the data at 370.3777 cm^{-1} and 7164965 indicates the data at $(370.3777 + \text{##DELTAX}) \text{ cm}^{-1}$. (The actual value can be obtained by multiplying the y-axis data by **##YFACTOR=**.)

JCAMP-DX defines other data formats, but Spectra Manager is only compatible with this data format.

(18) **##END=**

All JCAMP-DX blocks terminate with **##END=**.

4.1.3 User-defined labels

In addition to the labels reserved by JCAMP-DX, users can create new labels. These labels are called user-defined labels. User-defined labels start with a \$. Spectra Manager CFR cannot output user-defined labels.

4.1.4 Adding comments

(1) **##=**

Comments starting with this label can occupy any number of lines up to the next **##** data label flag.

(2) **\$\$**

Comments starting with **\$\$** can be placed anywhere on a line. This type of comment can only continue to the end of the line and does not designate the end of the LDR.

Reference Literature

1) R.S. McDonald, P.A. Wilks Jr., Appl. Spectrosc. 42, 151(1988)

4.2 Smoothing Methods

Smoothing is used to eliminate noise components from a spectrum in order to improve the signal-to-noise ratio. It is important that the optimum method is selected for a given waveform because smoothing may distort spectrum analyte bands. The [Spectra Analysis] program has the following four types of smoothing methods. All are based on the moving averaging method, in which appropriate weights are assigned to the data point of interest and surrounding points.

1. Simple moving average

This method is used when the spectrum bandwidth is wide, that is, when the signal component is sufficiently smooth compared with the noise. The simple moving average method uses a rectangular function as a weighting function and produces a large smoothing effect; however, a significant amount of signal distortion can also occur.

2. Savitzky-Golay method

This method can be used to eliminate noise from most spectra. It uses the weighting algorithm outlined by Savitzky and Golay¹⁾. The weighting function changes depending on the data width (convolution width) setting used for smoothing.

3. Adaptive smoothing method

This method is convenient for eliminating noise from sharp spectra, such as gas phase spectra. The spectrum is distorted only slightly in the peak area and the smoothing effect can be enhanced in the baseline area by choosing weighting coefficients based on the noise variance and local signal variance.

4. Binomial method

In this smoothing method, the weighting is calculated using binomial coefficients²⁾. Compared with the Savitzky-Golay method, the binomial method has a number of advantages, such as no distortion over the entire waveform, no phase (peak position) shift, and a short processing time.

Reference Literature

1) A. Savitzky, M. Golay, Anal. Chem. 36, 1627(1964).

2) P. Marchand, L. Marmet, Rev. Sci. Instrum. 54, 1034(1983).

4.2.1 Moving average method

The operating principle of the moving average method is based on a convolution calculation, similar to that for an analog filter. The output $y(t)$ of an analog filter is expressed by the following equation representing convolution of the input signal $x(t)$ and the filter impulse response $h(t)$.

$$y(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t - \tau) d\tau = \int_{-\infty}^{\infty} h(\tau) \cdot x(t - \tau) d\tau$$

In the moving average method, the input signal is expressed by n discrete values $x(i)$ (where $i = 1, 2, \dots, n$), and the smoothed value $y(i)$ is determined as shown below by using a weighting function, $w(j)$, consisting of $N (=2m + 1)$ discrete points (where, $j = -m, \dots, -1, 0, 1, \dots, m$).

$$y(i) = \frac{1}{W} \sum_{j=-m}^m w(j) \cdot x(i + j) \quad : i = m + 1, m + 2, \dots, n - m \quad (1)$$

$$W = \sum_{j=-m}^m w(j) \quad (2)$$

where W is a normalization constant.

4.2.2 Simple moving average method

This is the simplest smoothing method and uses a rectangular function as a weighting function. For a rectangle with a width of $N = 2m + 1$, the operation is described by the following expression, which is derived by setting n equation (2):

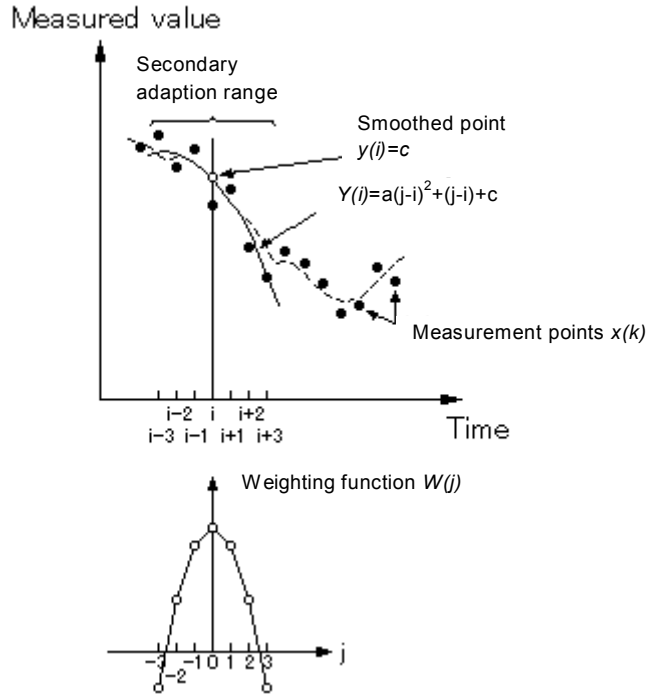
$$y(i) = \frac{1}{N} \sum_{j=-m}^m x(i + j) \quad : i = m + 1, m + 2, \dots, n - m$$

4.2.3 Savitzky-Golay method

This is a method of fitting the measured spectrum to a polynomial curve, based on the least squares error criterion in the neighborhood of each sample point. The type of weighting function varies with the order of the polynomial. A detailed table of values (weighting coefficients) for each discrete point of this weighting function was first reported by Savitzky and Golay. Thus, the smoothing operation can be carried out simply by performing a convolution using the coefficient table below.

Discrete point No.	25	23	21	19	17	15	13	11	9	7	5
-12	-253										
-11	-138	-42									
-10	-33	-21	-171								
-9	62	-2	-76	-136							
-8	147	15	9	-51	-21						
-7	222	30	84	24	-6	-78					
-6	287	43	149	89	7	-13	-11				
-5	342	54	204	144	18	42	0	-36			
-4	387	63	249	189	27	87	9	9	-21		
-3	422	70	284	224	34	122	16	44	14	-2	
-2	447	75	309	249	39	147	21	69	39	3	-3
-1	462	78	324	264	42	162	24	84	54	6	12
0	467	79	329	269	43	167	25	89	59	7	17
1	462	78	324	264	42	162	24	84	54	6	12
2	447	75	309	249	39	147	21	69	39	3	-3
3	422	70	284	224	34	122	16	44	14	-2	
4	387	63	249	189	27	87	9	9	-21		
5	342	54	204	144	18	42	0	-36			
6	287	43	149	89	7	-13	-11				
7	222	30	84	24	-6	-78					
8	147	15	9	-51	-21						
9	62	-2	-76	-136							
10	-33	-21	-171								
11	-138	-42									
12	-253										
Normalization coefficient	5175	805	3059	2261	323	1105	143	429	231	21	35

The coefficients in this table were obtained by fitting to quadratic and cubic polynomials. The following figure shows the principle of smoothing using seven discrete points.



In this example, it is assumed that within the range $i-m$ to $i+m$ (number of points is $2m+1$) about a given data point i (seven points in the above figure), the measured spectrum $y(j)$ can be expressed by the following quadratic polynomial:

$$y(j) = a(j-i)^2 + b(j-i) + c \quad : j = -m+i, \dots, -1+i, i, 1+i, \dots, m+i$$

Here, the coefficients a , b and c are determined by minimizing the sum of the squared errors between the polynomial and the measured data points, given by:

$$\sum_{j=-m+i}^{m+i} \{x(j) - y(j)\}^2$$

The smoothed value $y(i)$ at the center point i of the quadratic function is then given by c . The coefficient table contains the weighting coefficients used to determine the coefficient c . Therefore, the smoothed value $y(i)$ at point i is calculated using the $(2m+1)$ measurement values $x(j)$ and the coefficients $w(j)$ ($J = -m, \dots, -1, 0, 1, \dots, m$) from the table as follows:

$$y(i) = \frac{1}{W} \sum_{j=-m}^m x(i+j)w(j)$$

$$W = \sum_{j=-m}^m w(j)$$

4.2.4 Adaptive smoothing method

It is assumed that the measured data curve spectrum $x(i)$ can be expressed as the sum of a signal component $s(i)$ and a noise component $n(i)$:

$$x(i) = s(i) + n(i):$$

It is also assumed that $n(i)$ is random static noise (i.e., noise that does not change with time) having an average value of zero and a variance σ_n^2 , and that the signal is also static within a short period of time. The estimated value of the signal that satisfies the least squares error criterion is denoted by $\hat{s}(i)$. That is, $\hat{s}(i)$ is the value that minimizes $E\{(\hat{s}(i) - s(i))^2\}$ and is expressed as:

$$\hat{s}(i) = \frac{\sigma_x^2(i) - \sigma_n^2}{\sigma_x^2(i)} \{x(i) - \bar{x}(i)\} + \bar{x}(i)$$

$$\bar{x}(i) = E\{x(i)\}$$

$$\sigma_x^2 = E\{(x(i) - \bar{x}(i))^2\}$$

(Strictly speaking, $E\{\}$ represents an aggregate average, but here it is given as a local average that varies with time.)

The noise variance σ_n^2 is obtained from actual measurements. $\sigma_x^2(i)$ is determined by a simple moving average with a constant width N , and the estimated value of the true signal component can be obtained by calculating the variance dispersion $\sigma_x^2(i)$ of the measured data curve within this width.

4.2.5 Binomial method

This is a moving average method that is very simple to implement and produces no rippling effects after smoothing. Smoothing is performed by repeatedly applying the following equation a designated number of times:

$$y(i) = \frac{1}{4}(x(i-1) + 2 \times x(i) + x(i+1))$$

The table below gives the coefficients calculated using the weighting function $w(j)$ and the normalization constant W for different cycle numbers. In contrast to the Savitzky-Golay method based on the polynomial approximation, in this method the weighting coefficients cannot be negative. Thus, no rippling occurs in the neighborhood of abrupt changes.

Discrete point No.	12	11	10	9	8	7	6	5	4	3	2	1
-12	1											
-11	24	1										
-10	276	22	1									
-9	2024	231	20	1								
-8	10626	1540	190	18	1							
-7	42504	7315	1140	153	16	1						
-6	134596	26334	4845	816	120	14	1					
-5	346104	74613	15504	3060	560	91	12	1				
-4	735471	170544	38760	8568	1820	364	66	10	1			
-3	1307504	319770	77520	18564	4368	1001	220	45	8	1		
-2	1961256	497420	125970	31824	8008	2002	495	120	28	6	1	
-1	2496144	646646	167960	43758	11440	3003	792	210	56	15	4	1
0	2704156	705432	184756	48620	12870	3432	924	252	70	20	6	2
1	2496144	646646	167960	43758	11440	3003	792	210	56	15	4	1
2	1961256	497420	125970	31824	8008	2002	495	120	28	6	1	
3	1307504	319770	77520	18564	4368	1001	220	45	8	1		
4	735471	170544	38760	8568	1820	364	66	10	1			
5	346104	74613	15504	3060	560	91	12	1				
6	134596	26334	4845	816	120	14	1					
7	42504	7315	1140	153	16	1						
8	10626	1540	190	18	1							
9	2024	231	20	1								
10	276	22	1									
11	24	1										
12	1											
Normalization constant	16777216	4194304	1048576	262144	65536	16384	4096	1024	256	64	16	4

4.3 Application of the Self-deconvolution Method

The Fourier self-deconvolution method is used to precisely distinguish the peak positions of individual bands in measured spectra containing multiple bands with the same width. The following conditions apply when using this method.

- (1) The program assumes that the spectral band shapes are Lorentzian.
- (2) After processing, only the peak position of each band is meaningful. The reason for this is that vertical axis information is sacrificed in order to elucidate horizontal axis information.
- (3) The program cannot be applied to measured spectra consisting of bands with different widths. If the half width at half maximum of a band is σ' and that of the deconvolution filter is σ , excessive deconvolution will occur when $\sigma > \sigma'$ and insufficient deconvolution will occur when $\sigma < \sigma'$.
- (4) The signal-to-noise ratio of the measured spectrum will be reduced after processing because, as with differential operations, the deconvolution filter emphasizes high frequencies.

Principles

The program is an improved version of both the FSD (Fourier self-deconvolution)¹⁾ method proposed by Kauppinen et al. and the FIRO (finite impulse response operator)²⁾ method formulated by Jones and Shimokoshi. Since both methods can be explained based on the FSD principles, only the FSD method will be described here.

If the real spectrum is given by $E(\nu)$, the measured spectrum $M(\nu)$ is generally distorted by the instrument function $G(\nu)$ of the measurement system. If the system is linear, the following equation applies:

$$M(\nu) = G(\nu) * E(\nu) \quad (1)$$

where the $*$ symbol expresses a convolution integral. If $I_M(x)$ and $I_E(x)$ are taken to be the inverse Fourier transforms of $M(\nu)$ and $E(\nu)$, respectively, the equation becomes

$$I_E(x) = \frac{1}{F^{-1}\{G(\nu)\}} I_M(x) \quad (2)$$

where $F^{-1}\{G(\nu)\}$ is the inverse Fourier transform of $G(\nu)$. In other words, this gives the ideal inverse filter in the interferogram region. This filter is extremely sensitive to noise, so an apodization function $D(x, L)$ is introduced into equation (2) to cut the high-frequency noise using the FSD method

$$I_E(x) = \frac{D(x, L)}{F^{-1}\{G(\nu)\}} I_M(x) \quad (3)$$

where L is the frequency to be eliminated.

The instrument function $G(\nu)$ is assumed to be Lorentzian with a half width at half maximum σ . Therefore, equation (3) becomes

$$I_E(x) = \frac{D(x, L)}{\exp\{\pi\sigma|x|\}} I_M(x) \quad (4)$$

By taking the Fourier transform of equation (4), the deconvoluted profile $E(\nu)$ can be obtained as

$$E(\nu) = F \left\{ \frac{D(x, L)}{\exp\{\pi\sigma|x|\}} I_M(x) \right\} \quad (5)$$

These are the principles of the FSD method.

Equation (5) can be converted as follows:

$$E(\nu) = F \left\{ \frac{D(x, L)}{\exp\{\pi\sigma|x|\}} \right\} * M(\nu) \quad (6)$$

The FIRO method involves expanding the term $\exp\{\pi\sigma|x|\}$ in equation (6) in the spectrum region for various types of apodization functions $D(x, L)$.

In this program, in order to reduce the calculation time, an algorithm that combines the FSD and FIRO methods is used. The deconvolution filter that is ultimately used on the spectrum has the following form:

$$F \left\{ \frac{D(x, L)}{\exp\{\pi\sigma|x|\}} \right\} * D'(\nu, FL)$$

where $D(x, L)$ and $D'(\nu, FL)$ are taken to be Bessel functions, and L and FL are taken to be automatically determined values.

Reference literature

- (1) Appl. Spectrosc. 35, 271 (1981).
- (2) Appl. Spectrosc. 37, 59 (1983).

4.4 Kubelka-Munk Transform (K-M Transform)

The Kubelka-Munk equation expresses the relationship between the diffuse reflectance and the absorption/scattering coefficient. In JASCO [Spectra Manager], the function that transforms the reflectance to the extinction/diffusion coefficient using the Kubelka-Munk equation is called the “K-M Transform”. After this transform has been applied, the vertical data axis is K/M.

4.4.1 Principle of K-M Transform

The interaction between light and a diffusely reflecting sample is considered to be as follows. The surface and interior of the sample consist of many fine particles. When light is incident on the sample, it is repeatedly transmitted, absorbed, or reflected by the fine particles as it enters the sample. Figure 1 illustrates this process for a single particle layer with a thickness dx at an arbitrary depth x in the sample. Assuming K to be the extinction coefficient and S to be the diffusion coefficient, and taking the direction of the light into consideration, equations (1a) and (1b) express the variation of the transmitted light intensity i and reflected light intensity j .

$$d_i = -(K + S)i \times dx + Sj \times dx \quad \dots \quad (1a)$$

$$-d_j = -(K + S)j \times dx + Si \times dx \quad \dots \quad (1b)$$

Kdx is the attenuation due to absorption of light beam i , $Sidx$ is its attenuation due to scattering, $Kjdx$ is the attenuation due to absorption of light beam j , and $Sjdx$ is its attenuation due to scattering.

Kubelka and Munk solved the above differential equations and expanded them to include the measured quantity as a parameter. The Kubelka-Munk equation is expressed as follows:

$$\frac{K}{S} = \frac{(1 - R_\infty)^2}{2R_\infty} \quad \dots \quad (2)$$

R_∞ : R_∞ represents the reflectance for a sample thick enough for the effect of the base to be negligible.

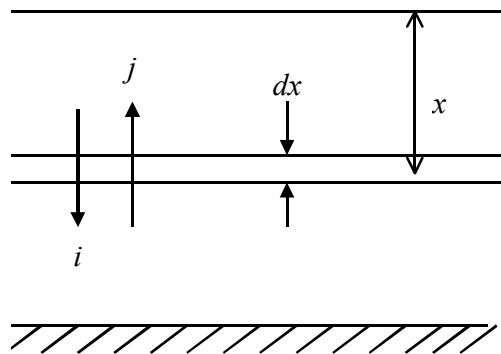


Figure 1 Repeated transmission, absorption and reflection of light incident on a sample

4.5 Kramers-Kronig Transform (K-K Transform)

The Kramers-Kronig transform calculates the wavelength dispersion of the refractive index n and extinction coefficient k from the specular reflectance spectrum of a sample.

4.5.1 Principle of the Kramers-Kronig Transform

In the following, normal incidence is considered. The reflected amplitude r and energy R can be expressed by the following equations:

$$\mathbf{r} = |\mathbf{r}| \exp(i\phi) = \frac{1 - (n + ik)}{1 + (n + ik)} \quad (1)$$

$$R = |\mathbf{r}|^2 = \frac{(n-1)^2 + k^2}{(n+1)^2 + k^2} \quad (2)$$

Here, ϕ is the phase difference upon reflection.

The refractive index n and extinction coefficient k can be expressed using the following equations:

$$n = \frac{1 - R}{1 + R - 2\sqrt{R} \cos \phi} \quad (3)$$

$$k = \frac{-2\sqrt{R} \sin \phi}{1 + R - 2\sqrt{R} \cos \phi} \quad (4)$$

The following Kramers-Kronig relation is applied to calculate the phase difference.

$$\phi(\omega) = \frac{2\omega}{\pi} P \int_0^\infty \frac{\ln \sqrt{R(\omega')}}{\omega'^2 - \omega^2} d\omega' \quad (5)$$

The integral in equation (5) contains a pole, and several methods exist to perform the integration.

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